



DM-OTC18T Overhead Tube Crane System

Operation Manual

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Important: Consult accompanying documentation as applicable.

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Notices and Conventions

Safety Conventions

Specific safety information is presented in this manual in the form of Warning and Caution statements. Pay close attention to these statements—they contain important information on avoiding hazards to you or the equipment.

Warning Statements

- Indicate hazards or unsafe practices that can cause severe injury, or product or property damage.
- Appear in **bold** type.
- Have a triangular symbol enclosing an exclamation point, with the word **Warning**.
- Precede the step or piece of information to which they refer.
- Appear as the following example:



Warning: This text describes special safety precautions to avoid unsafe practices that can result in severe injury or death.

Caution Statements

- Indicate hazards or unsafe practices that can cause minor injury, or product or property damage.
- Appear in **bold** type.
- Have a triangular symbol enclosing an exclamation point, with the word **Caution**.
- Precede the step or piece of information to which they refer.
- Appear as the following example:



Caution: This text describes special safety precautions to avoid unsafe practices that can result in injury or product or property damage.

Other Conventions

Important

Provides critical information that enhances understanding or ensures proper operation. While not a safety warning, ignoring this information may result in suboptimal performance, potential confusion, or unintended consequences.



Important: Important appear throughout the documentation to provide key explanations and recommendations.

Notes

Notes offer extra details, including clarifications, tips, and reminders.



Note: Notes appear throughout the documentation to provide additional explanations and recommendations.

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S1 Safety Information

S1.1 Introduction

Del Medical Inc. X-Ray equipment meets high standards of performance and reliability. We enforce strict quality control techniques to eliminate the potential for defects and hazards in our products.

These safety guidelines are intended to educate the operator on safety considerations for the proper operation and maintenance of the DM-OTC18T Overhead Tube Crane System.



Warning: Read and fully understand this manual before operating this equipment. Failure to follow instructions can result in serious injury or death.

S1.2 Statement of Liability

To prevent excess radiation exposure (primary or secondary) to patients or staff, this equipment must be operated and serviced by trained personnel knowledgeable in the required safety precautions. While this equipment is designed for safe use, improper operation can cause serious injury or equipment damage. Del Medical Inc. or its agents and representatives assume no responsibility for the following:

1. Injury or danger to any person from X-ray exposure.
2. Overexposure of patients or staff to X-radiation due to improper techniques or procedures.
3. Injury or danger from improper use of the adjustable column or tube head assembly.
4. Problems or hazards resulting from failure to maintain the equipment as specified in this manual.
5. Damage or injury caused by equipment that has been modified or used in any application other than that intended by the manufacturer.
6. Damage or injury caused by operation of this equipment by unauthorized or untrained persons.

7. Damage or injury arising from failure to follow the instructions and procedures in this manual or associated documents, or from failure to use caution when installing, operating, adjusting, or servicing this equipment.

S1.3 Safety Conventions Used in this Manual

Specific safety information is presented in this manual in the form of **Warning** and **Caution** statements. Pay close attention to these statements, they contain important information on avoiding hazards to you or the equipment.

Please refer to *"Notices and Conventions" on page 3* for more information regarding Safety Conventions.

S1.4 Equipment Safety Guidelines

The DM-OTC18T Overhead Tube Crane System is intended for **medical diagnostic purposes only**. Observe the following general safety precautions:

- Do not use this equipment for anything other than medical diagnosis.
- Observe all warnings and cautions, stated or implied, in this manual and the manuals for associated equipment.
- Comply with all safety labels on the equipment.
- Permit only properly trained and qualified personnel to operate this equipment. Because this equipment produces ionizing radiation, operators must meet all state and local requirements and regulations.
- Permit only qualified personnel to access internal parts.
- When planning for and installing this equipment, observe all local and national regulations, and establish safe exposure procedures.
- X-rays pose a health risk for both patients and operators. Take all measures to minimize radiation exposure to any persons.
- Routinely inspect all movable assemblies and parts of this equipment according to this manual.
- Provide protective earth grounding for all components in compliance with applicable regulations.

- This equipment contains no user-serviceable parts. For service assistance, contact your service representative.
- Do not defeat or bypass built-in equipment safety features.
- Do not operate this equipment in the presence of moisture.

The following warnings and cautions are specific to the DM-OTC18T Overhead Tube Crane System. Read them carefully—some are not obvious for typical equipment use:



Warning: To avoid the dangerous and cumulative effects of X-rays, minimize radiation exposure to any and all patients and staff.



Warning: Do not operate this equipment in an explosive atmosphere (such as anesthetic gas). This equipment is not suitable for use in the presence of a flammable anesthetic mixture with air, oxygen, or nitrous oxide.



Warning: To avoid injury when moving this equipment, at least 2 qualified people are required.



Warning: Any serious incident should be reported to the manufacturer and the competent authority of the member state in which the user is established, return equipment to the manufacturer promptly.

S1.5 Radiation Protection

Serious detrimental health effects can result from exposure to both short-term/high-level and long-term/low-level X-rays. Become familiar with the effects of both types of radiation exposure, and take appropriate measures to minimize exposure. Some effects of X-radiation are cumulative over a period of months or years. Most importantly, **avoid exposure to the primary beam at all times.**

X-Ray facilities typically use lead to shield the radiologist's workstation. The typical workstation partition is generally effective in reducing residual radiation to a safe level. Windows in the partition are typically glazed with lead glass and provide effective protection. Do not leave the protected area during exposure.

Any object in the path of the primary beam produces scattered radiation, which can result in a substantial radiation dose to unprotected persons. To minimize exposure to scattered radiation, use movable lead screens, and lead-impregnated gloves and aprons. These protective devices must contain a lead (or equivalent) thickness of at least 0.25 mm. Use such protective devices for all operators, observers, and/or servicing personnel exposed to radiation fields of five or more milli-Roentgens per hour.

The X-Ray system produces X-rays only when high voltage is applied to the X-ray tube. When the high voltage is removed, X-ray emission ceases immediately.

When the distance between the X-ray tube focal spot and the patient's skin is decreased, the radiation dose to the patient increases.

Ensure audio and visual communication between operator and patient even during exposure. This may require video cameras or an intercom.

To minimize risk, ensure that all operating personnel are educated in the hazards of radiation and understand the safety requirements for X-ray operation. Review the manuals for each component in the system.



Warning: Ensure exposure parameters are set to within safety limits.



Caution: Incorrectly positioning the X-ray tube and collimator can cause the X-ray field to be misaligned, resulting in an unacceptable image and the need to re-expose.

S1.6 Radiation Safety Regulation

Ensure that all operators of this equipment, and their supervisors, are fully acquainted with recommendations from the following authorities:

- Center for Devices and Radiological Health (CDRH)
- National Institute of Standards and Technology (NIST)
- National Council on Radiation Protection and Measurements (NCRP)
- International Commission on Radiological Protection (ICRP).

Current sources of information include the following:

- *National Council on Radiation Protection Report No. 102* (“Medical X-ray, Electron Beam and Gamma-Ray Protection for Energies up to 50 MeV [Equipment Design, Performance and Use]”).
- Current recommendations of the International Commission on Radiological Protection.

To learn more about shielding and licensing requirements for your area, visit www.crcpd.org (Conference of Radiation Control Program Directors), select Radiation Control Programs, and locate your state office.

S1.7 Radiation Monitoring

Monitoring radiation exposure to personnel provides a valuable assessment of safety measures, and can reveal inadequate or improper protection practices or serious exposure situations.

The most effective monitoring method is the use of instruments to measure the exposure (in rads). Take this measurement at all locations where the operator may be inadequately shielded during exposure.

Another, less accurate, method is to place film at strategic locations. After a specified period of time, develop the film to determine the amount of radiation. Fluorescent screens used in a darkened room can also detect excessive radiation.

Another common method is for personnel to wear film or aluminum oxide badges that track X-Ray dosage.

S1.8 Radiation Protection Survey

A radiation protection survey must be made by a qualified expert after every change in equipment or conditions that might increase the probability of personnel receiving more than the maximum permissible dose equivalent.

S1.9 Pediatric Use

Exercise special care when imaging patients outside the typical adult size range, especially smaller pediatric patients whose size does not overlap the adult size range (e.g., patients less than 110 lb [50 kg] in weight and 59 in [150 cm] in height).

Ionizing radiation exposure is of particular concern in pediatric patients because:

- For certain organs and tumor types, younger patients are more radiosensitive than adults (i.e., the cancer risk per unit dose of ionizing radiation is higher for younger patients).
- Use of equipment and exposure settings designed for adults of average size can result in excessive and unnecessary radiation of smaller patients.
- Younger patients have a longer expected lifetime over which the effects of radiation exposure may manifest as cancer.

To help reduce the risk of excessive radiation exposure, follow the ALARA (As Low As Reasonably Achievable) principle and seek to reduce radiation doses to only the amount necessary for clinically adequate images.

If available on your system, utilize the following features (and their pediatric-appropriate settings):

- Filters
- Grid or cabinet
- Protocols or exposure indices
- Variable focal spot size
- Automatic Exposure Control (AEC)
- DAP meter

Also, ensure that all testing and calibration procedures include measurements and adjustments for pediatric settings.

For additional guidance, refer to these resources:

- The Image Gently Campaign: <http://www.imagegently.org>.
- FDA resource on Pediatric X-ray Imaging: <http://www.fda.gov/Radiation-EmittingProduct-s/RadiationEmittingProductsandProcedures/MedicalImaging/ucm298899.htm>

S1.10 Restrictions on Use

The hardware specified for intended applications of the DM-OTC18T Overhead Tube Crane System has been selected, tested, and verified by Del Medical Inc. to meet applicable regulatory agency requirements for those countries where it is offered for sale with respect to its intended applications.



Warning: Do not install components or accessories not intended for use with this system. Failure to comply can cause injury or equipment damage.

You are responsible for ensuring that the application and use of this equipment does not compromise the patient contact rating of any equipment used in the vicinity of or in conjunction with the system.

S1.11 Electromagnetic Compatibility (EMC)

Medical electrical equipment requires special precautions with regard to EMC information in this manual. When installing and operating this equipment you must comply with the EMC information in this document. This equipment is intended for use by health care professionals only.

This equipment is designed to be used only in a clinical or hospital environment with regards to EMC.



Caution: The use of accessories and cables other than those specified (or supplied by Del Medical Inc. as replacements) may cause higher emission or lower immunity levels.



Caution: This equipment may cause radio interference or disrupt nearby equipment. You may need to take mitigation measures, such as re-orientating or relocating the equipment, or shielding the location.



Caution: Communications among wireless devices (e.g., mobile devices or printers) can affect medical equipment. Exercise caution when using such devices within the specified radius of this equipment.



Caution: Avoid using this equipment near, or stacked on, other equipment. If this is necessary, first verify normal operation for this configuration.



Caution: Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be no closer than 12 inches (30 cm) to any part of the system. Otherwise, degradation of the performance of the equipment could result.



Caution: This equipment may experience interference with other equipment.



Caution: The quality of the mains line should be typical for a hospital environment.



Caution: The system is not intended to be installed on a public mains network as defined in IEC 60601-1-2.



Caution: If used in an environment outside of what is prescribed in this manual, the equipment may experience procedural delays.

S1.11.1 Combining with other Products/Components

To ensure safety, use only products/components approved by Del Medical Inc. in combination with this equipment. To learn the current approval state of combinations and upgrades, contact your sales representative.



Caution: Avoid using this equipment near, or stacked on, other equipment. If this is necessary, first verify normal operation for this configuration.



Caution: This equipment might experience interference from other equipment, even if the other equipment complies with CISPR emission requirements.

S1.11.2 Guidance and Declarations for Electromagnetic Compatibility

The DM-OTC18T Overhead Tube Crane System which is part of the DMX System is classified as a Class A device per CISPR 11:2019 Edition 6.2.

The following tables show compliance test results.

S1.11.2.1 Emissions

The DM-OTC18T Overhead Tube Crane System is intended for use in the electromagnetic environment specified in the following table. You must ensure that it is used in such an environment.

Emissions Test	Compliance	Guidance
RF emissions EN 60601-1-2:, 6.1.1	CISPR 11 Class A	With normal operation, the DM-OTC18T Overhead Tube Crane System emits electromagnetic energy. Nearby electronic equipment may be affected.
Harmonic emissions IEC 60601-1-2:, 6.1.3.1; IEC 61000-3-2	Class A up to 40th Harmonic	
Voltage fluctuations/ Flicker emissions IEC 60601-1-2:, 6.1.3.2; IEC 61000-3-3	$d(t)$, d_{max} , d_c , P_{st} , and P_{lt}	

Table S-1: Emissions Test Table

S1.11.2.2 Immunity

Immunity Test	Test Level	Compliance Level	Guidance
Electro-static discharge (ESD) IEC 60601-1-2, 6.2.2/EN 61000-4-2	± 8 kV contact ± 15 kV air ± 6 kV indirect discharge via VCP	± 8 kV contact ± 15 kV air ± 6 kV indirect	Floors should be wood, concrete or ceramic tile. If floors are covered with carpet or other synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 60601-1-2, 6.2.4/EN 61000-4-4	± 2 kV for power supply lines	±2 kV to AC/DC power lines	Mains power quality should be that of a typical hospital environment.
	± 1 kV for input/output lines	± 1 kV for input/output lines	Mains power quality should be that of a typical hospital environment.
Surge IEC 60601-1-2, 6.2.5/EN 61000-4-5	± .5 kV differential and common mode	± .5 kV differential and common mode	Mains power quality should be that of a typical hospital environment.
	± 1 kV differential and common mode	± 1 kV differential and common mode	
	± 2 kV common mode	± 2 kV common mode	

Immunity Test	Test Level	Compliance Level	Guidance
RF induced conducted disturbances IEC 60601-1-2, 6.2.6/EN 61000-4-6	6 V m) in ISM bands between 0.15 MHz and 80 MHz 0.15-80 MHz 3 V (rms), 80% AM signal and I/O cables	6 V m) in ISM bands between 0.15 MHz and 80 MHz 0.15-80 MHz 3 V (rms), 80% AM signal and I/O cables	Mains power quality should be that of a typical hospital environment.
Power Frequency Magnetic Fields IEC 60601-1-2, 6.2.8.1/EN 61000-4-8	3 A/m, 50 and 60Hz, Continuous Field	N/A	The DM-OTC18T Overhead Tube Crane System is not susceptible to magnetic fields.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 60601-1-2, 6.2.7/EN 61000-4-11	0% <i>UT</i> ; 0.5 cycle At 0°, 45°, 90°, 135°, 180°, 225°, 270° and 315° 0% <i>UT</i> ; 1 cycle and 70% <i>UT</i> ; 25 cycles Single phase: at 0° 0% <i>UT</i> ; 250 cycle	0% <i>UT</i> ; 0.5 cycle At 0°, 45°, 90°, 135°, 180°, 225°, 270° and 315° 0% <i>UT</i> ; 1 cycle and 70% <i>UT</i> ; 25 cycles Single phase: at 0° 0% <i>UT</i> ; 250 cycle	Mains power quality must be that of a typical hospital environment.

Immunity Test	Test Level	Compliance Level	Guidance
Conducted RF, IEC 61000-4-6 Radiated RF, IEC 61000-4-3	3 V _{rms} , 150kHz to 80MHz	3 V 10 V/m	Separate portable and mobile RF communications equipment from the DM-OTC18S and cables, by the following minimum distance, d, in meters: $d = 1.1 / \sqrt{P}$  Warning: This equipment has been tested for radiated RF immunity only at selected frequencies, and use nearby of emitters at other frequencies could result in improper operation

Immunity Test	Test Level	Compliance Level	Guidance
			Discreet Licensed Radios were used to induce 10V/m in the frequencies of 136-174MHz, 222-225MHz, 400-520MHz, 462.55-462.725MHz, 467.5625-467.7125MHz , 925.610677-927.698745MHz , 912.75-917.1MHz, 2402.48-2409MHz Field strengths from fixed transmitters, as determined by an electromagnetic site survey, should be less than the compliance level in each frequency range. Interference may occur in the vicinity of equipment marked with the  symbol.
Note: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption a and reflection from structures, objects and people.			
Table S-2: Immunity Test Table			

The DM-OTC18T Overhead Tube Crane System is intended for use in the electromagnetic environment specified in the following table. Ensure that it is used in such an environment.



Note: Field strengths from transmitters, such as base stations for wireless/cordless telephones, land mobile radios, amateur radio, or AM/FM/TV stations, cannot be predicted with accuracy. To assess the electromagnetic environment affected by fixed RF transmitters, consider an electromagnetic site survey. If the measured field strength exceeds the applicable RF compliance level, monitor the equipment to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating this equipment. Over the frequency range of 150 kHz to 80 MHz, field strengths should be less than 1 V/m.

S1.11.2.3 Separation

This equipment is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. You can help prevent electromagnetic interference by maintaining a minimum distance between transmitting equipment and this system per the following recommendations and per the maximum output power of the communications equipment.



Note: This equipment may use components from other manufacturers. See this document, the manufacturer's document, or a Del Medical wiring diagram for proper cabling.

S1.12 Grounding

Provide protective earth grounding for all components. All associated components must be grounded to a common bus according to national and local electrical codes. Ensure that the earth grounding connections between this equipment and the host system are maintained at all times.

S1.12.1 Wiring Safety Information

- During installation, make sure that all protective ground wire connections are completed before the equipment is turned on.
- It is required that a separate line disconnect and current limiting device be provided for table power.

- Regulations of professional associations concerning safety and accident prevention must be observed.

S1.13 Product Life and Disposal

The device has an estimated life of 10 years from the date of purchase. This may vary depending on the product use, product maintenance, and environmental conditions. When disposing of the system, you must comply with local regulations due to the presence of potentially hazardous materials. The device can contain electronic components, such as soldered circuit boards (which may contain lead), tinned copper wire, transistors, diodes, resistors, capacitors, etc. The cabinet is made from steel and/or aluminum. Please contact Del Medical Inc. for more information.



Note: Do not incinerate this product or place it in trash destined for landfills. Disposal of this product must adhere to local ordinances and regulations. Contact local authorities for proper disposal and/or recycling options for all of the components listed above.

S1.14 Hazardous Materials

The X-Ray tube, PCBs, and collimator used with this equipment contain lead and oil. Refer to the equipment manuals or contact Del Medical Inc. for additional information regarding use and disposal of hazardous materials.

S1.15 Acceptance Testing for Regulatory Compliance

Required safety- and compliance-related installation actions include the following activities:

- installation inspection per documented criteria
- installation approval
- creation and delivery of an acceptance report document



Note: Please see the Installation or Installation and Service manual for information on the Acceptance Testing manual.

1 Introduction

1.1 Overview

This manual provides Operation information for the DM-OTC18T Overhead Tube Crane System. Only components related to the DM-OTC18T Overhead Tube Crane System are covered.

For information regarding other components not part of the DM-OTC18T Overhead Tube Crane System, please see the appropriate documentation.

1.2 Description

The DM-OTC18T Overhead Tube Crane System offers versatility and ease of handling for general purpose radiography and is ideally suited for modern hospitals, urgent care centers, clinics, and private practices.

1.3 Application and Usage

1.3.1 Intended Use and Operating Principle

The intended use of this stationary general radiographic X-ray system is to position the X-ray tube and receptor in a desired manner to produce diagnostic X-ray images of human anatomy. The image acquisition component is responsible for setting the desired exposure parameters and the generator component is responsible for adapting the parameters to the appropriate high voltage which is sent to the tube component to create the X-ray. The imaging receptor captures the X-ray and sends to the image acquisition component to create an image. The resultant images are evaluated by a radiologist within the diagnostic process prior to the development of a treatment plan.

The DM-OTC18T Overhead Tube Crane System is intended to be used for general radiographic studies such as those listed in *"4 Image Quality" on page 139* and not intended for mammography, fluoroscopy, etc.

To ensure the images are acceptable diagnostic quality, please review *"4 Image Quality" on page 139*.

1.3.2 Intended Patient Population

Intended patients are people of all ages determined to benefit from diagnostic radiographic procedures common to hospitals, satellite hospitals, ambulatory care centers and group practice facilities, under the direct supervision of a doctor.

1.3.3 Intended User

This stationary general radiographic X-ray system must be operated by a radiologic technologist (as required by state law) and serviced by properly trained field service engineers (as determined by their employers) who are familiar with the safety precautions required, as well as the hazards of radiation.

At least two qualified, trained service engineers as determined by their employers, are required when installing and/or moving equipment in order to prevent injury, strain or, equipment damage.

1.4 DM-OTC18T System Overview

The DM-OTC18T Overhead Tube Crane System comes standard with the tubecrane, collimator, radiographic table, wallstand, generator, image receptor, and acquisition software. For information regarding other components not part of the DM-OTC18T Overhead Tube Crane System, please see the appropriate documentation.

1.4.1 Standard Components

Below shows the standard component models that are included with the DM-OTC18T

Component	Model	Description
Tubecrane	OTC18T	Overhead tube crane with motorized vertical movements and touchscreen display. Includes the DM-PSI power system integration box.

Component	Model	Description
Table	EV800	Elevating radiographic table
	EV800T	Elevating radiographic table with motorized receptor movement
	MT500	Mobile table
	No Table	
Wall Stand	VT300	Tilting wallstand
	VS300	Vertical wallstand
	No wallstand	
Collimator	AL02	Automatic positive beam limitation (PBL) collimator
	ML03	Manual collimator
Generator	CM Series RF Series	X-Ray generator with various power output options.
Tube	Various	See "Table 1-2: DM-OTC18T System component options" on the next page.
Table 1-1: DM-OTC18T System component options		

1.4.2 Component Options

Component	Available Options
Tube	Canon Electron Devices E7254
	Canon Electron Devices E7255
	Varex Imaging RAD14
	Varex Imaging RAD60
	Varex Imaging RAD92
	Siemens SV150/40/80C-100L
	Siemens SV150/40/80C-100

Table 1-2: DM-OTC18T System component options



Note: See "1.4.2.1 X-Ray Tube Compatibility" on the next page for more information on compatible tubes

1.4.2.1 X-Ray Tube Compatibility

The table below describes the X-ray tubes that have been fully tested and are certified with the DM-OTC18T Overhead Tube Crane System.

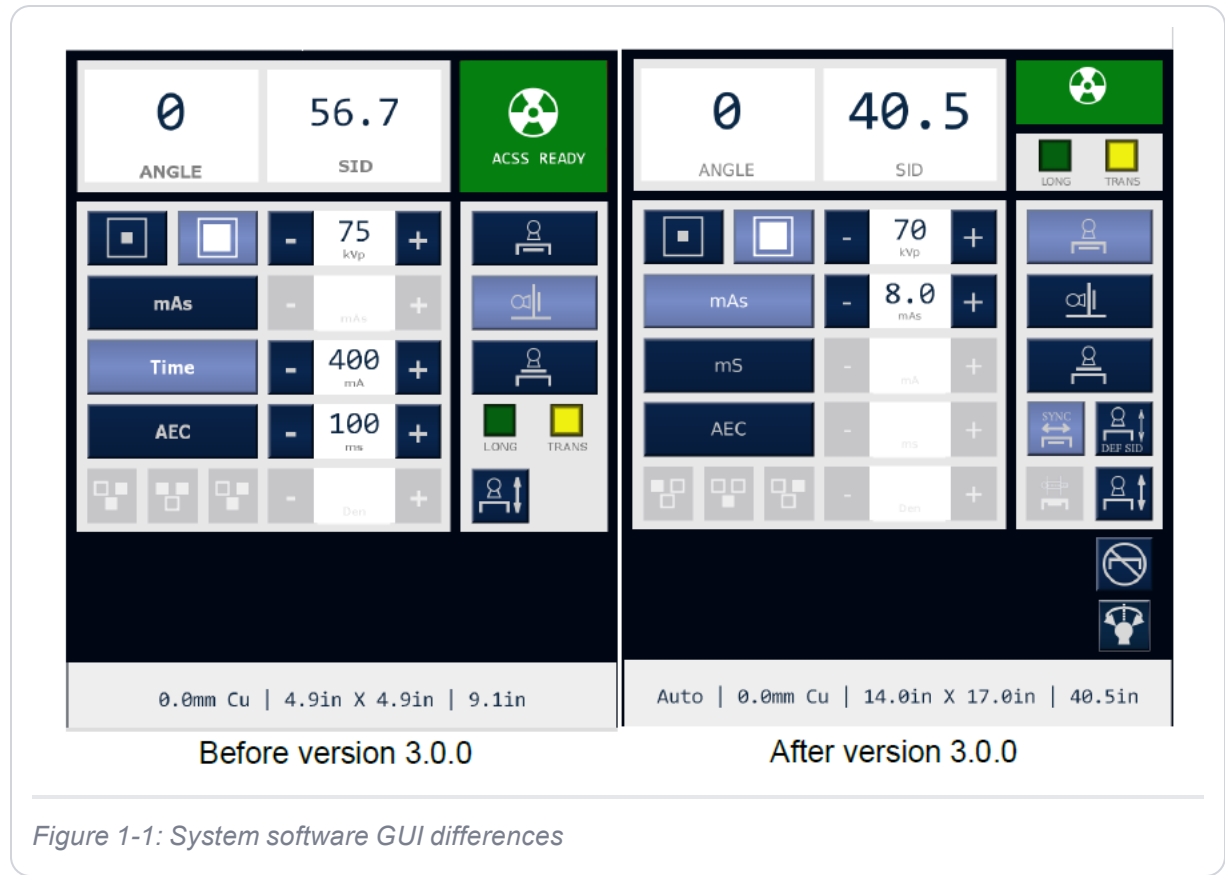
Manufacturer	Tube Type	Focal Spots (mm)	Anode Heat Units (kHU)	Anode Diameter (in)
Canon Electron Tubes and Devices	E7254	0.6 - 1.2	400	4
Canon Electron Tubes and Devices	E7255	0.6 - 1.2	300	4
Varex Imaging	RAD14	0.6 - 1.2	300	3
Varex Imaging	RAD60	0.6 - 1.2	400	4
Varex Imaging	RAD92	0.6 - 1.2	600	4
Siemens	SV150/40/80C-100L	0.6 - 1.0	600	4

Table 1-3: X-ray Tube Compatibility Chart

1.4.3 System Software Versions

In an effort of continuous improvement, the system software is periodically changed. The scope of this manual covers all software versions. Some features described in this document may be dependent on the software installed. Some features were added in version 3.0.0. This manual will describe any differences as versions before, at, or later than version 3.0.0.

The simplest way to know if the software version is before, at, or later than version 3.0.0 is by the graphical user interface (GUI) on the touchscreen. *"Figure 1-1: System software GUI differences" on the next page* describes the difference in the look of the GUI.



A functional description of the GUI can be found in "2 Operation" on page 79.



Note: The system version defined in this chapter only relates to the DM-OTC18T tube support. This version number in no way relates to any software version of the image acquisition system or computer. For information on versions of the image acquisition system, see the appropriate documentation.

1.4.4 DM-OTC18T System Components Summary

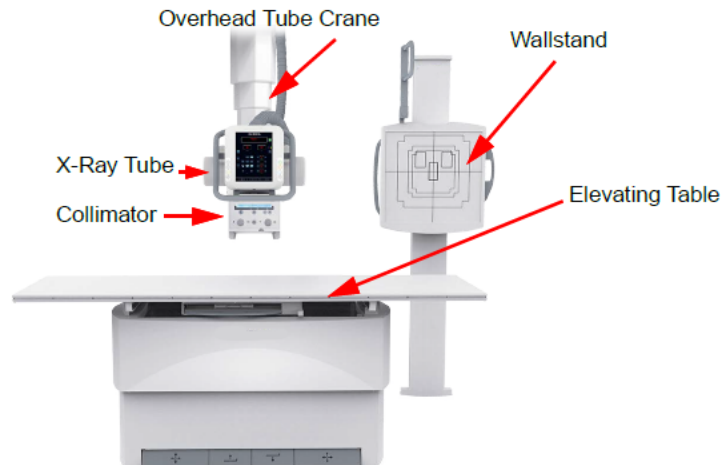


Figure 1-2: DM-OTC18T System Components

1.4.5 Table Accessories

The following accessories are compatible.

- Hand Grips
- Lateral Receptor Holder



Figure 1-3: Hand Grips



Figure 1-4: Lateral Receptor Holder

1.4.6 Image Acquisition System

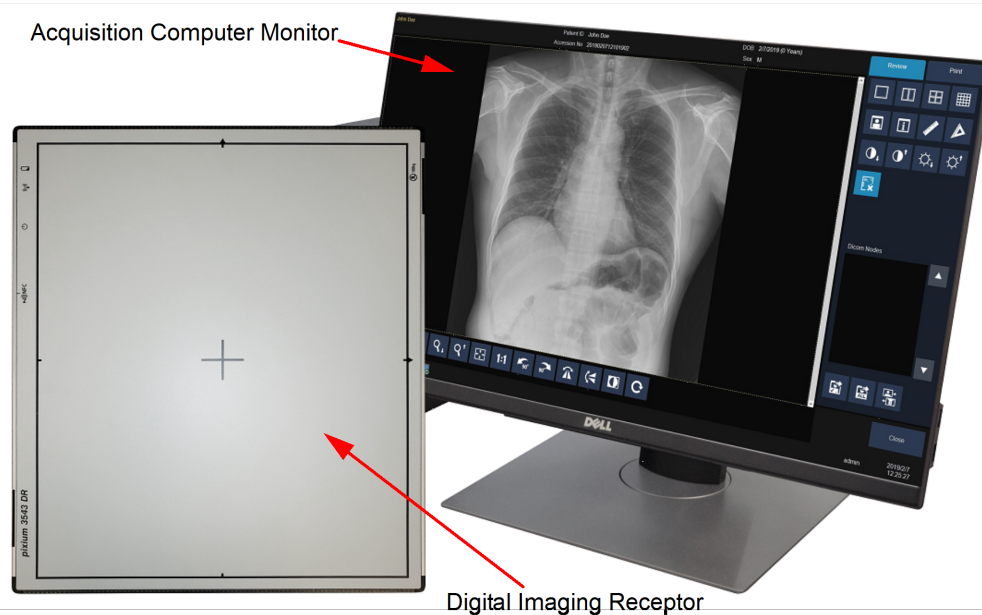


Figure 1-5: Image acquisition system components



Note: The Image Acquisition System includes an isolated uninterpretable power supply designed to isolate the imaging components from the supply mains.

1.5 DM-OTC18T System Component Overview

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1.5.1 OTC18T Tube Crane Diagram

1.5.1.1 OTC18T User Interface Overview

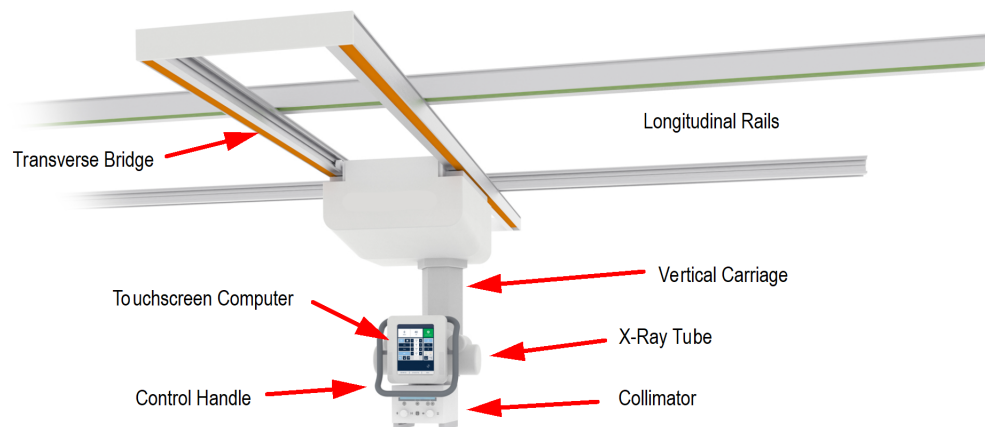


Figure 1-6: OTC18T Diagram

1.5.2 MT500-5145 Mobile Table

The MT500 features a heavy-duty steel frame for maximum stability. The table is 28" high (71 cm) and can support patients weighing up to 500 lb (227 kg). The stain-resistant fiber-resin tabletop is 26" wide and 80" long.

The large-diameter casters provide smooth movement and are less sensitive to variations in floor surfaces, thresholds, or small objects. All four casters can swivel 360 degrees and have reliable foot locks that are easy to apply and release.

The optional Grid Cabinet can travel the entire length of the table.

The following table is included with the DM-OTC18T Overhead Tube Crane System:

- 110-5145

1.5.2.1 MT500-5145 Mobile Table Options

Grid Cabinet

Non-motorized, can include a 14x17 receptor tray

Rotating Tray Cabinet

Accommodates and includes the Del Medical Inc. Rotating Tray

The following options are included:

Grid (Available for Purchase)

Several options are available, contact Del Medical Inc. for more information.



Note: Detailed information on the function of these components can be found in *"1.1 RT100 Controls" on page 1*

1.5.3 EV800 Elevating Table

The EV800 features a four-way, floating table top with 55" (140 cm) of longitudinal travel and 9" (24 cm) of transverse travel.

Electromechanical locks maintain the desired position. You can move the table top by first pressing the lock release treadle. The cabinet can also be released for movement by pressing a button the cabinet handle.

The following tables are included with the DM-OTC18T Overhead Tube Crane System:

- 110-5171
- 110-5183

1.5.3.1 EV800-5171 Elevating Table Options

Grid Cabinet

Non-motorized, can include a 14x17 receptor tray

Bucky Cabinet

Motorized grid movement, includes a 14x17 receptor tray

Rotating Tray Cabinet

Accommodates and includes the Del Medical Inc. Rotating Tray

Flat Panel 17x17 Fixed DR Detector Cabinet

Please see detector documentation for information

The following options are included:

PBL (Positive Beam Limitation)

For auto receptor size sensing.

Ionization Chamber

For automatic exposure control (AEC); ionization chambers are available. Please see any corresponding documentation for specific information.

Differential SID

For auto table height sensing to maintain SID

Four Table Top Options (EV800/EV800T Only)

36"x86", 32"x86", 36"x78", and 32"x78".

Grid (Available for Purchase)

Several options are available, contact Del Medical Inc. for more information.

1.5.3.2 EV800-5183 Elevating Table Options

Grid Cabinet with Flex Detector Tray (FLEXTRAY-5185)

Please refer to the Grid Cabinet with Flex Detector Tray Manuals for information on use and detector compatibility. See 8000-FLEXTRAY-5185-OP .



Note: This is also referred to as the "**Rotate Tray Cabinet with Optional Charging**", please be aware as these might be used interchangeably throughout this documentation



Important: For Charging Systems, use **only** the supplied detector. Using another detector can damage the FLEXTRAY-5185.

The following options are included:

Ionization Chamber

For automatic exposure control (AEC); ionization chambers are available. Please see any corresponding documentation for specific information.

Differential SID

For auto table height sensing to maintain SID

Four Table Top Options (EV800/EV800T Only)

36"x86", 32"x86", 36"x78", and 32"x78".

Grid (Available for Purchase)

Several options are available, contact Del Medical Inc. for more information.



Note: Detailed information on the function of these components can be found in "*2.5 EV800 Controls*" on page 92

1.5.4 EV800T Elevating Table with Motorized Receptor

The EV800T features a four-way, floating table top with 55" (140 cm) of longitudinal travel and 9" (24 cm) of transverse travel.

Electromechanical locks maintain the desired position. You can move the table top by first pressing the lock release treadle. The cabinet can also be released for movement by pressing a button the cabinet handle.

The following tables and options are included with the DM-OTC18T Overhead Tube Crane System:

- 110-5168
- 110-5184

1.5.4.1 EV800T-5168 Elevating Table with Motorized Receptor Options

Grid Cabinet

Non-motorized, can include a 14x17 receptor tray

Rotating Tray Cabinet

Accommodates and includes the Del Medical Inc. Rotating Tray

Flat Panel 17x17 Fixed DR Detector Cabinet

Please see detector documentation for information

The following options are included:

Auto Tracking (FMT18T & OTC18T Only)

For automatic tubehead movement to match the vertical position of the receptor cabinet.

PBL (Positive Beam Limitation)

For auto receptor size sensing.

Ionization Chamber

For automatic exposure control (AEC); ionization chambers are available. Please see any corresponding documentation for specific information.

Differential SID

For auto table height sensing to maintain SID

Four Table Top Options (EV800/EV800T Only)

36"x86", 32"x86", 36"x78", and 32"x78".

Grid (Available for Purchase)

Several options are available, contact Del Medical Inc. for more information.

1.5.4.2 EV800T-5184 Elevating Table with Motorized Receptor Options

Grid Cabinet with Flex Detector Tray (FLEXTRAY-5185)

Please refer to the Grid Cabinet with Flex Detector Tray Manuals for information on use and detector compatibility. See 8000-FLEXTRAY-5185-OP .



Note: This is also referred to as the "**Rotate Tray Cabinet with Optional Charging**", please be aware as these might be used interchangeably throughout this documentation



Important: For Charging Systems, use **only** the supplied detector. Using another detector can damage the FLEXTRAY-5185.

The following options are included:

Auto Tracking (FMT18T & OTC18T Only)

For automatic tubehead movement to match the vertical position of the receptor cabinet.

Ionization Chamber

For automatic exposure control (AEC); ionization chambers are available. Please see any corresponding documentation for specific information.

Differential SID

For auto table height sensing to maintain SID

Four Table Top Options (EV800/EV800T Only)

36"x86", 32"x86", 36"x78", and 32"x78".

Grid (Available for Purchase)

Several options are available, contact Del Medical Inc. for more information.



Note: Detailed information on the function of these components can be found in *"2.6 EV800T Controls" on page 97.*

1.5.5 VS300 Vertical Wallstand

The VS300 features a smoothly counterbalanced receptor that can be easily moved, angled, and locked into place manually. The VS300 offers a full range of motion, from standing knee exposures to standing head and neck exposures on the tallest of patients.

The following wallstands are options included with the DM-OTC18T Overhead Tube Crane System:

- 110-5154
- 110-5186

1.5.5.1 VS300-5154 Vertical Wallstand Options

Grid Cabinet

Non-motorized, can include a 14x17 receptor tray

Bucky Cabinet

Motorized grid movement, includes a 14x17 receptor tray

Rotating Tray Cabinet

Accommodates and includes the Del Medical Inc. Rotating Tray

Flat Panel 17x17 Fixed DR Detector Cabinet

Please see detector documentation for information

The following options are included:

Auto Tracking (FMT18T & OTC18T Only)

For automatic tubehead movement to match the vertical position of the receptor cabinet.

PBL (Positive Beam Limitation)

For auto receptor size sensing.

Ionization Chamber

For automatic exposure control (AEC); ionization chambers are available. Please see any corresponding documentation for specific information.

Solid State Measurement Chamber (VS300-5154 Only)

Claymount SSMC 601; for automatic exposure control (AEC) with the fixed DR detector cabinet

Fixed Grid

The grid cabinet ,Bucky and rotating cabinet accommodate a fixed 17x17 grid that is typically shipped already installed, but can be changed in the field.

CAN BUS

For use with the OTC18M/T or FMT18M/T systems.

Grid (Available for Purchase)

Several options are available, contact Del Medical Inc. for more information.

1.5.5.2 VS300-5186 Vertical Wallstand Options

Grid Cabinet with Flex Detector Tray (FLEXTRAY-5185)

Please refer to the Grid Cabinet with Flex Detector Tray Manuals for information on use and detector compatibility. See 8000-FLEXTRAY-5185-OP .



Note: This is also referred to as the "**Rotate Tray Cabinet with Optional Charging**", please be aware as these might be used interchangeably throughout this documentation



Important: For Charging Systems, use **only** the supplied detector. Using another detector can damage the FLEXTRAY-5185.

The following options are included:

Auto Tracking (FMT18T & OTC18T Only)

For automatic tubehead movement to match the vertical position of the receptor cabinet.

Ionization Chamber

For automatic exposure control (AEC); ionization chambers are available. Please see any corresponding documentation for specific information.

CAN BUS

For use with the OTC18M/T or FMT18M/T systems.

Grid (Available for Purchase)

Several options are available, contact Del Medical Inc. for more information.



Note: Detailed information on the function of these components can be found in *"2.7 VS300 Controls" on page 104.*

1.5.6 VT300 Tilting Wallstand

The VT300 features a smoothly counterbalanced receptor that can be easily moved, angled, and locked into place manually. The VT300 offers a full range of motion, from standing knee exposures to standing head and neck exposures on the tallest of patients.

The following wallstands are options included with the DM-OTC18T Overhead Tube Crane System:

- 110-5157
- 110-5187

1.5.6.1 VT300-5157 Tilting Wallstand Options

Grid Cabinet

Non-motorized, can include a 14x17 receptor tray

Bucky Cabinet

Motorized grid movement, includes a 14x17 receptor tray

Rotating Tray Cabinet

Accommodates and includes the Del Medical Inc. Rotating Tray

Flat Panel 17x17 Fixed DR Detector Cabinet

Please see detector documentation for information

The following options are included:

Auto Tracking (FMT18T & OTC18T Only)

For automatic tubehead movement to match the vertical position of the receptor cabinet.

Receptor Extension Adapter (VT300 Only)

To extend the receptor cabinet an additional 8 inches from the column.

PBL (Positive Beam Limitation)

For auto receptor size sensing.

Ionization Chamber

For automatic exposure control (AEC); ionization chambers are available. Please see any corresponding documentation for specific information.

CAN BUS

For use with the OTC18M/T or FMT18M/T systems.

Grid (Available for Purchase)

Several options are available, contact Del Medical Inc. for more information.

1.5.6.2 VT300-5187 Tilting Wallstand Options

Grid Cabinet with Flex Detector Tray (FLEXTRAY-5185)

Please refer to the Grid Cabinet with Flex Detector Tray Manuals for information on use and detector compatibility. See 8000-FLEXTRAY-5185-OP .



Note: This is also referred to as the "Rotate Tray Cabinet with Optional Charging", please be aware as these might be used interchangeably throughout this documentation



Important: For Charging Systems, use **only** the supplied detector. Using another detector can damage the FLEXTRAY-5185.

The following options are included:

Auto Tracking (FMT18T & OTC18T Only)

For automatic tubehead movement to match the vertical position of the receptor cabinet.

Receptor Extension Adapter (VT300 Only)

To extend the receptor cabinet an additional 8 inches from the column.

PBL (Positive Beam Limitation)

For auto receptor size sensing.

Ionization Chamber

For automatic exposure control (AEC); ionization chambers are available. Please see any corresponding documentation for specific information.

CAN BUS

For use with the OTC18M/T or FMT18M/T systems.

Grid (Available for Purchase)

Several options are available, contact Del Medical Inc. for more information.



Note: Detailed information on the function of these components can be found in "2.8 VT300 Controls" on page 105.

1.6 Specifications

Description	Component	Value or Range
Input Power Generator	Generator	See "1.0.1 Generator Power Input Specifications" on page 1
Input Power (DM-PSI)	DM-PSI	115 / 230-240 VAC 15A 50 / 60 Hz
Input Power System	OTC18T Overhead Tube Crane (Supplied by DM-PSI)	24 VDC 8 A (from DM-PSI)

Description	Component	Value or Range
Input Power Component	EV800-5171 Elevating Table (Supplied by DM-PSI)	120 / 230 / 240 VAC (from DM-PSI) 10A 50 / 60 Hz
Input Power Component	EV800-5183 Elevating Table (Supplied by DM-PSI)	120 / 230 / 240 VAC (from DM-PSI) 10A 50 / 60 Hz
Input Power Component	EV800T-5168 Elevating Table with Motorized Receptor (Supplied by DM-PSI)	120 / 230 / 240 VAC (from DM-PSI) 10A 50 / 60 Hz
Input Power Component	EV800T-5184 Elevating Table with Motorized Receptor (Supplied by DM-PSI)	120 / 230 / 240 VAC (from DM-PSI) 10A 50 / 60 Hz
Input Power Component	VS300-5154 Vertical Wall-stand (Supplied by DM-PSI)	Standard: 24 VDC (from DM-PSI)
Input Power Component	VS300-5186 Vertical Wall-stand (Supplied by DM-PSI)	Standard: 24 VDC (from DM-PSI) w/ Charging Option: 24 VDC (from DM-PSI) & 110/240 VAC (from DM-PSI)
Input Power Component	VT300-5157 Tilting Wall-stand (Supplied by DM-PSI)	Standard: 24 VDC (from DM-PSI)
Input Power Component	VT300-5187 Tilting Wall-stand (Supplied by DM-PSI)	Standard: 24 VDC (from DM-PSI) w/ Charging Option: 24 VDC (from DM-PSI) & 110/240 VAC (from DM-PSI)

Description	Component	Value or Range
External Heat Generation	All	Negligible
Fuse Type Component	EV800 Elevating Table	F1, FL1, FL2, 10A, 250V, Time Delay
Fuse Type Component	EV800T Elevating Table	F1, FL1, FL2, 10A, 250V, Time Delay
Fuse Type Component	VS300 Vertical Wallstand	F2, 5A, Time Delay
Fuse Type Component	VT300 Tilting Wallstand	F2, 5A, Time Delay
Fuse Type Generator	Generator	See generator service manual
Fuse Type System	OTC18T Overhead Tube Crane	F1, 3A, 250V, Time Delay
Environmental Intended Use	All	Indoor use only
Temperature Limits	Tube crane, table, wallstand, and collimator	Transit/Storage: –40°F to +158°F (–40°C to +70°C) Operating: +50°F to +95°F (+10°C to +35°C)
	Generator	Transit/Storage: –4°F to +158°F (–20°C to +70°C) Operating: +50°F to +104°F (+10°C to +40°C)

Description	Component	Value or Range
Relative Humidity Limits	Tube crane, table, and wall-stand	Transit/Storage: 10% to 100% Operating: 10%-80% Non-Condensing
	Generator	Transit/Storage: 5% to 95% Operating: 20%-80% Non-Condensing
	Collimator	Transit/Storage: 0% to 95% Operating: 20%-75% Non-Condensing
Operating Atmospheric Limits	Tube crane, table, and wall-stand	800 hPa to 1060 hPa
	Generator and collimator	700 hPa to 1060 hPa
Storage Atmospheric Limits	Tube crane, table, and wall-stand	500 hPa to 1060 hPa
	Generator and collimator	500 hPa to 1060 hPa

Description	Component	Value or Range
Weight	Tube crane	507 lb (230 kg)
	10 ft (3 m) transverse bridge	154 lb (70 kg)
	13 ft (4 m) transverse bridge	198 lb (90 kg)
	Longitudinal rails	59 lb (26.5 kg)
	EV800T Table	Table Base: 563lb (256 kg) 36"x86" Table Top: 146 lb (66kg) 32"x86" Table Top: 132 lb (60kg) 36"x78" Table Top: 141 lb (64kg) 32"x78" Table Top: 130 lb (59kg)
	Generator	122 lbs (55.3 kg) - 155 lbs (70.3 kg)
	Collimator	19.4 lbs (8.8 kg)
Weight Capacity	EV800 Table	 800 lb / 363 kg (Centered)
Weight Capacity	EV800T Table	 800 lb / 363 kg (Centered)
Pull Force	VS300 Wallstand (P-Handle Accessory)	35 lb / 16 kg
	VT300 Wallstand (P-Handle Accessory)	35 lb / 16 kg
WIFI Frequency (when applicable)	Image acquisition wireless panel and wireless access point	2.4 GHz or 5.0 GHz
SID Display Accuracy	Tubecrane	± 2% of SID

Description	Component	Value or Range
Mode of Operation	EV800 Table	Momentary
	EV800T Table	Momentary
	VS300 Wallstand	Momentary
	VT300 Wallstand	Momentary
Auto-Tracking Accuracy	Tube crane, table, wallstand	$\pm 2\%$ of SID

Table 1-4: System Specifications

1.7 Applicable Standards

This equipment complies with the following regulatory requirements and design standards:

- Conforms to AAMI Standard ES60601-1, CSA Standard C22.2 No. 60601-1, and IEC 60601-1, and IEC 60601-2-54.
- Type of protection against electric shock: Class I
- Degree of protection against electric shock: Type B Applied Parts
- Degree of protection against harmful ingress of water: Ordinary equipment.
- Equipment not suitable for use in presence of a FLAMMABLE ANESTHETIC MIXTURE WITH AIR OR WITH OXYGEN OR NITROUS OXIDE.

1.8 Label Icons

This section covers symbols typically used on labels and on the device.

Symbol	Definition
	Potential hazard to operators, service personnel or equipment.
	Dangerous voltage is present and accessible.
	A protective earth terminal or ground.
 	This indicates the need for the user to consult the Operation manual or Instructions for Use.
	IEC 60601-1 Type B classification. Type B equipment provides a degree of protection against electric shock, particularly regarding allowable leakage currents and reliability of the protective earth connection (if present).
	Dispose of this equipment according to local laws and regulations. Because this equipment contains electronic components, dispose of it separately from household waste. When the equipment reaches its end of life, contact local authorities to learn about disposal and recycling options.

Symbol	Definition
See "1.9 Identification Labels" on the next page for ETL logo	This item is an ETL Listed Device
	Date of manufacture indicates when the Medical Device was manufactured
	Indicates the medical device manufacturer, this information can be found on page 2 and the back page.
	This device emits laser radiation.
	This product is RoHS compliant.
	Weight limit symbol, usually follows with a maximum weight allowed (placed on fixed table.) Please see table controls in "2 Operation" on page 79 for more information.

Table 1-5: Commonly Found Label Icons

1.9 Identification Labels

Labels will include the following manufacturing and certification information:

- The full name and address of the manufacturer of the device
- The place, month, and year of manufacture
- The model number and serial number of the device

The certification label also states that the component complies with either “21CFR, Subchapter J” or the applicable DHHS standards under the Radiation Control for Health and Safety Act of 1968 (or its equivalent).

A label may combine both manufacturing and certification information.

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1.9.14 Other Labels	75

1.9.1 System Labels

1.9.1.1 DM-OTC18T System Certification and Compliance Information

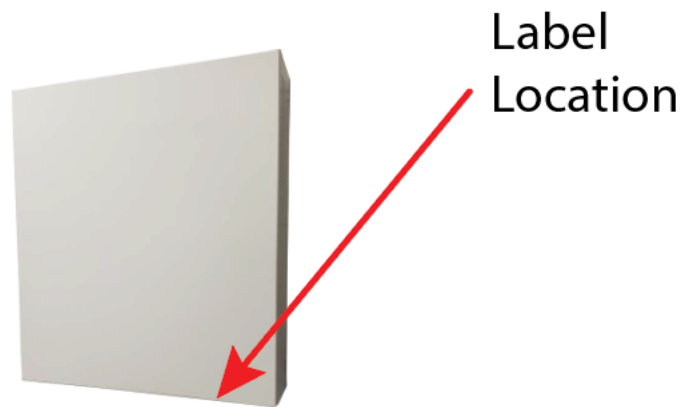


Figure 1-7: DM-OTC18T system label location bottom right of PSI

MANUFACTURED BY: DEL MEDICAL INC. 241 COVINGTON DRIVE BLOOMINGDALE, IL 60108 U.S.A.		(REF) MODEL NUMBER: PART NUMBER: _____	
		(SN) SERIAL NUMBER: _____ MANUFACTURED: _____	
ETL CLASSIFIED			
 Intertek 3076414			
MEDICAL ELECTRICAL EQUIPMENT		COMPLIES WITH FDA RADIATION PERFORMANCE STANDARDS 21 CFR SUBCHAPTER J IN EFFECT AT DATE OF MANUFACTURE LABEL P/N 408-5374P_	

Figure 1-8: DM-OTC18T System certification and compliance information

1.9.1.2 DM-PSI Certification and Compliance Information

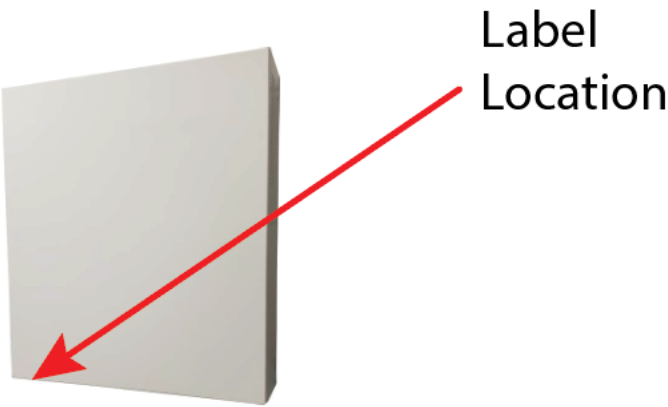


Figure 1-9: DM-PSI Label Location, bottom left of PSI

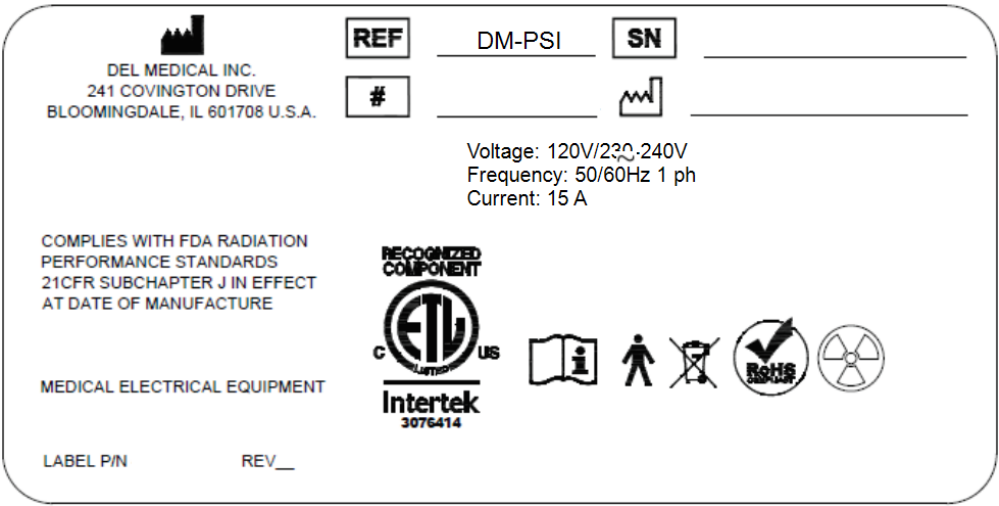


Figure 1-10: DM-PSI certification and compliance information

1.9.2 OTC18T Component Label



Figure 1-11: OTC18T Label Location Behind on Right Side

 DEL MEDICAL INC. 241 COVINGTON DRIVE BLOOMINGDALE, IL 601708 U.S.A.	REF	OTC18T	SN	
	#	110-5152G_		
COMPLIES WITH FDA RADIATION PERFORMANCE STANDARDS 21CFR SUBCHAPTER J IN EFFECT AT DATE OF MANUFACTURE	RECOGNIZED COMPONENT		VOLTAGE: 24 V --- MAX CURRENT: 8 A	
			     	
MEDICAL ELECTRICAL EQUIPMENT				
LABEL P/N 408-5337P1 REV_	UDI			

Figure 1-12: OTC18T certification and compliance information

1.9.3 Touchscreen Computer Label

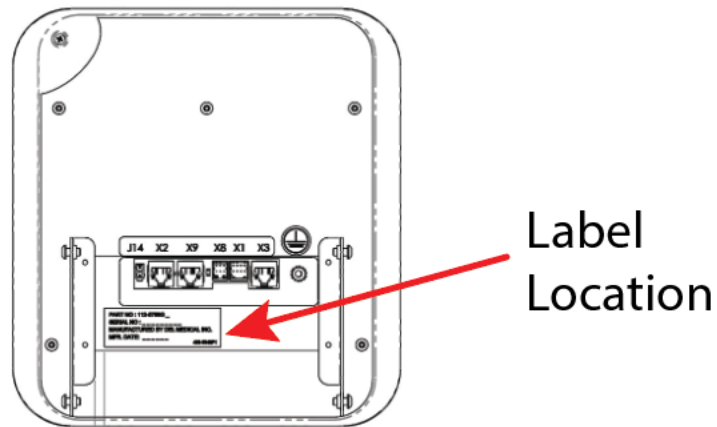
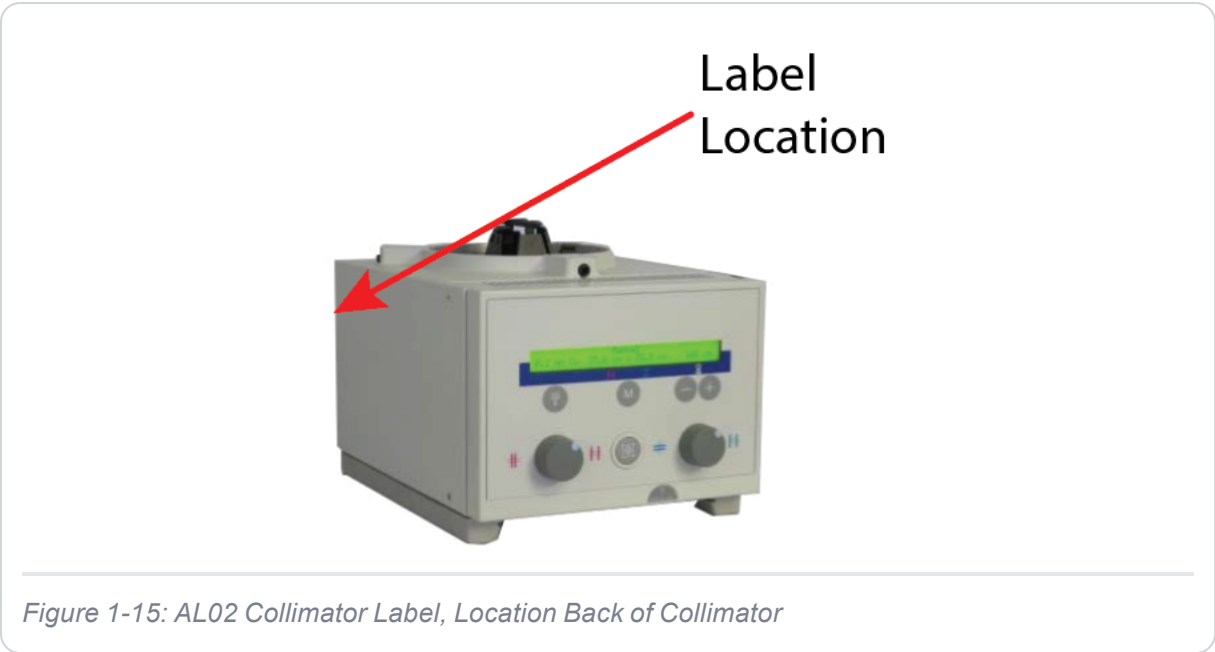


Figure 1-13: Touchscreen Label, Location Back of Touchscreen

PART NO : 112-5758G _
SERIAL NO : _ _ _ _ _
MANUFACTURED BY DEL MEDICAL INC.
MFR. DATE: _ _ _ _ _ **408-5342P1**

Figure 1-14: Touchscreen certification and compliance information

1.9.4 AL02 Collimator Label



1.9.5 ML03 Collimator Label

Label
Location



Figure 1-17: ML03 Collimator Label, Location Back of Collimator

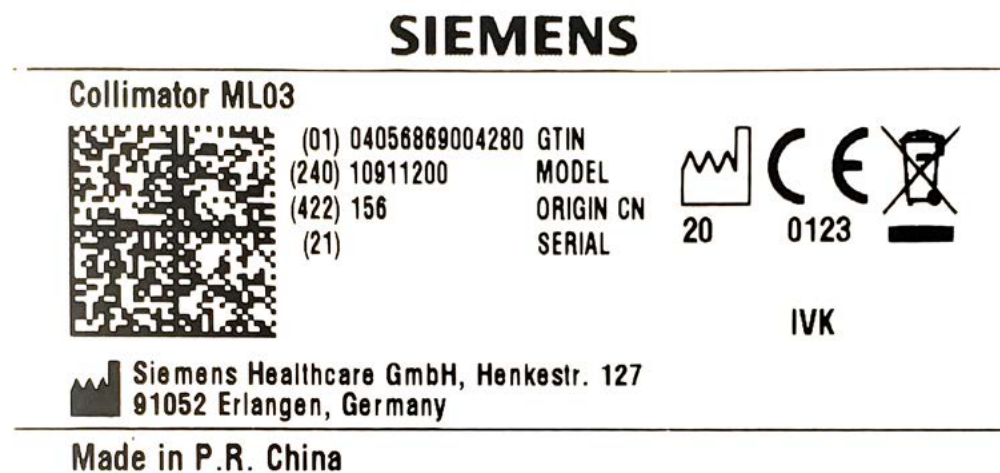


Figure 1-18: ML03 Collimator certification and compliance information

1.9.6 EV800-5171 Label

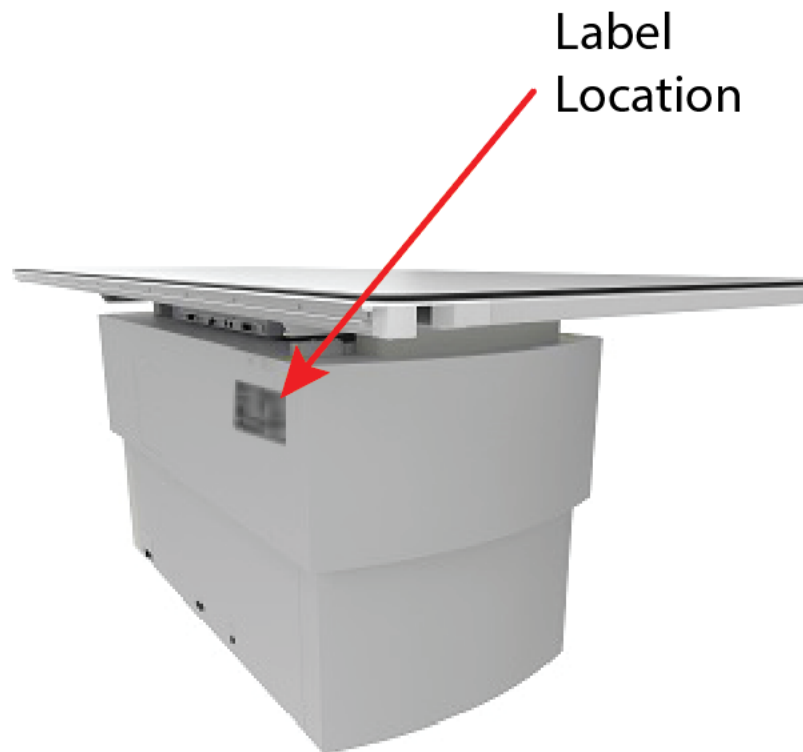
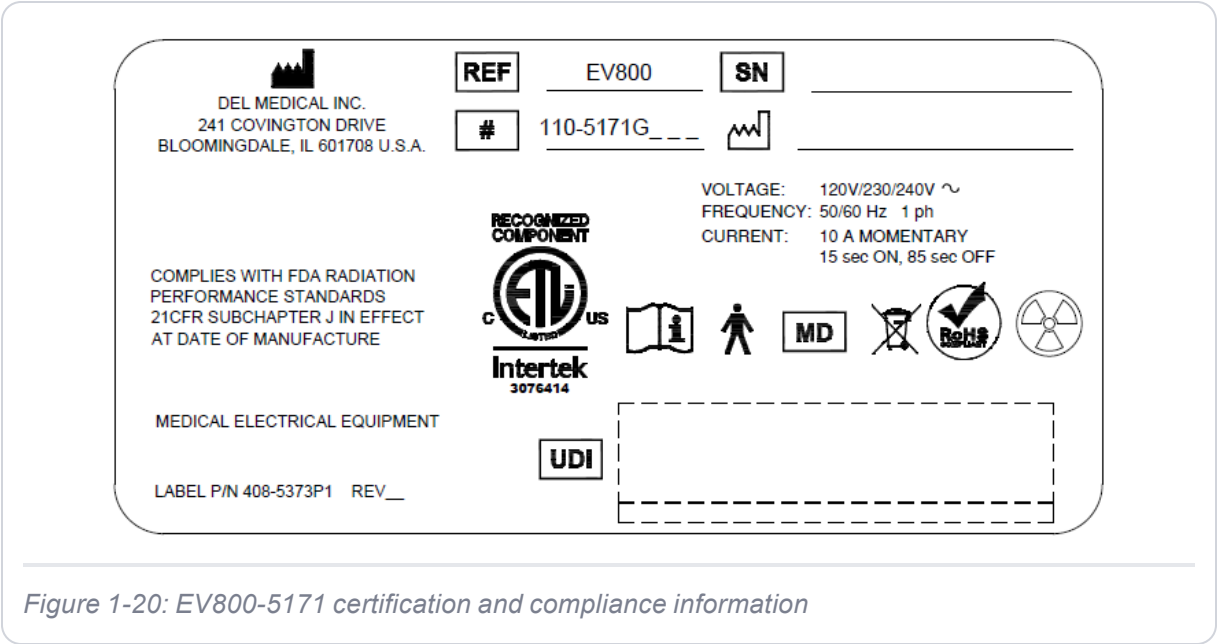


Figure 1-19: EV800-5171 Elevating Table Back View and Label Location



1.9.7 EV800-5183 Label

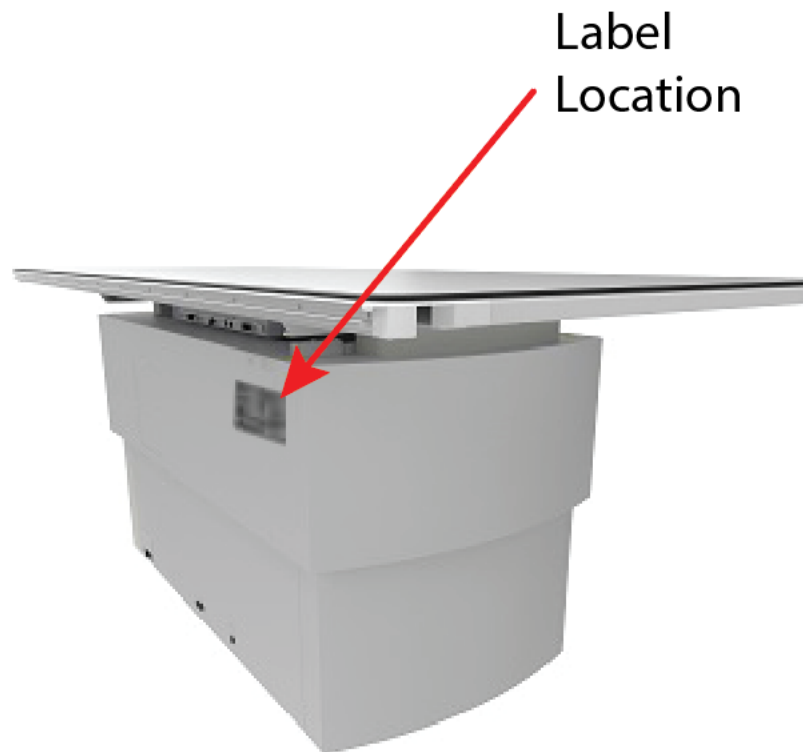


Figure 1-21: EV800-5183 Elevating Table Back View and Label Location

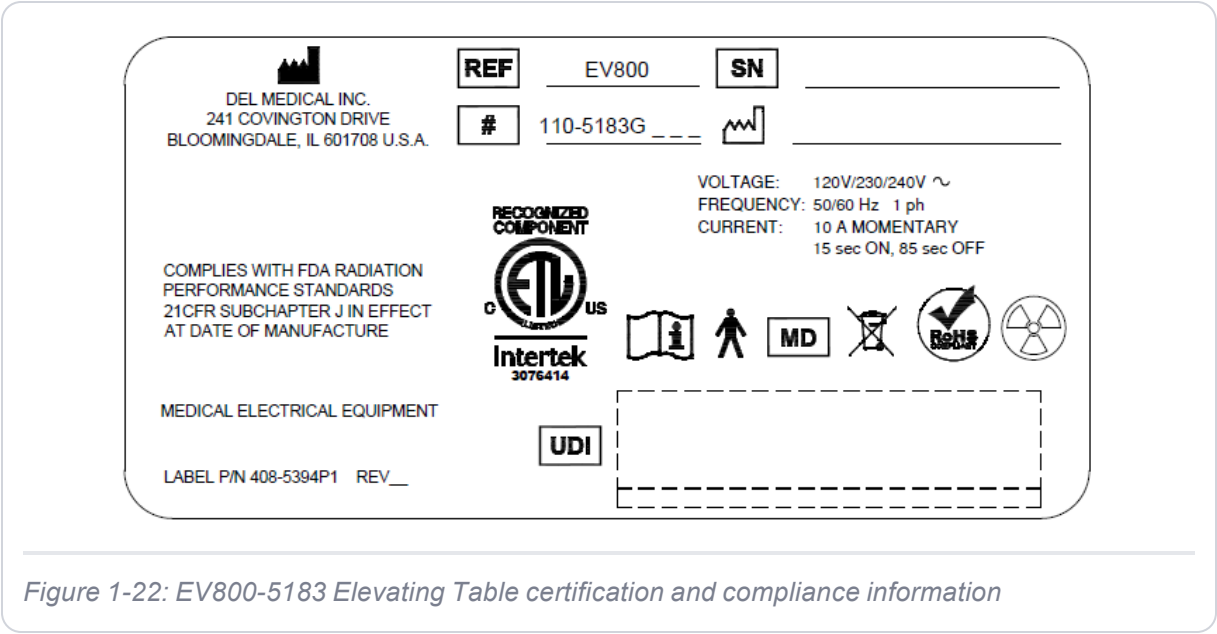


Figure 1-22: EV800-5183 Elevating Table certification and compliance information

1.9.8 EV800T-5168 Table Label

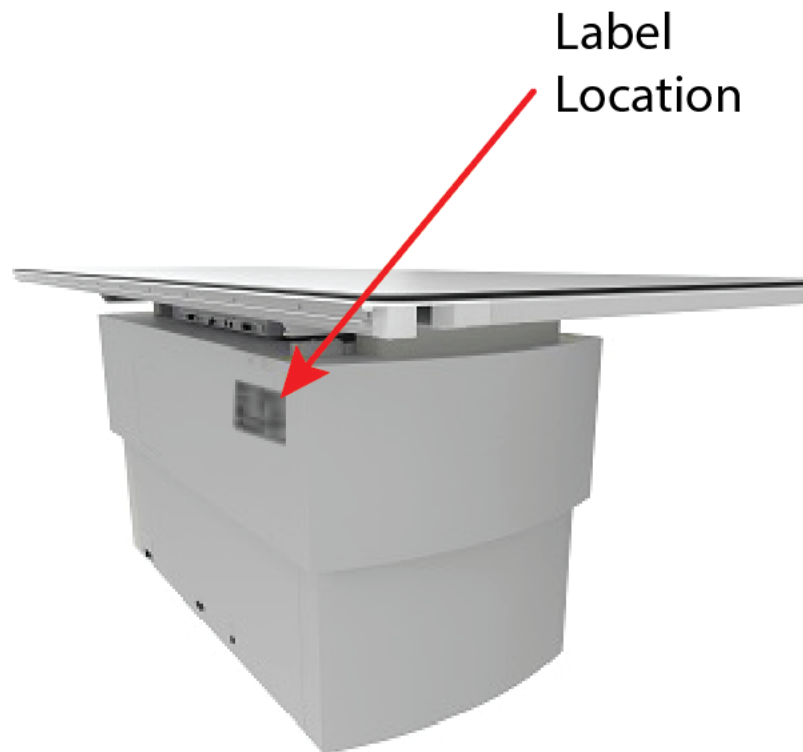


Figure 1-23: EV800T-5168 Elevating Table with Motorized Receptor Back View and Label Location

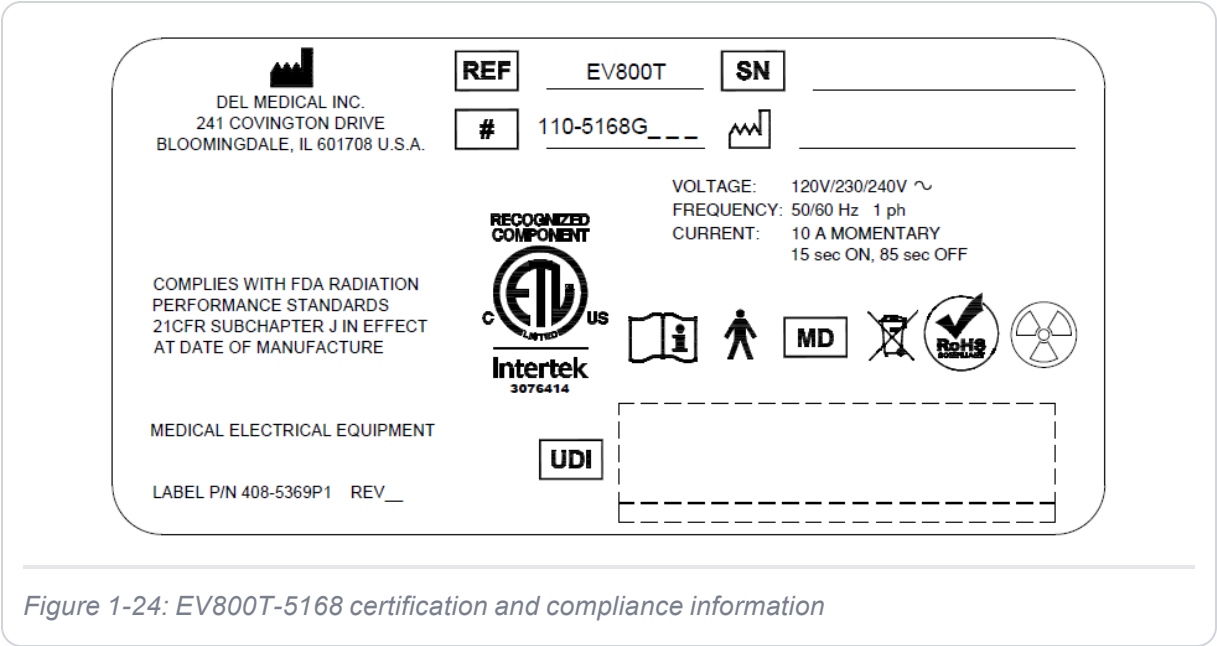


Figure 1-24: EV800T-5168 certification and compliance information

1.9.9 EV800T-5184 Label

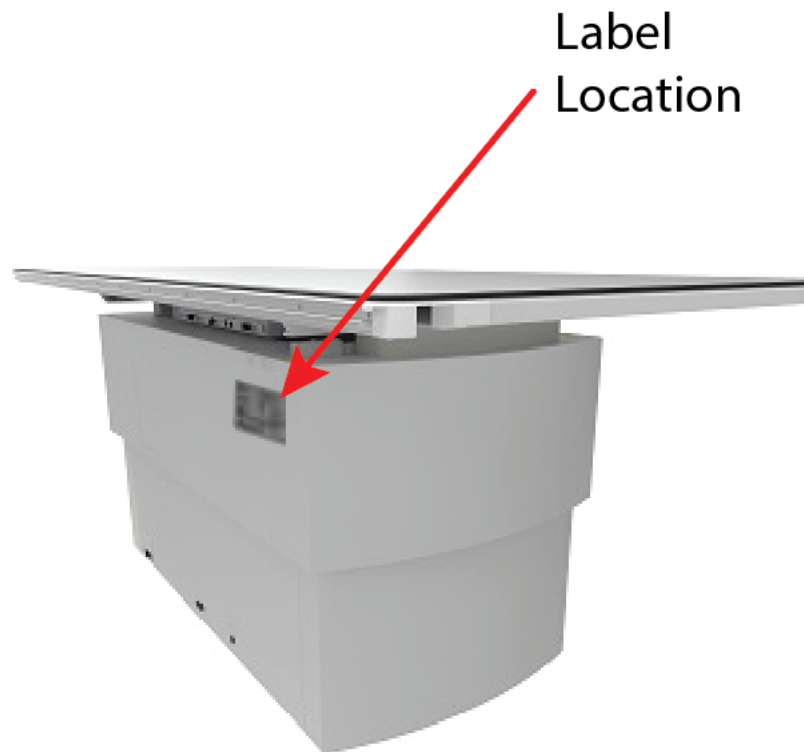
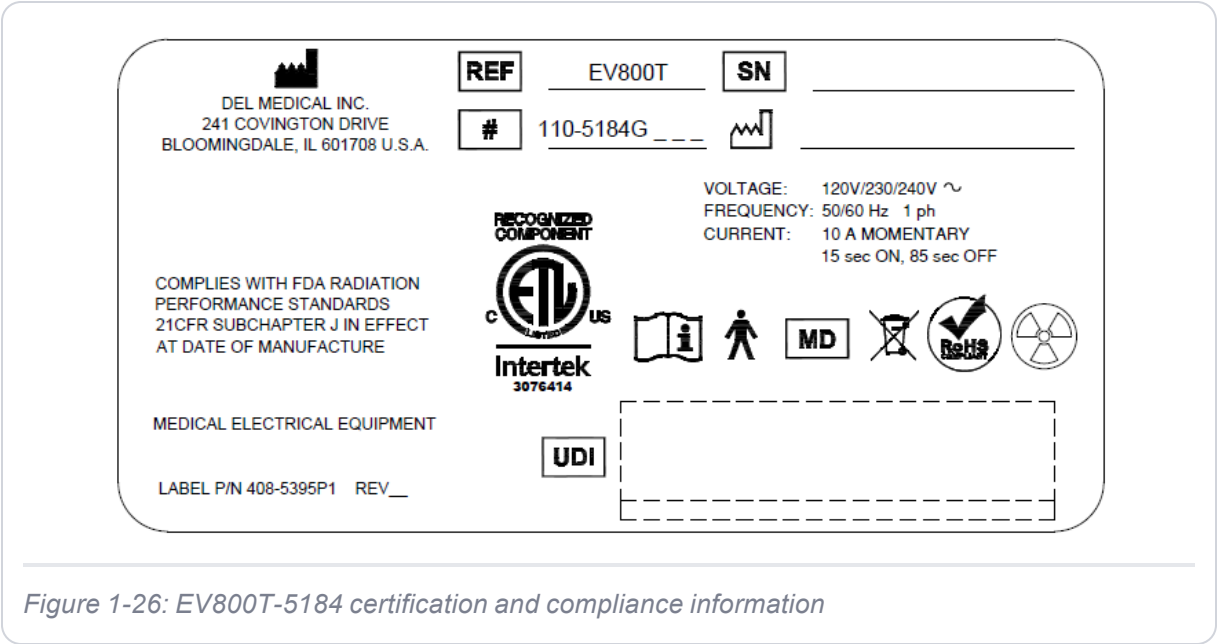


Figure 1-25: EV800T-5184 Elevating Table with Motorized Receptor Back View and Label Location



1.9.10 VS300-5154 Vertical Wallstand Label





DEL MEDICAL INC.
241 COVINGTON DRIVE
BLOOMINGDALE, IL 601708 U.S.A.

REF

VS300

SN

#

110-5154G

COMPLIES WITH FDA RADIATION
PERFORMANCE STANDARDS
21CFR SUBCHAPTER J IN EFFECT
AT DATE OF MANUFACTURE

1.2 A @ 24 V

RECOGNIZED
COMPONENT



Intertek
3076414



MD



MEDICAL ELECTRICAL EQUIPMENT

UDI

LABEL P/N 408-5341P1 REV__

Figure 1-28: VS300-5154 Vertical Wallstand certification and compliance information

1.9.11 VS300-5186 Vertical Wallstand Label

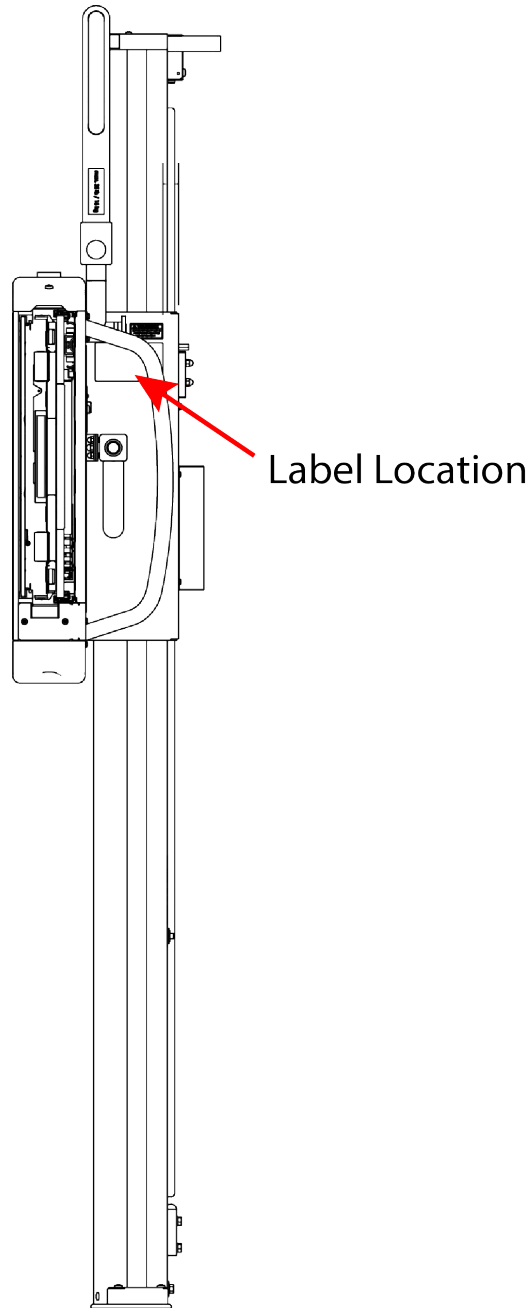


Figure 1-29: VS300-5186 Vertical Wallstand Side View and Label Location



DEL MEDICAL INC.
241 COVINGTON DRIVE
BLOOMINGDALE, IL 601708 U.S.A.

REF

VS300

SN

#

110-5186



COMPLIES WITH FDA RADIATION
PERFORMANCE STANDARDS
21CFR SUBCHAPTER J IN EFFECT
AT DATE OF MANUFACTURE

RECOGNIZED
COMPONENT



Intertek
3076414

VOLTAGE: 24V === @ 4.0A



MD



MEDICAL ELECTRICAL EQUIPMENT

UDI

LABEL P/N 408-5398P1 REV__



DEL MEDICAL INC.
241 COVINGTON DRIVE
BLOOMINGDALE, IL 601708 U.S.A.

REF

VS300

SN

#

110-5186



COMPLIES WITH FDA RADIATION
PERFORMANCE STANDARDS
21CFR SUBCHAPTER J IN EFFECT
AT DATE OF MANUFACTURE

RECOGNIZED
COMPONENT



Intertek
3076414

CURRENT: 110/240 ~ 50/60Hz @ 0.15/ 0.08A
VOLTAGE: 24V === @ 4.0A



MD



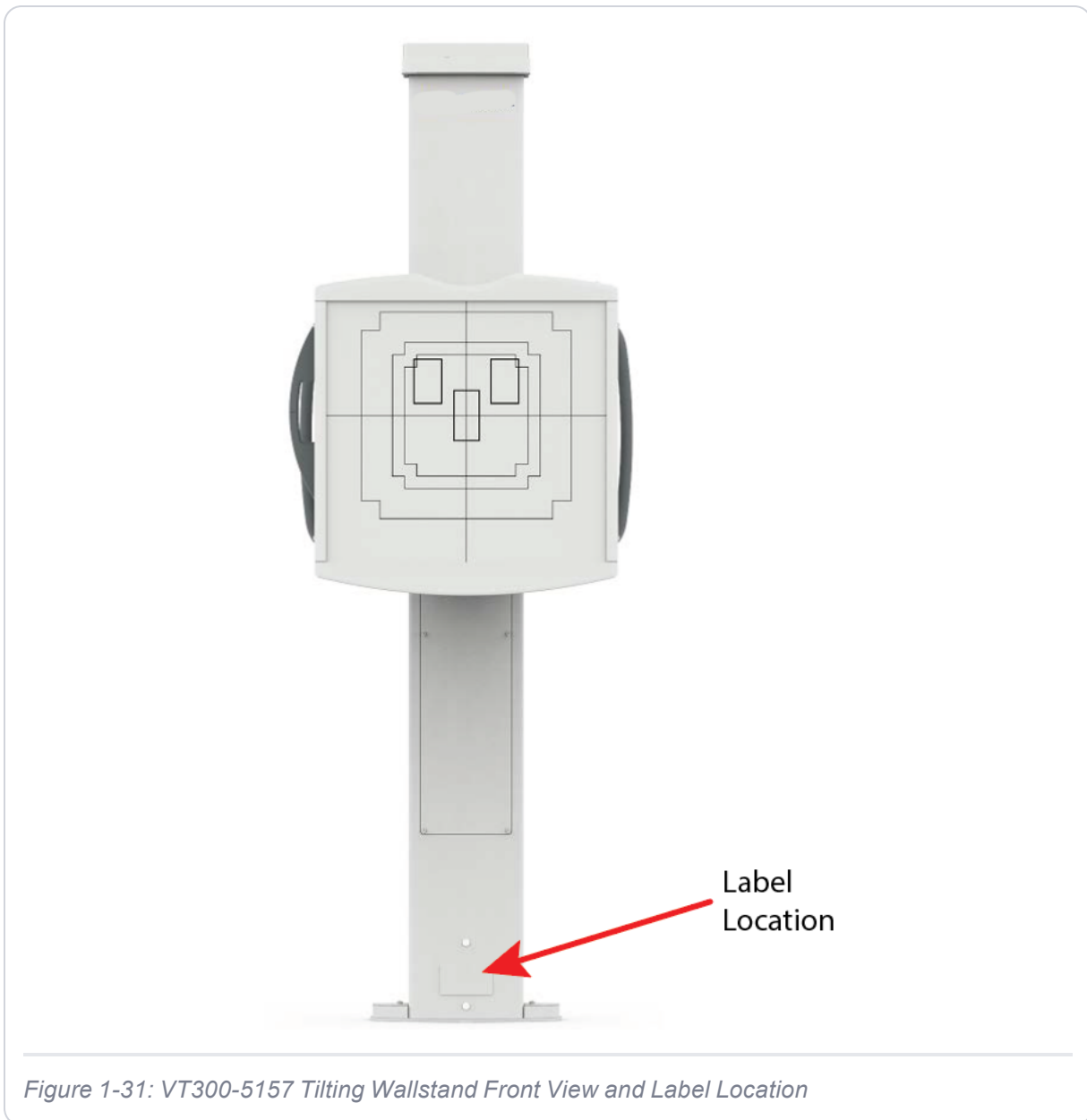
MEDICAL ELECTRICAL EQUIPMENT

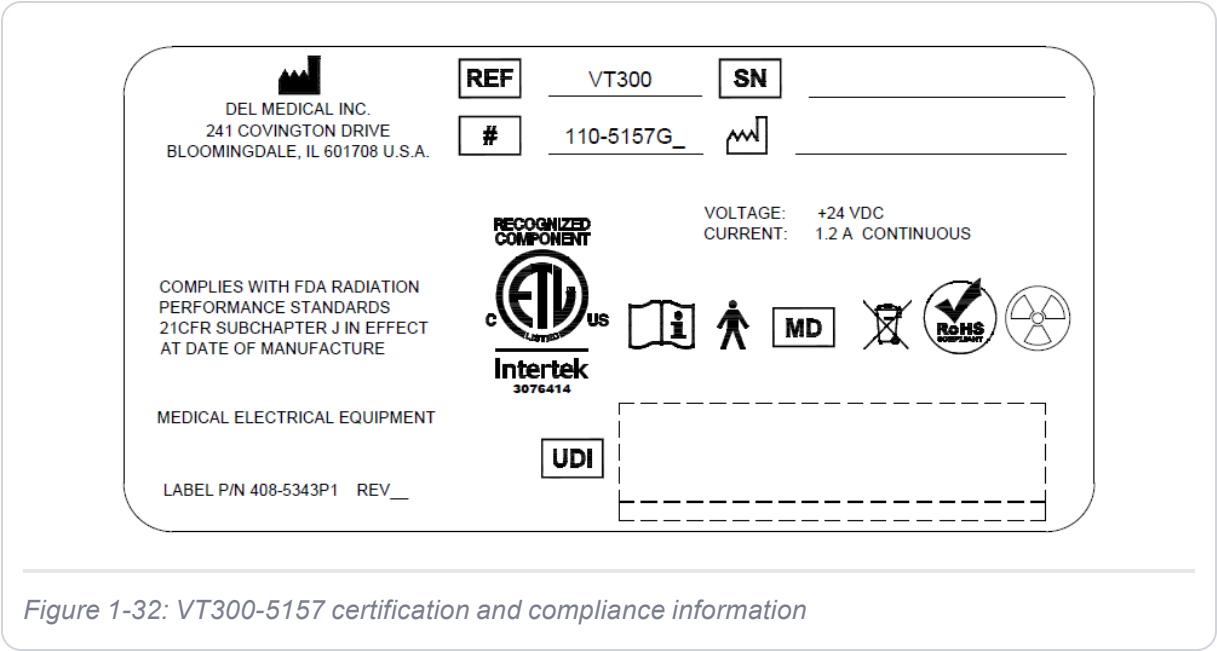
UDI

LABEL P/N 408-5398P1 REV__

Figure 1-30: VS300-5186Standard vs Charging, certification and compliance information

1.9.12 VT300-5157 Tilting Wallstand Label





1.9.13 VT300-5187 Tilting Wallstand Label

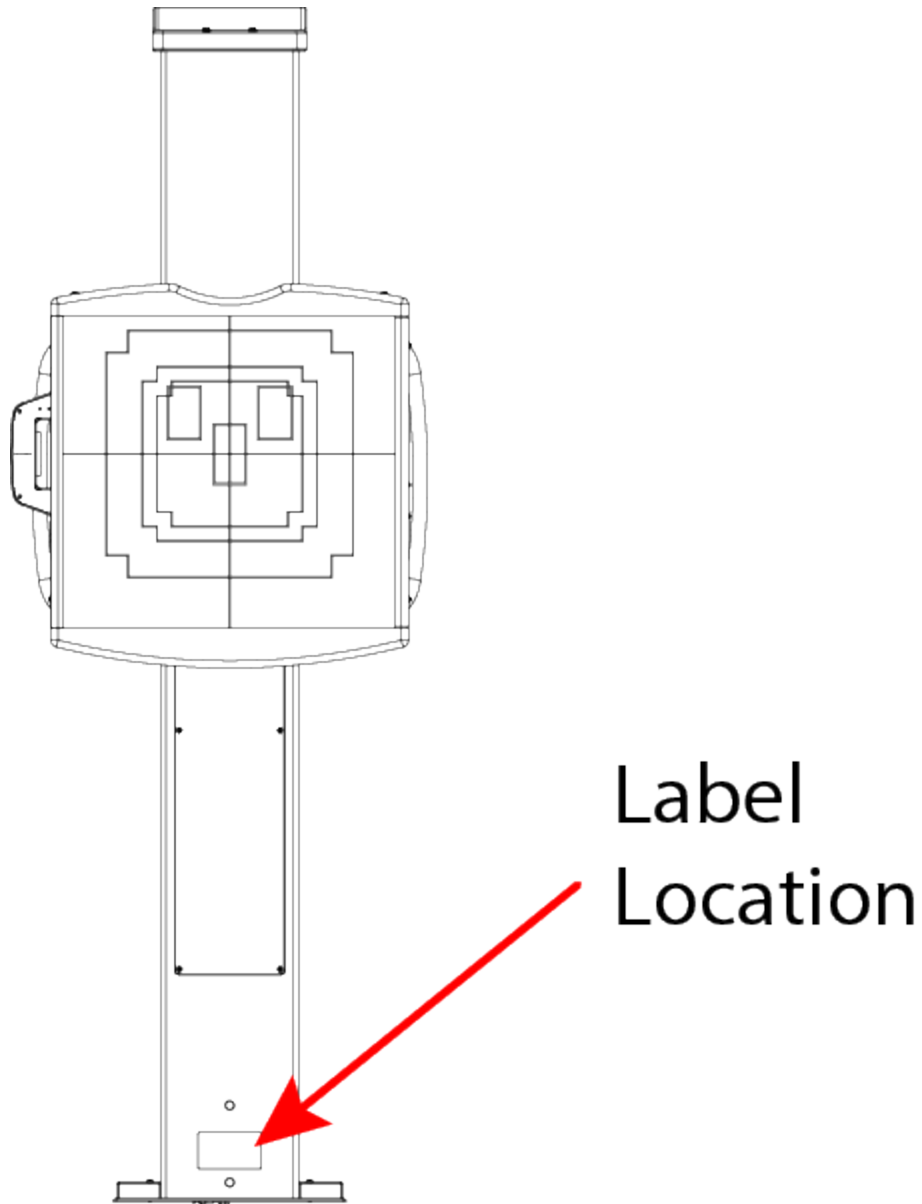


Figure 1-33: VT300-5187 Tilting Wallstand Front View and Label Location



DEL MEDICAL INC.
241 COVINGTON DRIVE
BLOOMINGDALE, IL 601708 U.S.A.

REF

VT300

SN

#

110-5187



COMPLIES WITH FDA RADIATION
PERFORMANCE STANDARDS
21CFR SUBCHAPTER J IN EFFECT
AT DATE OF MANUFACTURE

RECOGNIZED
COMPONENT



US

Intertek
3076414

VOLTAGE: 24V === @ 4.0A



MD



MEDICAL ELECTRICAL EQUIPMENT

UDI

LABEL P/N 408-5399P1 REV__



DEL MEDICAL INC.
241 COVINGTON DRIVE
BLOOMINGDALE, IL 601708 U.S.A.

REF

VT300

SN

#

110-5187



COMPLIES WITH FDA RADIATION
PERFORMANCE STANDARDS
21CFR SUBCHAPTER J IN EFFECT
AT DATE OF MANUFACTURE

RECOGNIZED
COMPONENT



US

Intertek
3076414

CURRENT: 110/240 ~ 50/60Hz @ 0.15/ 0.08A
VOLTAGE: 24V === @ 4.0A



MD



MEDICAL ELECTRICAL EQUIPMENT

UDI

LABEL P/N 408-5399P1 REV__

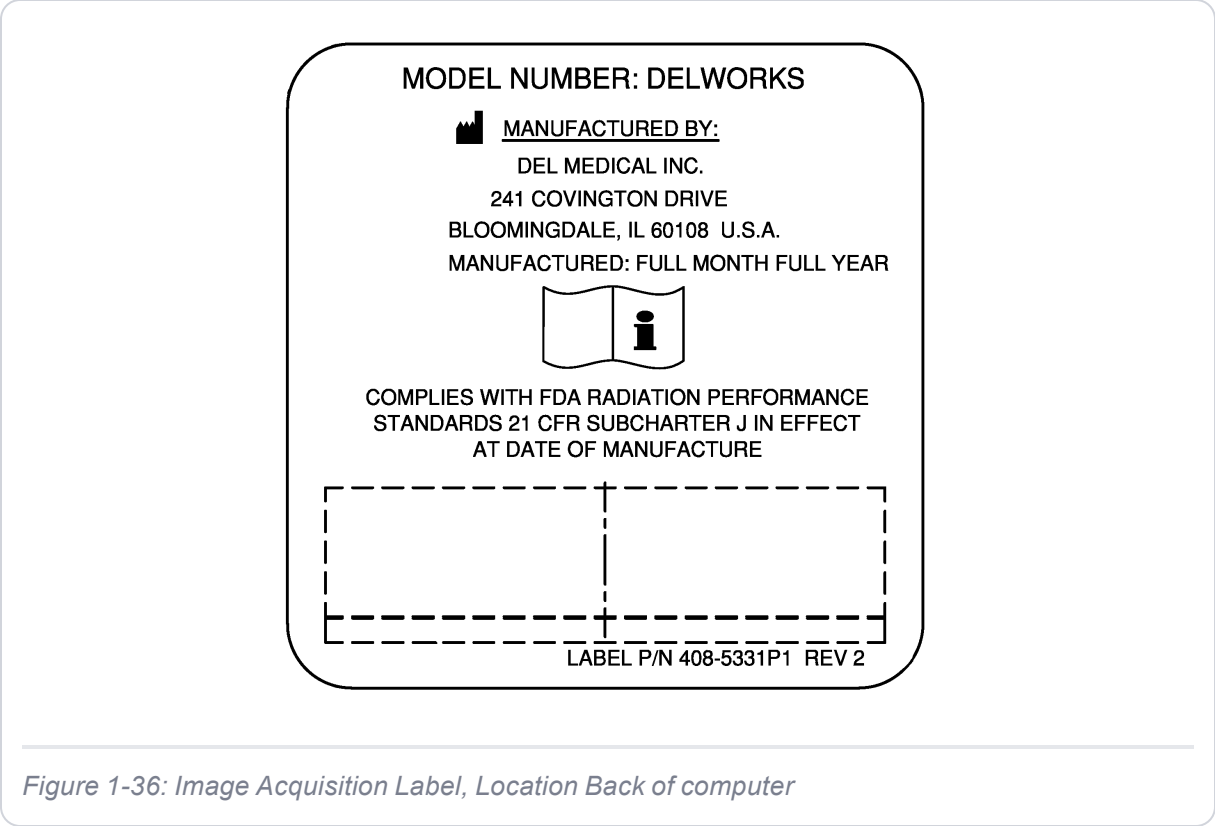
Figure 1-34: VT300-5187Standard vs Charging, certification and compliance information

1.9.14 Other Labels

1.9.14.1 Image Acquisition Computer Label



Figure 1-35: Image Acquisition Label, Location Back of computer



1.9.14.2 Generator Label Location

See accompanying documentation

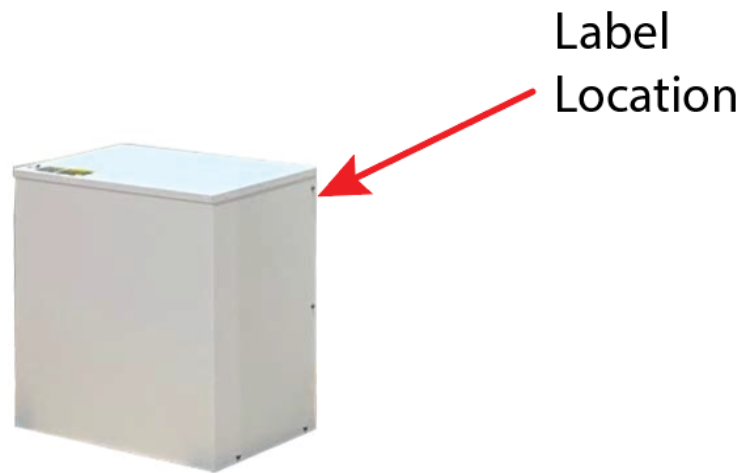


Figure 1-37: Generator Label, Location Back of Generator

RoHS	MANUFACTURED FOR	
	DEL MEDICAL IMAGING CORP.	
	BLOOMINGDALE, IL. 60108	
	BY	
	Communications & Power Industries Canada Inc.	
	45 River Drive, Georgetown, Ontario, Canada, L7G 2J4	
	MADE IN CANADA	
MODEL NO.: _____		
SERIES: _____		
PART NO.: _____		
SERIAL NO.: _____		
MFG. DATE: _____		
VOLTS AC: _____		
	MOMENTARY	NOMINAL
AMPS AC: _____		
FREQ: _____ PHASE: _____		
COMPLIES WITH IEC 60601-2-54: 2009		
& CDRH RADIATION PERFORMANCE		
STANDARD 21 CFR, SUBCHAPTER J		

Figure 1-38: Generator certification and compliance information

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2 Operation

2.1 Introduction

This chapter describes the aspects of operation and safety for the DM-OTC18T Overhead Tube Crane System.

2.2 Safety Precautions

Before attempting to operate the DM-OTC18T, familiarize yourself with the safety concerns listed in any accompanying documentation and in *"S1 Safety Information" on page 11*.



Warning: DMOTC18T is intended to be used as part of a system for the intended generation of X-rays for medical diagnosis. Do not use DMOTC18T for anything other than medical diagnosis.



Warning: Take precaution when touching or making contact with the X-ray tube. Depending on usage, the tube may be very hot and could cause burns.



Warning: No foreign objects that can attenuate or scatter the X-ray beam are allowed between the X-ray tube and the wall stand face during exposure. Failure to observe this may result in serious injury



Warning: No foreign objects that can attenuate or scatter the X-ray beam are allowed between the X-ray tube and the table during exposure. Failure to observe this may result in serious injury



Warning: X-rays generate a potential risk for both patients and operators. For this reason, the application of X-rays for a given medical purpose must aim at the minimization of radiation exposure to any persons.



Warning: Those persons responsible for the application must have the specific knowledge according to legal requirements and regulations and must establish safe exposure procedures for this kind of system.



Warning: Those persons responsible for the planning and installation of this equipment must observe all appropriate regulations.

2.3 Powering On/Off

All of the components with the exception of the image acquisition system are all powered on when the generator is powered on. The following subsections describe how to power on/off the generator depending on which console is installed with the system. Only follow the instructions for the console installed.

For information on powering on/off the image acquisition system, see the documentation related to the image acquisition system.

2.3.1 CM Series Mini Console



Figure 2-1: Mini Console Power On/Off

2.3.2 CM Series Full Membrane Console



Figure 2-2: Console Power On/Off

2.3.3 RF Series Control Room Module

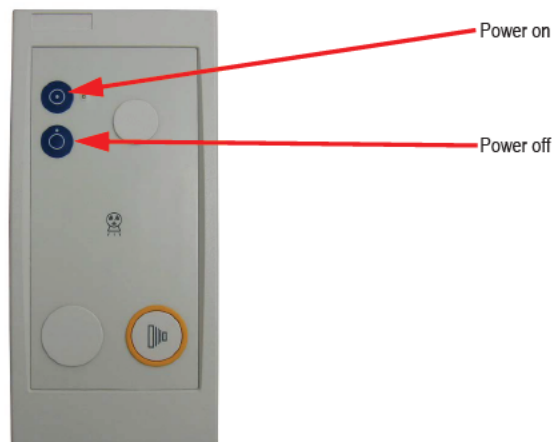


Figure 2-3: Control Room Module Power On/Off

2.3.4 RF Series Button Console



Figure 2-4: Button Console Power On/Off

2.3.5 RF Series Touchscreen Console



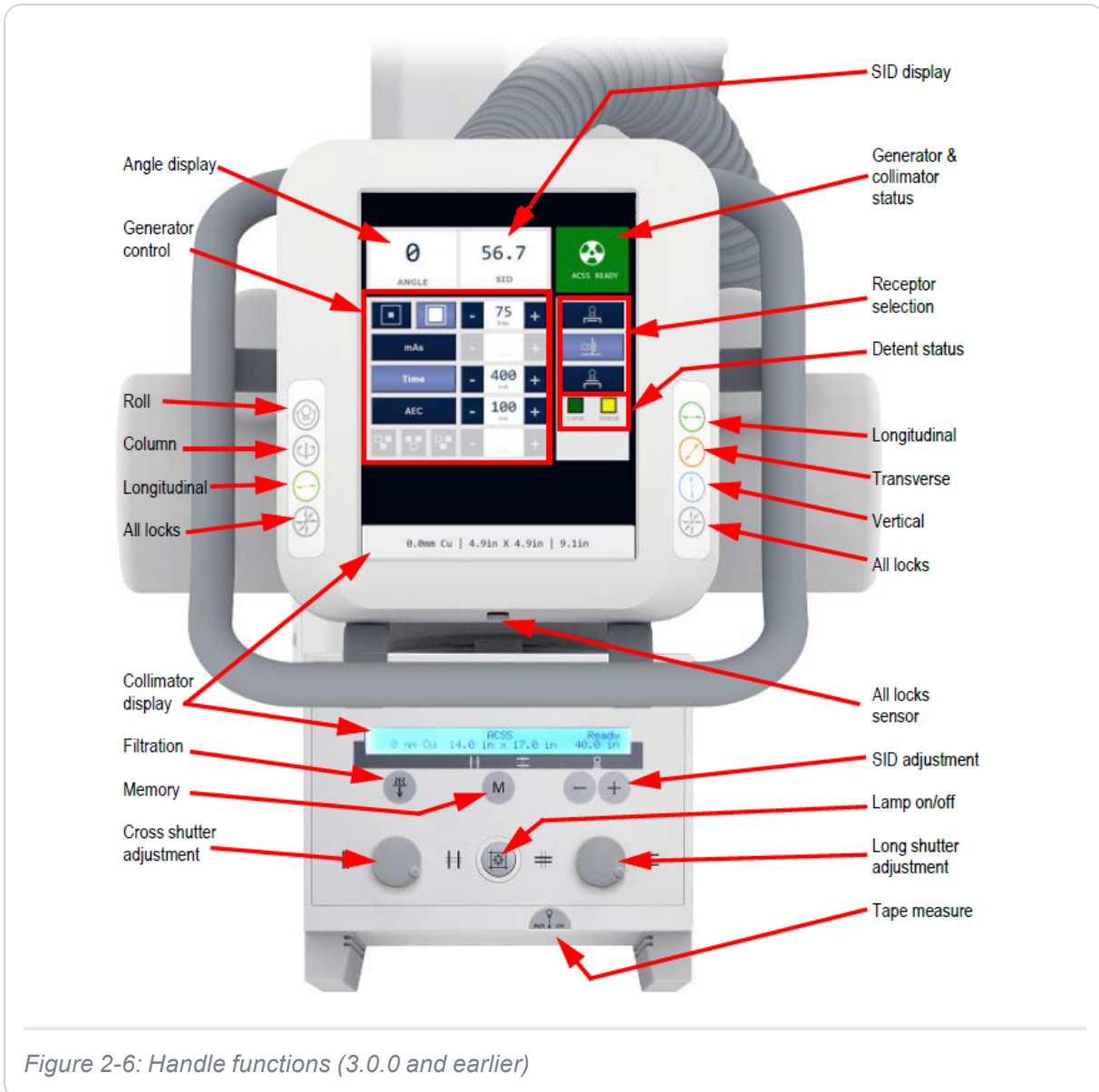
Figure 2-5: Touchscreen Console Power On/Off

2.4 Equipment Controls

This section describes the controls for each component part of the DM-OTC18T Overhead Tube Crane System.

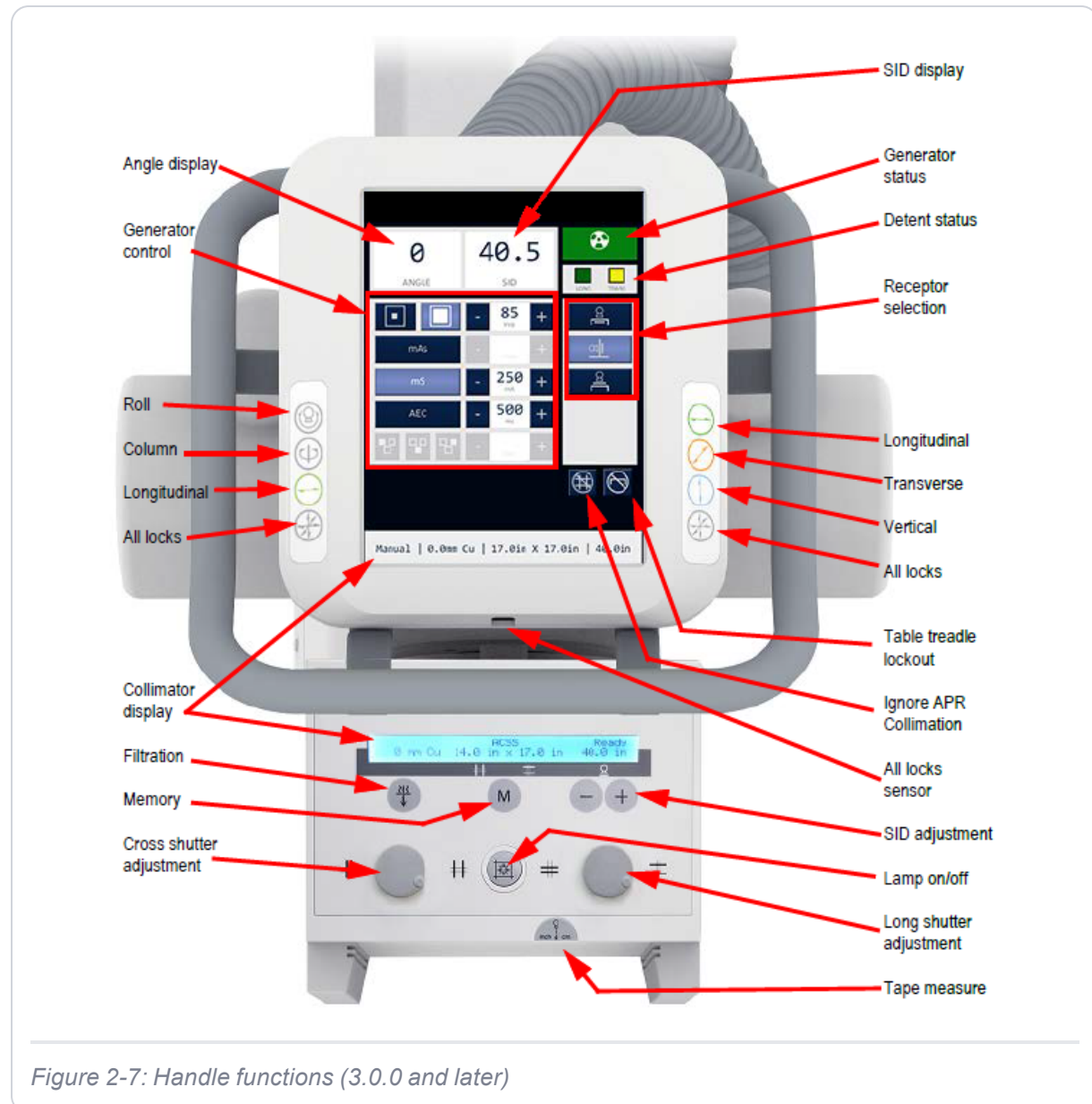
2.4.1 User Interface Overview Version 3.0.0 and Earlier

"Figure 2-6: Handle functions (3.0.0 and earlier)" below shows the handle assembly of the DM-OTC18T Overhead Tube Crane System with software version earlier than 3.0.0. The paragraphs that follow describe each control.



2.4.2 User Interface Overview Version 3.0.0 and Later

"Figure 2-7: Handle functions (3.0.0 and later)" below shows the handle assembly of the DM-OTC18T Overhead Tube Crane System (with software version later than 3.0.0). The paragraphs that follow describe each control.





Note: Automatic collimator functions shown. Not all functions available when a manual collimator is installed

2.4.3 Button Functions

This sub-section describes functions of the physical overlay and push buttons on the DM-OTC18T user handle.

Icon	Function	Description
	Roll	Rotation about the tube's horizontal axis is called roll. Press and hold this button to roll the tube-head to the desired position, then release the button to lock the angle. Roll angle is shown in the inclinometer, and there are detents at 90 and -90 degrees.
	Column	Press and hold this button to rotate the vertical column up to 110 degrees in either direction, for parking or certain cross table techniques. There are detents at 90 and -90 degrees.
	Vertical	Press and hold this button to move the tubehead up or down without disturbing other positioning.
	Transverse	Press and hold this button to move the tubehead in or out from the column without disturbing other positioning.
	Longitudinal	Press and hold this button to move the column left or right without disturbing other positioning.

Icon	Function	Description
	All Locks	Press this button for free movement in vertical, longitudinal, and transverse directions. This can also be achieved using the all locks sensor on the handlebar. See <i>"Figure 2-7: Handle functions (3.0.0 and later)"</i> on page 84
	Motorized Up	Press this button to drive the tube head vertically up.
	Motorized Down	Press this button to drive the tube head vertically down.

Table 2-1: Button Functions

2.4.4 AL02 Automatic Collimator Button Functions

Icon	Function	Description
	Collimator Filter	Press this button to cycle through the different copper filtration options. It will cycle between 0.0mm Cu, 0.1mm Cu, 0.2mm Cu, and 0.3mm Cu.
	Memory	When pressed, this button will apply to the collimator the sizes it was opened at during the last exposure.
	Lamp On/Off	Pressing this button will either turn the collimator lamp and laser on or off depending on the state of the lamp.

Icon	Function	Description
	SID Adjustment	These buttons will adjust the collimation size based on the SID.  Note: This function is only available when the collimator is in manual mode.

Table 2-2: AL02 Collimator Button Functions

2.4.5 ML03 Manual Collimator Button Functions

Icon	Function	Description
	Lamp On/Off	Pressing this button will either turn the collimator lamp and laser on or off depending on the state of the lamp.

Table 2-3: ML03 Collimator Button Functions

2.4.6 Touchscreen Functions

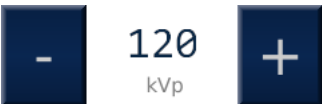
This sub-section describes the functions of the touchscreen controls. When a touch button is inactive, it will display in a dark blue color. When a touch button is active, it will display in a light blue color. See "*Figure 2-8: Active and inactive touch buttons*" on the next page as an example.

 **Note:** Not all functions described are touch buttons

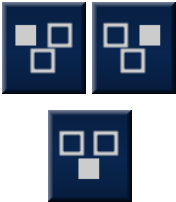


Figure 2-8: Active and inactive touch buttons

Icon	Function	Description
90 ANGLE	Angle Display	Displays the angle of the tube. This is not selectable from the touchscreen and only displays based on the rotate of the tube.
40.0 SID	SID Display	Displays the current SID to the selected receptor. This is not selectable from the touchscreen and only displays based on the physical distance. This will only display based on the receptor selected, angle, and detents.
	Status Display (Green)	Indicates the system is ready for an exposure..
	Status Display (Red)	Indicates the positioning is not correct for the selected receptor or the detector is not in the tray.
	Large Focus	Press to select the large focal spot.

Icon	Function	Description
	Small Focus	Press to select the small focal spot.
	mAs	Sets the generator to mAs (2 point) mode.
	ms	Sets the generator to time (3 point) mode.
	AEC	Sets the generator to Automatic Exposure Control mode.
	kVp Selection	Sets the kVp on the generator.
	mAs Selection	Sets the mAs on the generator.  Note: Only available when mAs mode is selected.
	mA Selection	Sets the mA on the generator.  Note: Only available when ms or AEC modes are selected.

Icon	Function	Description
	ms Selection	Sets the time on the generator.  Note: Only available when ms mode is selected.
	Density Selection	Sets the AEC density reference value  Note: Only available when AEC mode is selected.
	Table Receptor	Selects the table receptor (receptor 1)
	Wall Receptor	Selects the wallstand receptor (receptor 3)
	Free Receptor	Selects the table top or free receptor (receptor 2)  Note: No SID will be displayed unless over the table

Icon	Function	Description
	AEC Chamber Cell Selection	These buttons select the active cell(s) for the AEC chamber. Any combination of cells can be selected together as long as at least one cell is active. Note: Only available when AEC mode is selected
	Table Tracking	This button activates the auto tracking feature for the table receptor. Note: This button is only available when the table receptor, or table-top receptor is selected.
	Default SID	Automatically drives the tube crane to the default SID for the table of 40 inches (100cm). Note: This button is only available when the table receptor or table-top receptor are selected.

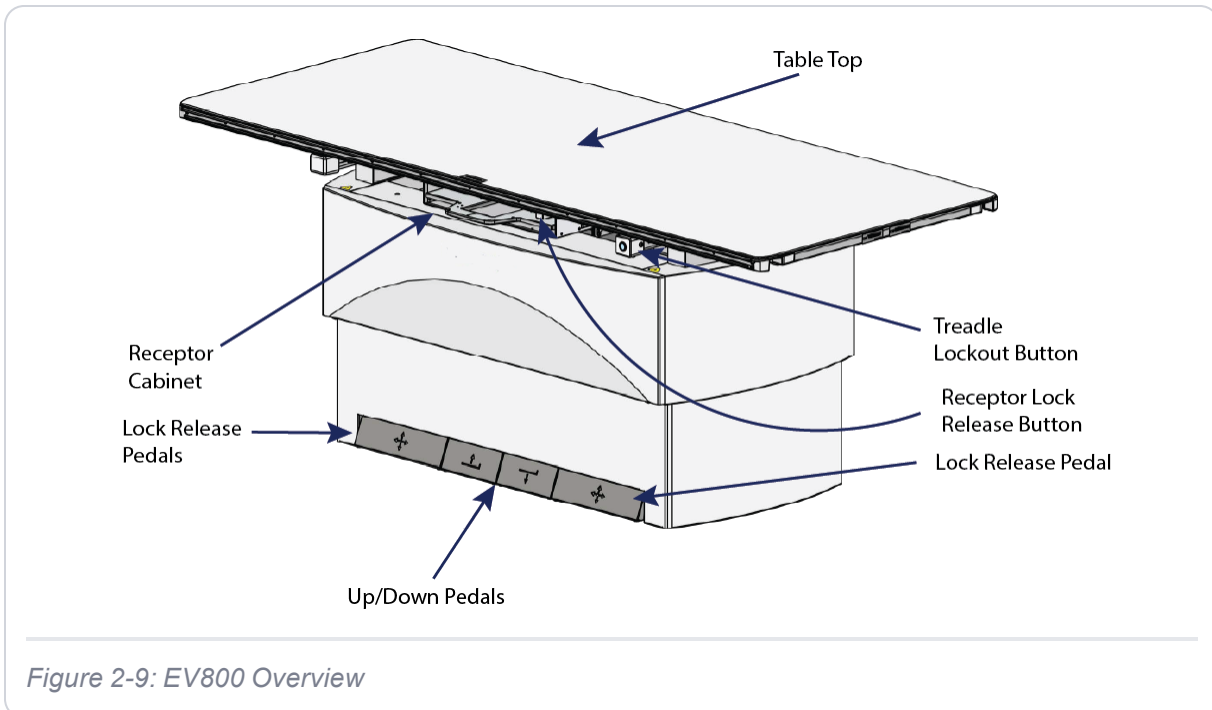
Icon	Function	Description
	Sync	When active the table receptor is synced with the logintudinal movement of the tube crane. Note: This button is only available when the table receptor, or table-top receptor is selected.
	Wall tracking	This indicator will display when the auto tracking feature for the wallstand is active.
	Table Treadle Dis- able	When active the table foot treadle is disabled. (version 3.0.0 and later only).
	Ignore APR Collimator	When active the system will ignore any collimator undersizing sent from the DR workstation. (versions GUI 3.5.0 / SCU 3.4.0 and later with auto collimator only).

Table 2-4: Touchscreen Button Functions

2.5 EV800 Controls

Shows the controls and functions of the EV800 Table

**Note:** This section covers the EV800-5183 Elevating Table and EV800-5171 Elevating Table.



Lock Release Treadle

To release the table top and move it front to back or side to side, step on the lock release treadle. The transverse range is 9.75" (24 cm). The side-to-side range is 55.38" (140 cm).

To lock the table top into position, remove your foot from the treadle.

Up Treadle

To raise the table top, press the up treadle. The table pauses briefly at midpoint (or you may need to release and press the treadle again to continue).

Down Treadle

To lower the table top, press the down treadle. The table pauses briefly at midpoint (or you may need to release and press the treadle again to continue).



Note: The lock release treadle can be disabled by pressing the treadle disable button on the touchscreen. See *"Table Treadle Disable"* on the previous page

Receptor Cabinet Lock Release

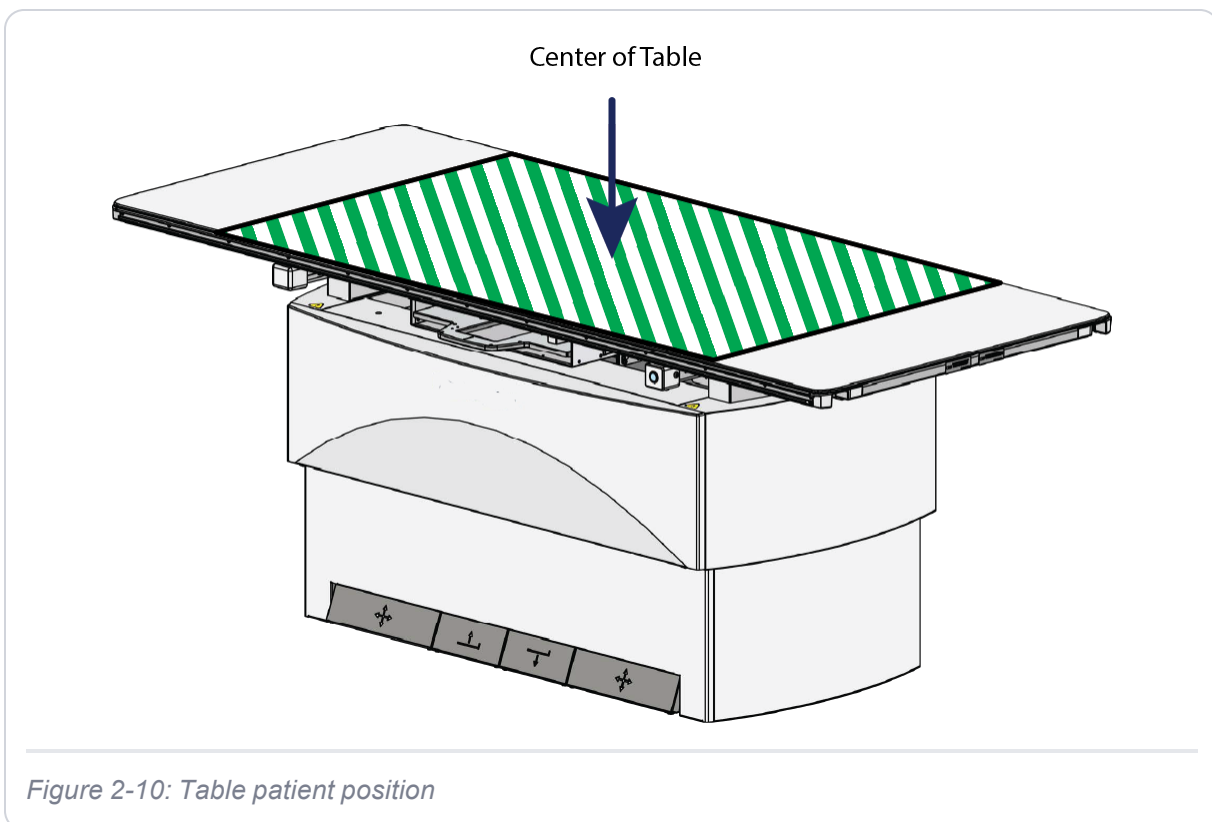
To move the cabinet left or right, press and hold the cabinet carriage lock release button at the front of the cabinet. The travel range of the cabinet is 11.4" (29 cm). To secure the cabinet in place, release the button.

Power Indicator

The green power light indicates that power to the table is on and the table is ready to operate.

Table Top

The surface which the patient is positioned. It is important that the patient is loaded/unloaded and positioned in the center of the table top.



Total weight allowed is measured with a weight  symbol label found on the top of the front cover, below the table-top of the table. Please refer to the specifications table found in "[1 Introduction](#)" on page 27 for more information on weight capacity.



Warning: Do not place any objects within the vertical motion area above or below the table. Objects in such areas (see *"2.5.1 EV800 Crush Risk Areas" on the next page*) can present crush hazards for the patient or operator and can also damage the table.



Warning: During downward motion, the table will stop if it encounters an object (resistance of 100 lbs [44 kg] of upward force). This feature is designed to protect the table, not a person or object.



Warning: Do not touch the table's lock release treadles while the patient is getting on or off the table. This may cause a fall and serious injury.

It is recommended to disable the foot pedals by activating the treadle disable  feature on the touchscreen or the button on the table any time a patient is being loaded or unloaded from the table.



Caution: Loading/unloading a patient on the edges of the table top could cause permanent damage to the table top or other table mechanics.



Caution: When the system is turned off or there is no power applied to the table, the table top locks will remain engaged. Do not try to force the table top to move while it is powered off.

2.5.1 EV800 Crush Risk Areas



Warning: Do not place any body parts or objects within the vertical motion area above or below the table. Objects in such areas (*"Figure 2-11: Vertical crush risk area" below*) can present crush hazards for the patient or operator and can also damage the table.

During downward motion, the table will stop if it encounters an object (resistance of 100 lbs [44 kg] of upward force). This feature is designed to protect the table, not a person or object.

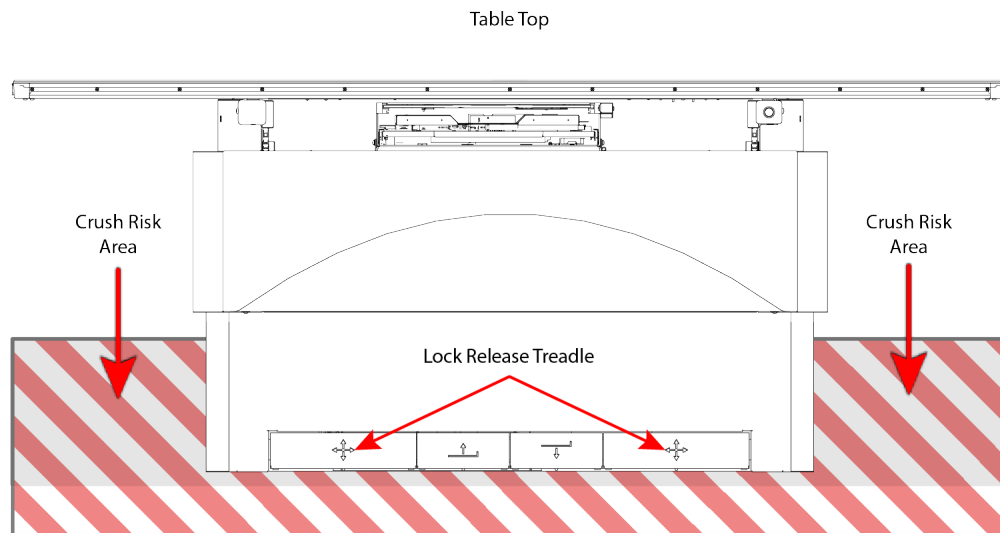
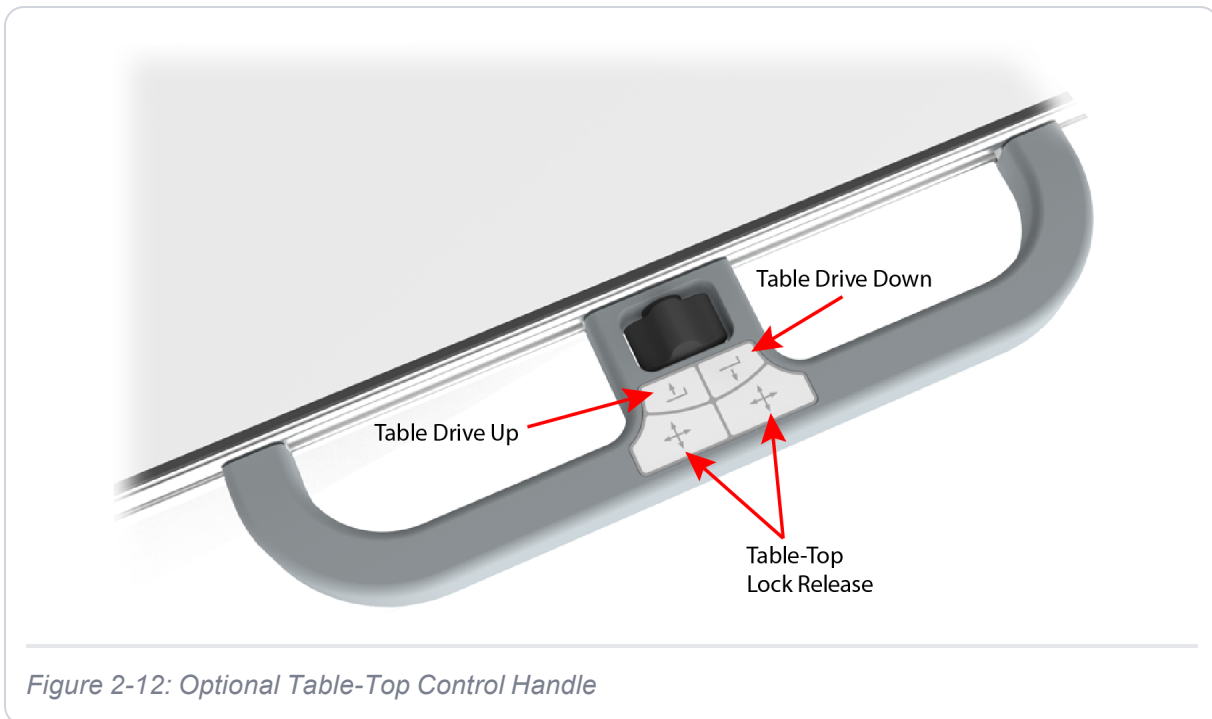


Figure 2-11: Vertical crush risk area

2.5.2 Table-Top Control Handle (Optional)



Note: When the table treadle disable button  is active on the touchscreen the foot pedals are disabled but the handswitch controls are still active.



Caution: The EV800 Table uses a combination of permanent and electromagnetic locks to lock the table top in place. If power is lost or the power to the system is turned off, the table top will remain locked in place. Do not attempt to move the table top without power.

2.6 EV800T Controls

Shows the controls and functions of the EV800T table.



Note: This section covers the EV800T-5184 Elevating Table with Motorized Receptor and EV800T-5168 Elevating Table with Motorized Receptor.

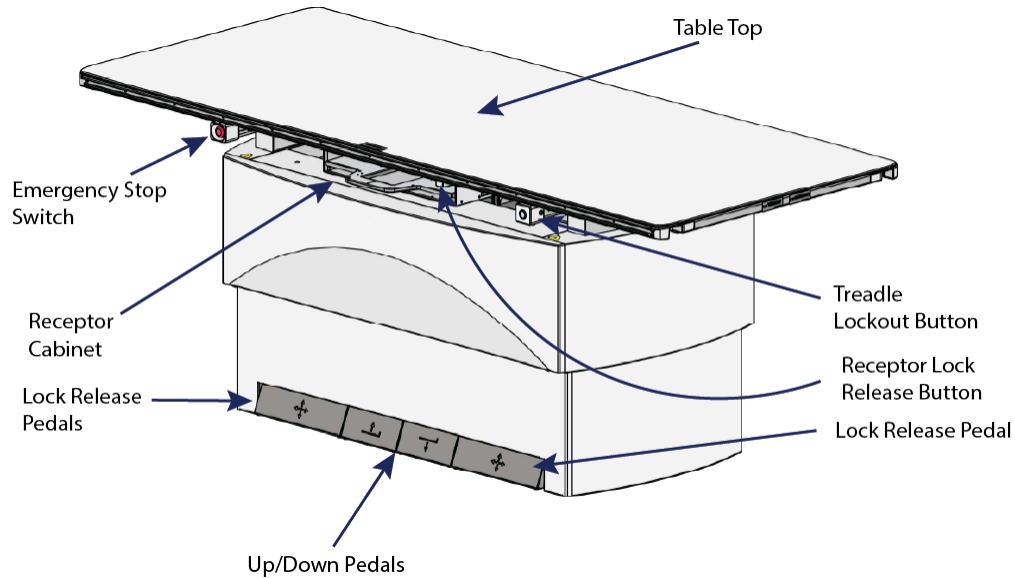


Figure 2-13: EV800T Overview

Lock Release Treadle

To release the table top and move it front to back or side to side, step on the lock release treadle. The transverse range is 9.75" (24 cm). The side-to-side range is 55.38" (140 cm).

To lock the table top into position, remove your foot from the treadle.

E-Stop Switch

Pressing the e-stop will disable any robotic movement motors. The e-stop does not disable the table vertical drive. To reset the switch after it has been pressed, rotate the cap clockwise.

Up Treadle

To raise the table top, press the up treadle. The table pauses briefly at midpoint (or you may need to release and press the treadle again to continue).

Down Treadle

To lower the table top, press the down treadle. The table pauses briefly at midpoint (or you may need to release and press the treadle again to continue).



Note: The lock release treadle can be disabled by pressing the treadle disable button on the touchscreen. See *"Table Treadle Disable" on page 92*

Receptor Cabinet Lock Release

To move the cabinet left or right, press and hold the cabinet carriage lock release button at the front of the cabinet. The travel range of the cabinet is 11.4" (29 cm). To secure the cabinet in place, release the button.

Power Indicator

The green power light indicates that power to the table is on and the table is ready to operate.

Table Top

The surface which the patient is positioned. It is important that the patient is loaded/unloaded and positioned in the center of the table top.

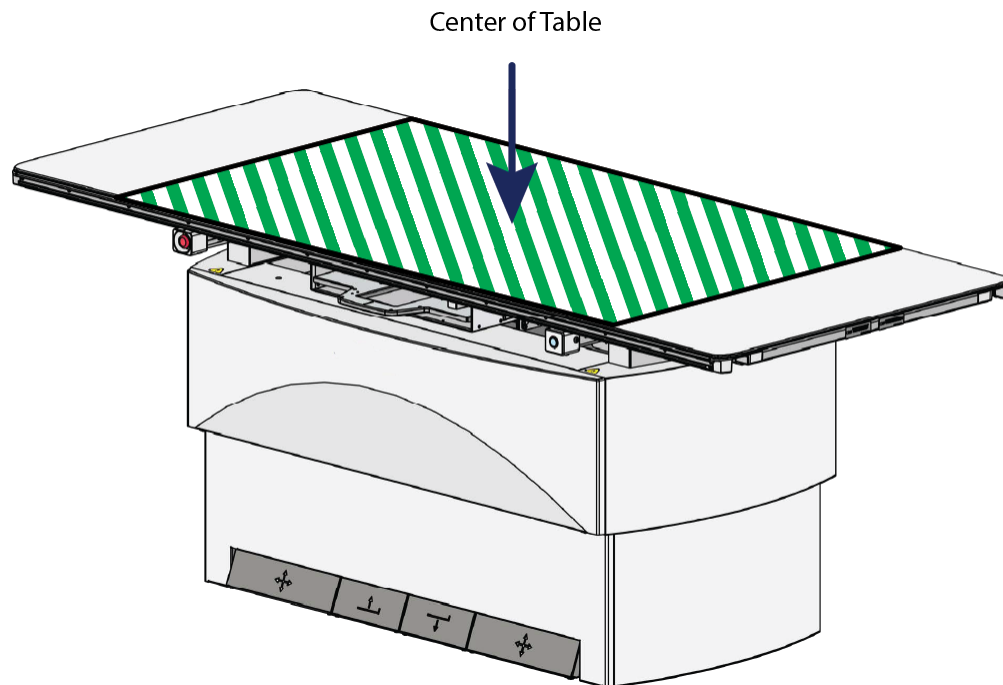


Figure 2-14: Table patient position

Total weight allowed is measured with a weight  symbol label found on the top of the front cover, below the table-top of the table. Please refer to the specifications table found in "[1 Introduction](#)" on page 27 for more information on weight capacity.



Warning: Do not place any objects within the vertical motion area above or below the table. Objects in such areas (see "[2.5.1 EV800 Crush Risk Areas](#)" on page 96) can present crush hazards for the patient or operator and can also damage the table.



Warning: During downward motion, the table will stop if it encounters an object (resistance of 100 lbs [44 kg] of upward force). This feature is designed to protect the table, not a person or object.



Warning: Do not touch the table's lock release treadles while the patient is getting on or off the table. This may cause a fall and serious injury.

It is recommended to disable the foot pedals by activating the treadle disable  feature on the touchscreen or the button on the table any time a patient is being loaded or unloaded from the table.



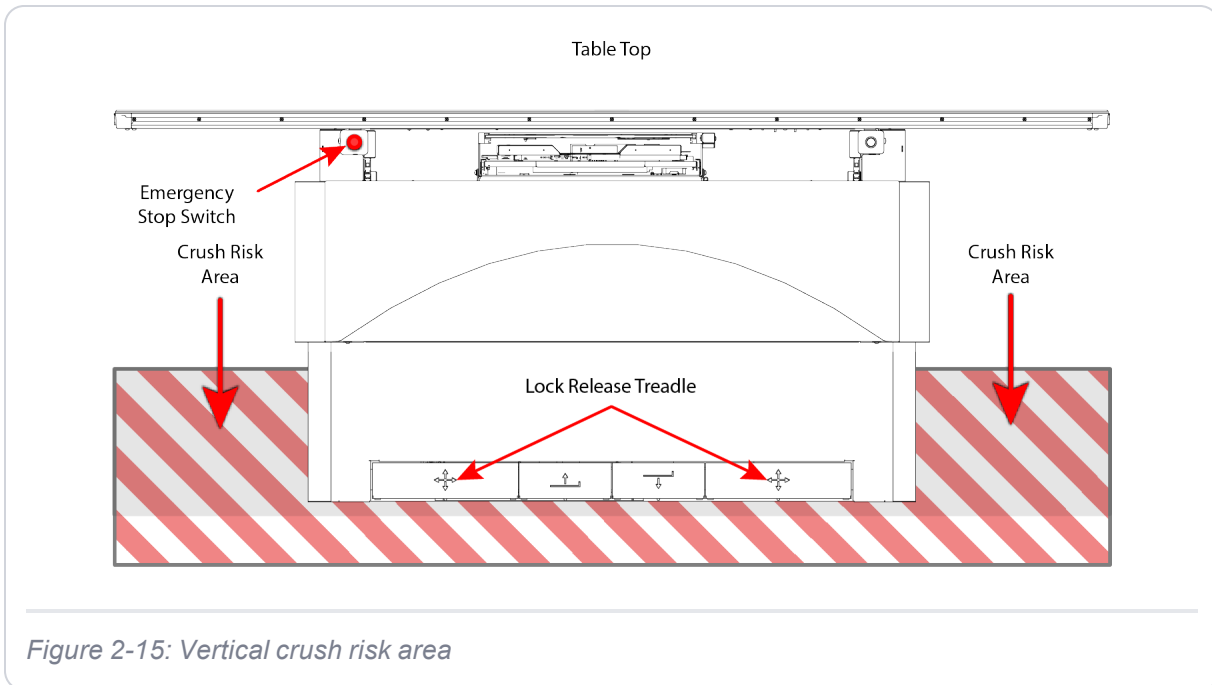
Caution: Loading/unloading a patient on the edges of the table top could cause permanent damage to the table top or other table mechanics.

2.6.1 EV800T Crush Risk Areas



Warning: Do not place any body parts or objects within the vertical motion area above or below the table. Objects in such areas (*"Figure 2-15: Vertical crush risk area" on the next page*) can present crush hazards for the patient or operator and can also damage the table.

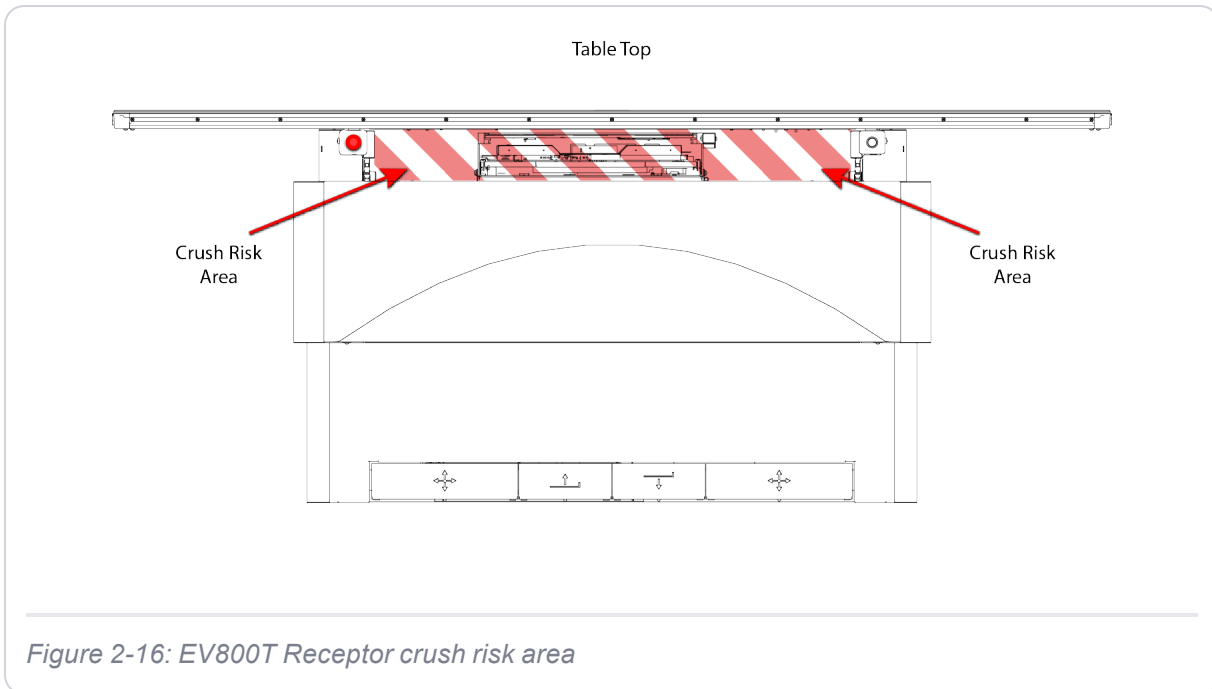
During downward motion, the table will stop if it encounters an object (resistance of 100 lbs [44 kg] of upward force). This feature is designed to protect the table, not a person or object.



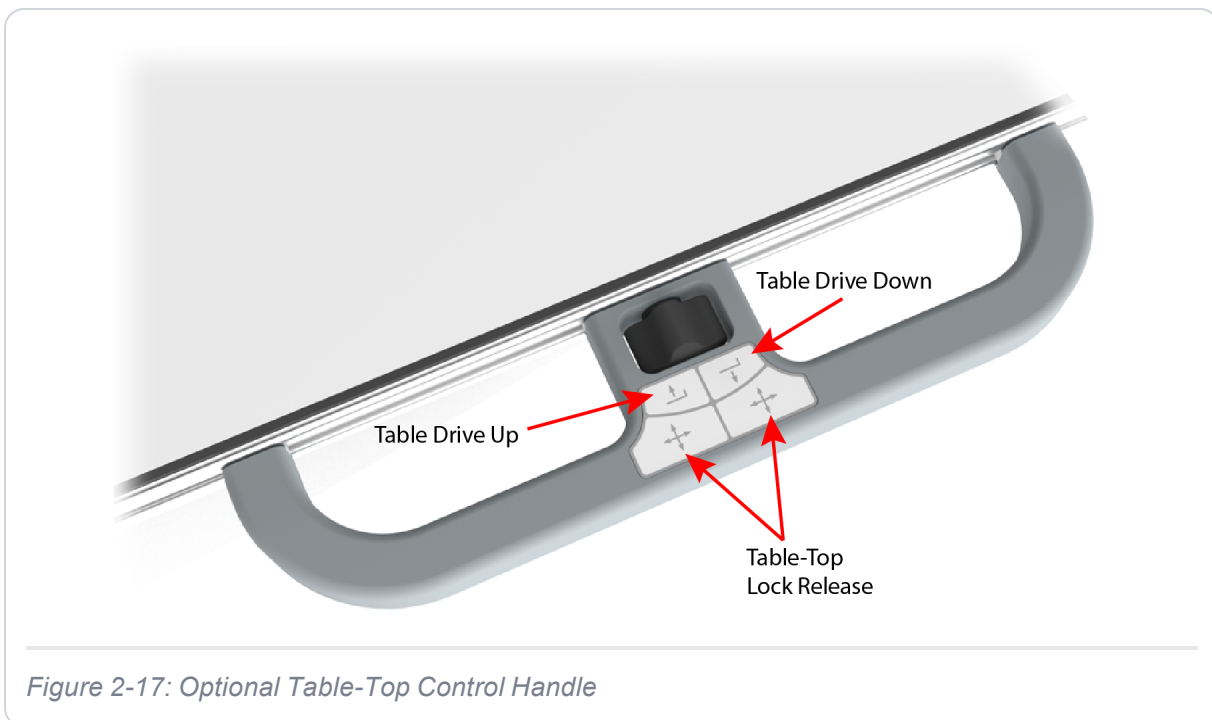
Caution: When the system is turned off or there is no power applied to the table, the table top locks will remain engaged. Do not try to force the table top to move while it is powered off.



Warning: Do not touch the table's lock release treadles while the patient is getting on or off the table. This may cause a fall and serious injury. It is recommended to disable the foot pedals by activating the treadle disable feature on the touchscreen or the button on the table any time a patient is being loaded or unloaded from the table.



2.6.2 Table-Top Control Handle (Optional)





Note: When the table treadle disable button  is active on the touchscreen the foot pedals are disabled but the handswitch controls are still active.



Caution: The EV800T table uses a combination of permanent and electromagnetic locks to lock the table top in place. If power is lost or the power to the system is turned off, the table top will remain locked in place. Do not attempt to move the table top without power.

2.7 VS300 Controls

Shows the controls and functions of the VS300-5154 Vertical Wallstand and VS300-5186 Vertical Wallstand

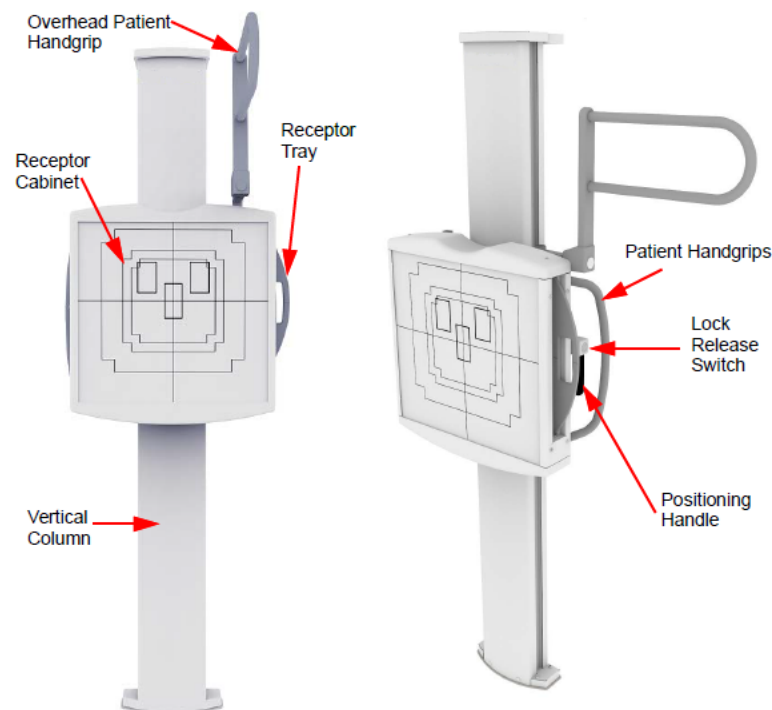


Figure 2-18: VS300 Overview

2.8 VT300 Controls

Shows the controls and functions of the VT300-5157 Tilting Wallstand and VT300-5187 Tilting Wallstand.

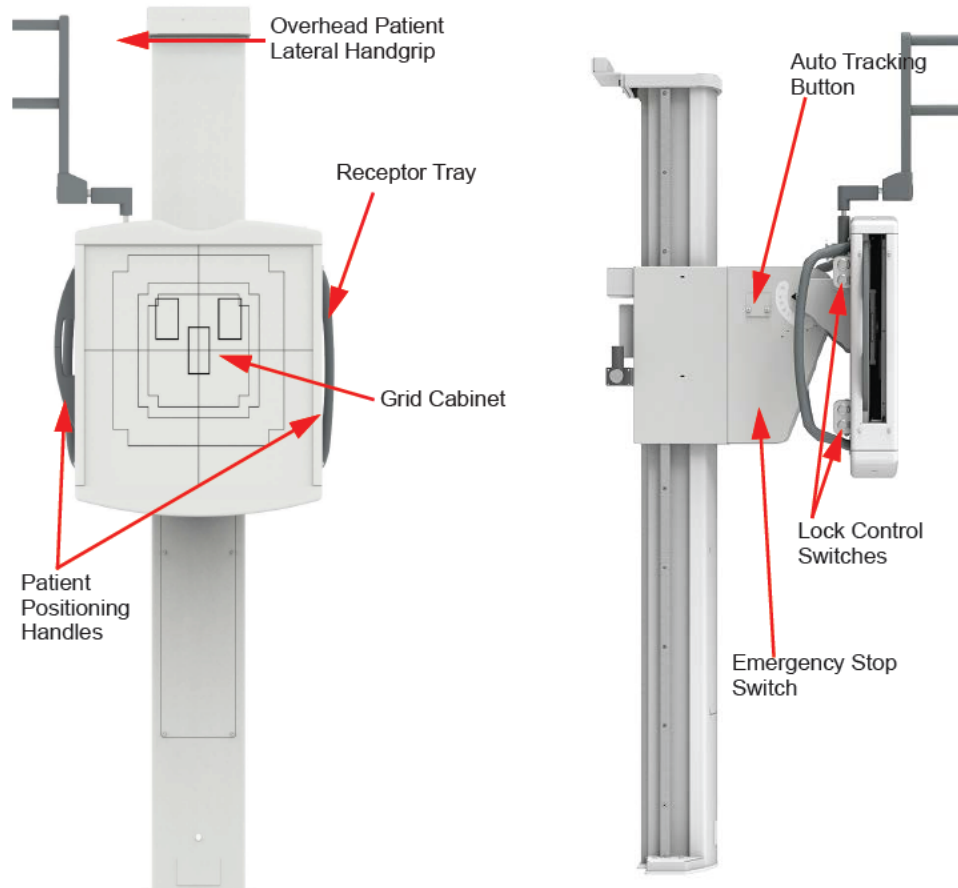


Figure 2-19: VT300 overview

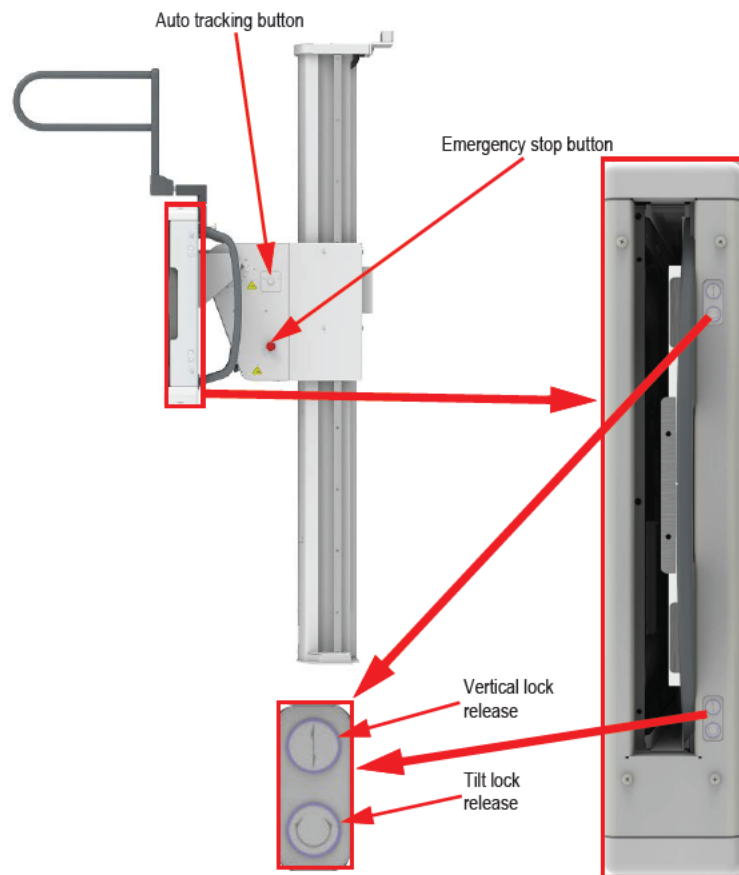
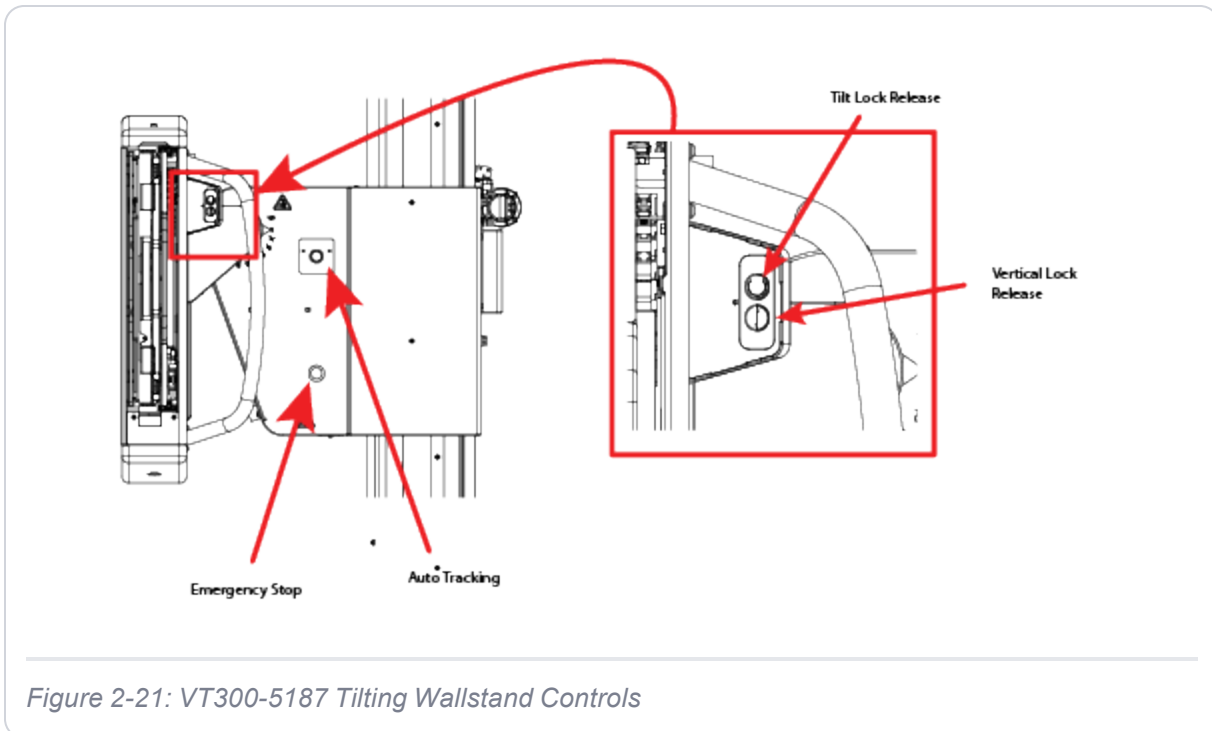


Figure 2-20: VT300-5157 controls



Note: Lock release switches located on both sides of cabinet.

2.9 Overhead Patient Hand Grip

The overhead patient hand grip is purposed to aid in the positioning of the patient depending on the study being performed. The overhead patient hand grip can be rotated on its horizontal and vertical axes.



Caution: Do not allow a patient to hang on the overhead patient hand grip. The brake assembly of the device is specified to hold 35 lbs. of force. Anything over 35 lbs. will cause the receptor cabinet to slip and move downward unexpectedly. On the VT300, use the lateral handgrip only when the receptor is tilted in the vertical position.

The overhead patient hand grip can pivot where it mounts to the wall stand. This is controlled by a spring loaded pull pin. Pull the pin out and pivot the overhead patient hand grip to operate. There are detents at 90 degree intervals. The spring loaded pin will lock into these

detents. The overhead patient hand grip can also be moved to the opposite side of the wall stand. Pull the spring loaded pin to its outmost position and hold it there. Then pull up on the P handle assembly. Repeat this process in reverse to replace the assembly. **"2.9 Overhead Patient Hand Grip" on the previous page.**

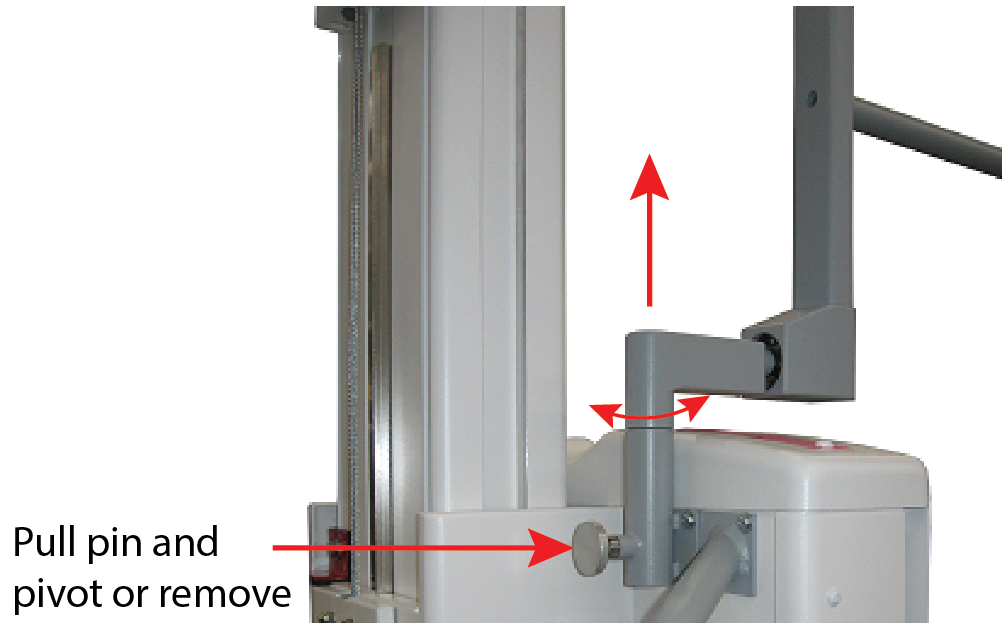


Figure 2-22: Overhead patient hand grip pivot

The P handle can also rotate a full 360 degrees independently from the wall stand mount. Pull the P handle away from the wall stand by grabbing near the rotation point. Rotate to desired position. There are predefined stop points. If your desired position doesn't lock, rotate the handle slightly in either direction to achieve a locked position. It is recommended to use two hands while rotating the P handle. See **"Figure 2-23: Patient overhead handgrip rotation" on the next page**

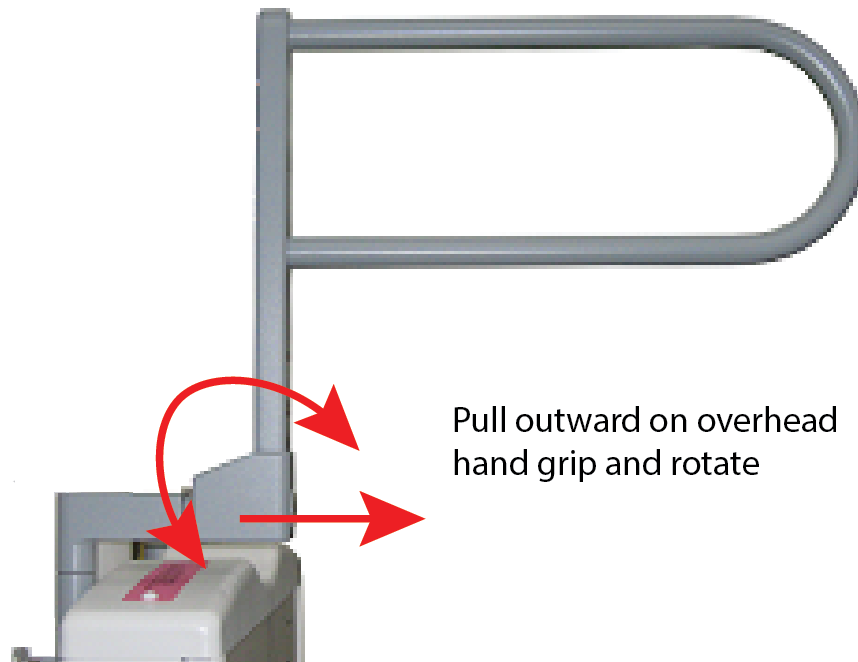


Figure 2-23: Patient overhead handgrip rotation

2.10 Tracking Motor E-Stop

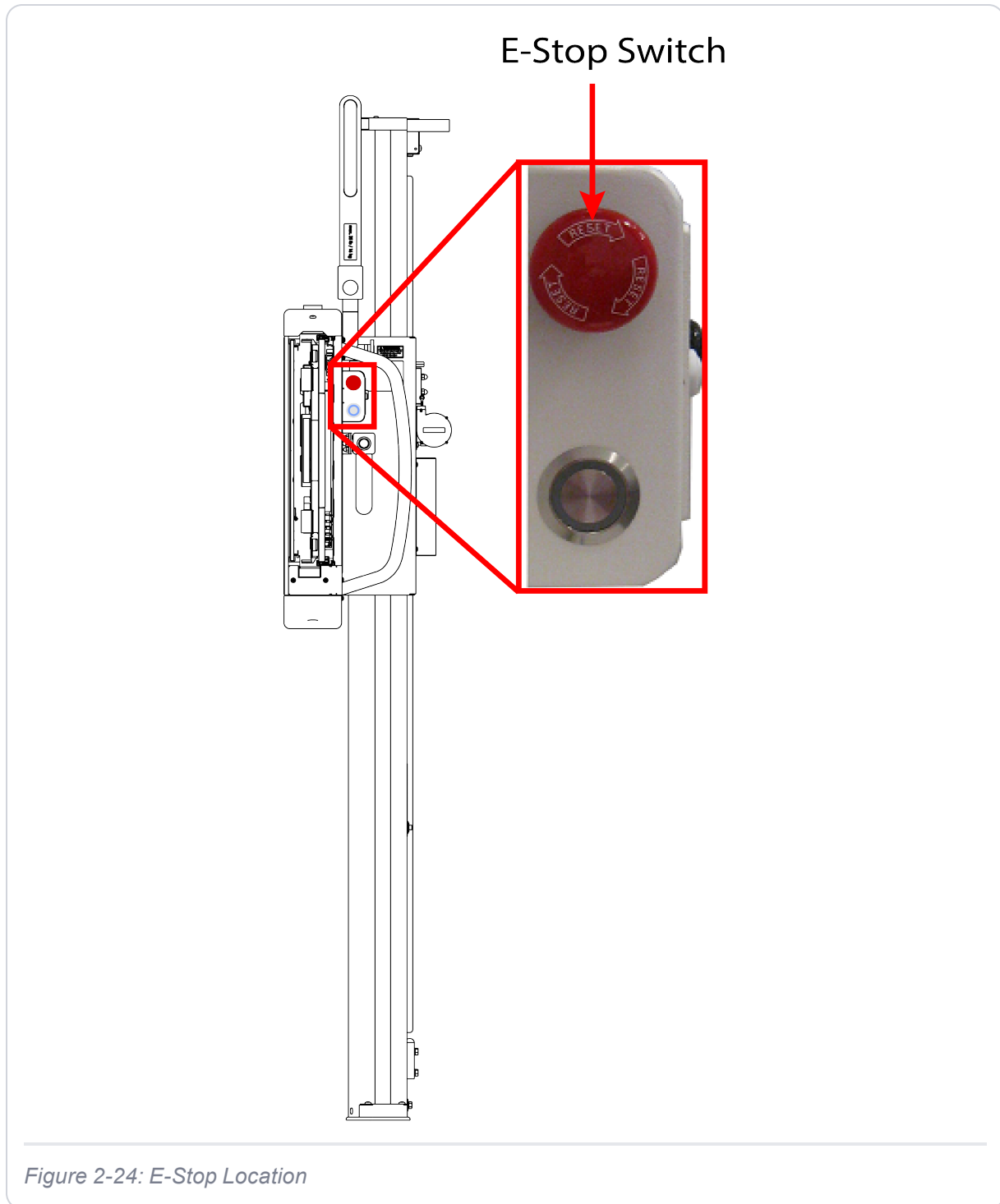


Note: The tracking motor E-Stop is only available when the wallstand is used in conjunction with the OTC18T or FMT18T.

When the wallstand is used with the OTC18T tracking overhead tube crane or FMT18T tracking floor mount tube stand, an E-Stop is available on the wall stand. This E-Stop functions to disable the drive motor in the OTC18T or FMT18T.

The E-Stop button is located on the load side of the wallstand. See *"Figure 2-24: E-Stop Location" on the next page* for the location of the E-Stop button.

When an E-Stop is active, an indication will be displayed on the touchscreen of the OTC18T or FMT18T as shown in *"Figure 2-26: E-Stop Indication on the OTC18T/FMT18T Display" on page 112*.



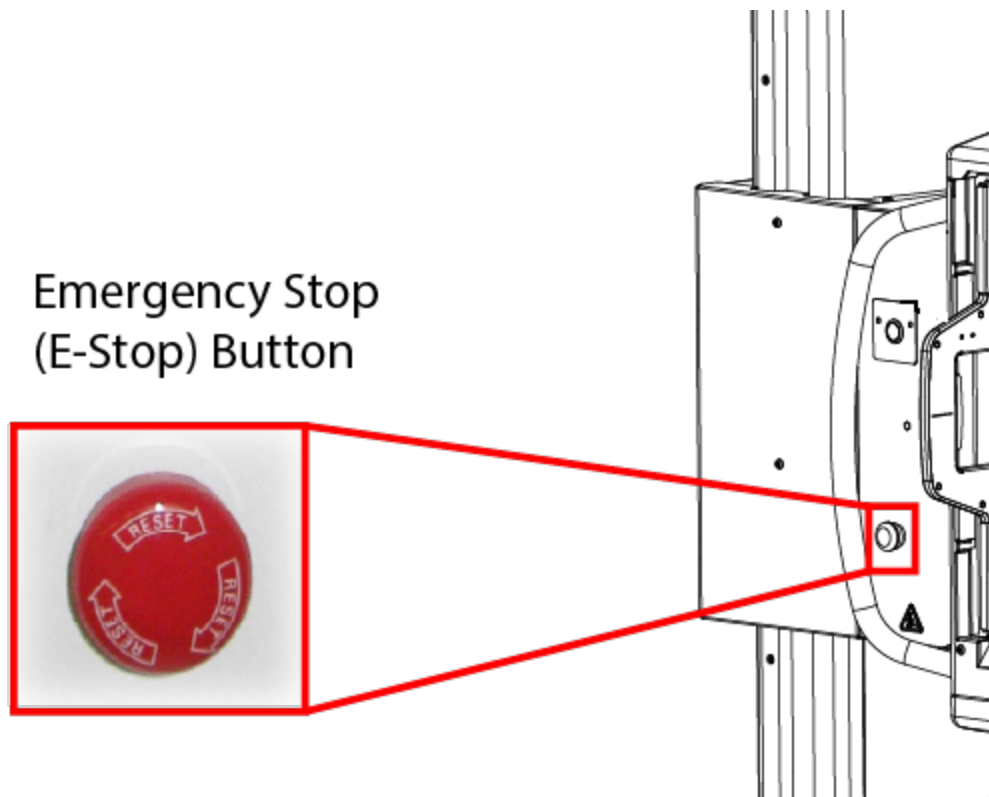
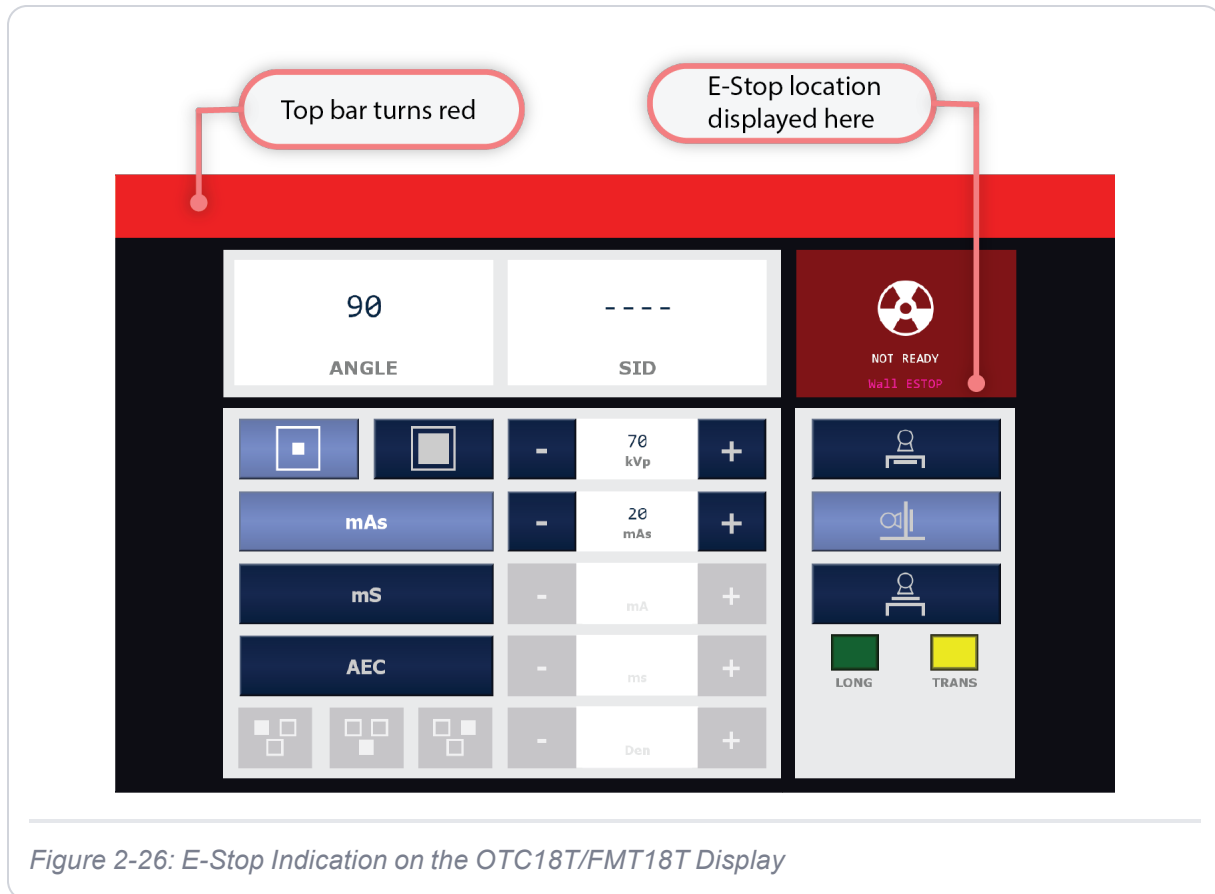


Figure 2-25: E-Stop Location



Note: Actual screen images may differ slightly based on the software version of the touchscreen.



Caution: The tracking motor E-Stop only disengages the drive motor of the tube support (OTC18T / FMT18T). The E-Stop does not remove power from any equipment and other motor drives will still be active (such as an elevating table).

2.11 Exam Positioning

The following sections are a general guide to positioning the equipment for a desired exam. The DM-OTC18T Overhead Tube Crane System has detents (auto stops) in the transverse, longitudinal, and vertical directions. These detents tell the system when it is positioned in certain areas. To allow exposure to the wall or table receptors, the tube crane must be engaged

in the proper transverse detent. See *"2.11.1 Transverse and Longitudinal Detents" below* for more information.

2.11.1 Transverse and Longitudinal Detents

To lock the DM-OTC18T in a transverse and / or longitudinal detent, only the transverse button  and / or the longitudinal button  can be used. Press and hold the button while moving the DM-OTC18T toward the desired detent. As the tube crane approaches the detent, a series of audible beeps can be heard. Once the tube crane is in position with the detent, it will automatically lock in place.

The DM-OTC18T touchscreen has detent status indicators for the longitudinal and transverse detents (*"2.4 Equipment Controls" on page 82*). These indicators will illuminate when the specific detent is active. See *"Figure 2-27: Detent status indicators" below* as an example of the indicators.



Figure 2-27: Detent status indicators

2.11.2 Table Exam

Perform the following steps to set the equipment in position for a table exam. The steps can be performed in any order, but all steps must be completed properly in order to allow an exposure.

- Ensure the detector is placed in the table tray and the tray is fully inserted into the receptor cabinet.
- Rotate the tube head so that the angle shows 0°.
- Position the tube crane so that it is engaged in the transverse detent over the center of the table. When in detent, the transverse indicator will illuminate. 

- Select the table receptor either by the acquisition software or by the touchscreen. The table receptor button will show as active. 
- Move the tube crane in the longitudinal direction so that it is in a position within the limits of the table receptor.

Once the previous steps are completed, the display of the DM-OTC18T will display a green status  and the SID between the focal spot and receptor plane will be displayed. If the status remains red, check the previous steps.

Use the receptor lock release to manually center the receptor.

Set the vertical SID by maneuvering the tube crane and / or table vertically or use the auto tracking feature.

See *"3 Acquisition Software" on page 135* to complete the exam.

2.11.3 Wall Exam

Perform the following steps to set the equipment in position for a wall exam. The steps can be performed in any order, but all steps must be completed properly in order to allow an exposure.

- Ensure the detector is placed in the table tray and the tray is fully inserted into the receptor cabinet.
- Rotate the tube head so that the angle shows 90°.
- Position the tube crane so that it is engaged in the transverse detent over the center of the wallstand. When in detent, the transverse indicator will illuminate. 
- Select the wallstand receptor either by the acquisition software or by the touchscreen. The wallstand receptor button will show as active. 
- If desired, move the tube crane to one of the longitudinal detents. When in detent, the longitudinal indicator will illuminate. 



Note: The longitudinal detents are typically set at 40" (100cm) and / or 72" (180cm) but can also be set at different intervals. The system can be configured to display or not display SID between detents. Longitudinal detents are typically not required by the system to allow exposure.

Once the previous steps are completed, the display of the DM-OTC18T will display a green status  and the SID between the focal spot and receptor plane will be displayed. If the status remains red, check the previous steps.

Use the receptor lock release to manually center the receptor, or engage the sync button the DM-OTC18T display  and the receptor will automatically center. See *"2.12 Auto Tracking" on the next page* for more information on this feature.

Center the receptor by maneuvering the tube crane and / or wallstand vertically. You can also press the auto tracking button to activate auto tracking  and the wallstand will automatically center. See *"2.12 Auto Tracking" on the next page* for more information on this feature.

See *"3 Acquisition Software" on page 135* to complete the exam.

2.11.4 Table Top Exam

Perform the following steps to set the equipment in position for a table top exam. The steps can be performed in any order, but all steps must be completed properly in order to allow an exposure.

- Rotate the tube head so that the angle shows 0°.
- Select the table top receptor either by the acquisition software or by the touchscreen. The table top receptor button will show as active.
- Position the tube crane so that it is positioned over the table. When the DM-OTC18T is positioned within the table top calibration limits, the SID to the table top will be displayed.



Note: No longitudinal detents are present over the table.

Once the previous steps are completed, the display of the DM-OTC18T will display a green status  and the SID between the focal spot and receptor plane will be displayed. If the status remains red, check the previous steps.

Set the vertical SID by maneuvering the tube crane and / or table vertically or use the auto tracking feature.

See *"3 Acquisition Software" on page 135* to complete the exam.

2.11.5 Free Exam

To perform a free exam (detector anywhere i.e. chair, floor, etc.) position the tube crane at any location outside of the table top calibration limits and activate the table top receptor



2.12 Auto Tracking

The DM-OTC18T Overhead Tube Crane System includes an automatic tracking feature. This feature automatically moves the tube vertically and the table receptor to the center position of the wall or table receptors.

2.12.1 Table Vertical (SID) Auto Tracking

To engage table auto tracking, follow the steps described in *"2.11.2 Table Exam" on page 113* or *"2.11.4 Table Top Exam" on the previous page*, and ensure the indicator is green and the SID is displayed. Press the table auto track button the touch screen display to activate auto tracking.

When auto tracking is active to the table, the tube stand will vertically follow the table at the SID that is set.

This will also function when set to table top and the DM-OTC18T is positioned within the table top calibration limits.

2.12.2 Table Receptor Auto Tracking

To engage table receptor auto tracking, follow the steps described in *"2.11.2 Table Exam" on page 113*.

Press the table receptor sync button if not already engaged.

2.12.3 Wallstand Vertical Auto Tracking

To engage table auto tracking, follow the steps described in *"2.11.3 Wall Exam" on page 114* and ensure that the indicator is green and the SID is displayed. Press the auto track button on the side of the wallstand to activate auto tracking. Once activated the ring around the button will illuminate and the symbol  will appear on the touchscreen.

When auto tracking is active to the wallstand, the tube stand will vertically follow the wallstand. This also functions when the wallstand is tilted to 90° as long as the DM-OTC18T is set in the wall center longitudinal detent and the tube is rotated to 0°.

2.13 Receptor Cabinet

Several different receptor cabinet and tray options are available. Each tray is supplied with operating instructions. Follow the instructions provided for the tray that is included with the system. The following devices contain the following receptor cabinet option(s):

Grid Cabinet

Non-motorized, can include a 14x17 receptor tray

Compatible devices:

- EV800-5171
- EV800T-5168
- VS300-5154
- VT300-5157

Bucky Cabinet

Motorized grid movement, includes a 14x17 receptor tray

Compatible devices:

- EV800-5171
- VS300-5154
- VT300-5157

Rotating Tray Cabinet

Accommodates and includes the Del Medical Inc. Rotating Tray

Compatible devices:

- EV800-5171
- EV800T-5168
- VS300-5154
- VT300-5157

Flat Panel 17x17 Fixed DR Detector Cabinet

Please see detector documentation for information

Compatible devices:

- EV800-5171
- EV800T-5168
- VS300-5154
- VT300-5157

Grid Cabinet with Flex Detector Tray (FLEXTRAY-5185)

Please refer to the Grid Cabinet with Flex Detector Tray Manuals for information on use and detector compatibility. See 8000-FLEXTRAY-5185-OP .

Compatible devices:

- EV800-5183
- EV800T-5184
- VS300-5186
- VT300-5187



Note: This is also referred to as the "Rotate Tray Cabinet with Optional Charging", please be aware as these might be used interchangeably throughout this documentation

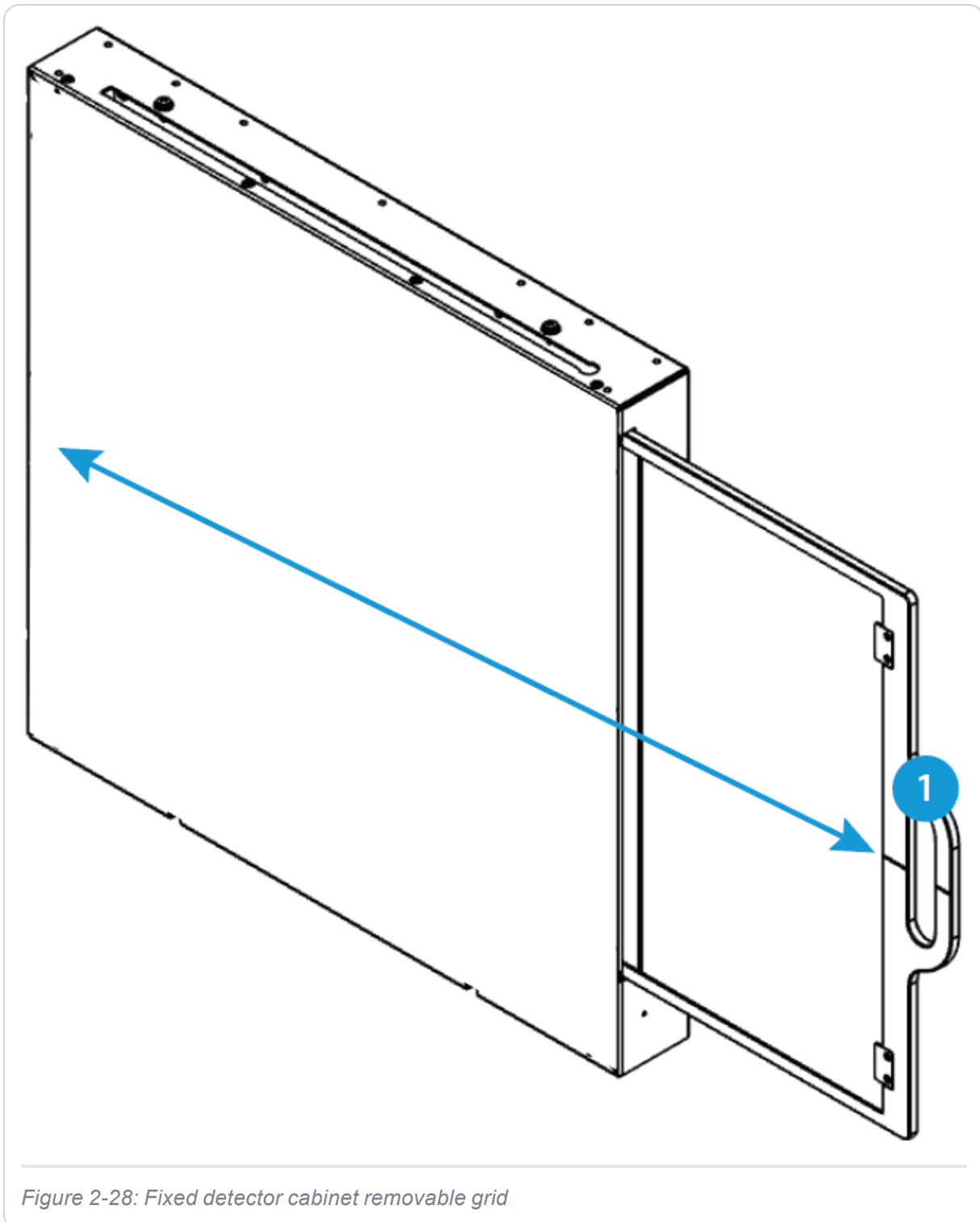


Important: For Charging Systems, use **only** the supplied detector. Using another detector can damage the FLEXTRAY-5185.

2.14 Fixed Detector Cabinet Removable Grid

If the device has a fixed detector cabinet, the detector is permanently installed and cannot be removed.

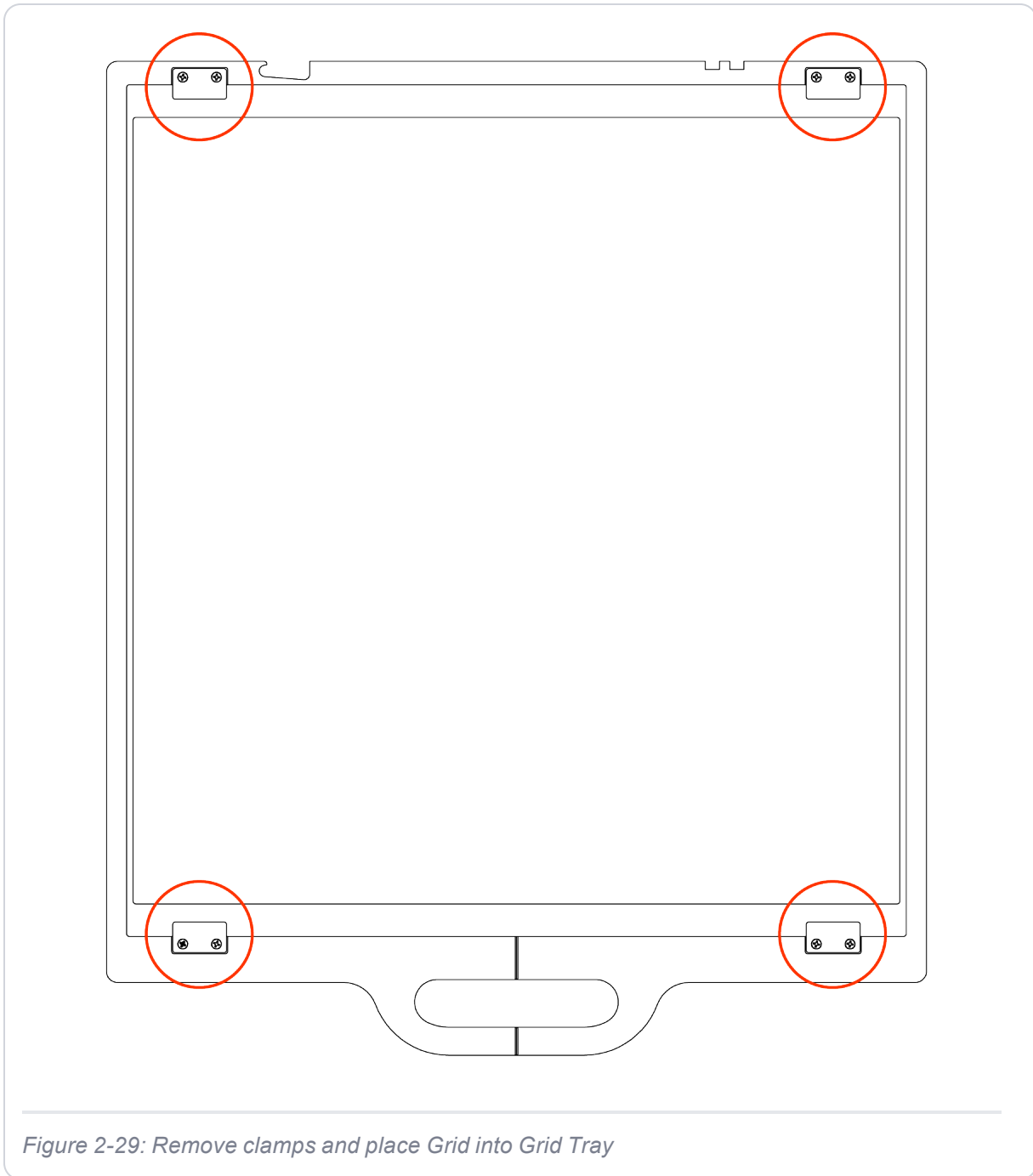
The fixed detector cabinet features a user removable grid. *"Figure 2-28: Fixed detector cabinet removable grid" on the next page*, item 1 describes the grid tray handle. Once removed, the user can replace the grid with a Phillips Screwdriver (PH1).



2.14.1 Changing the Grid

Instructions:

1. Once pulled out, locate the 4 tabs marked in red and place your Grid Tray after removing the clamps. There are two screws per clamp, use a Phillips Screwdriver (PH1) to remove the screws.



2. Once placed, screw back on the clamps. You can now reinsert the Grid Tray into the Rotate Tray Cabinet, ensure an audible click occurs after completing the installation.

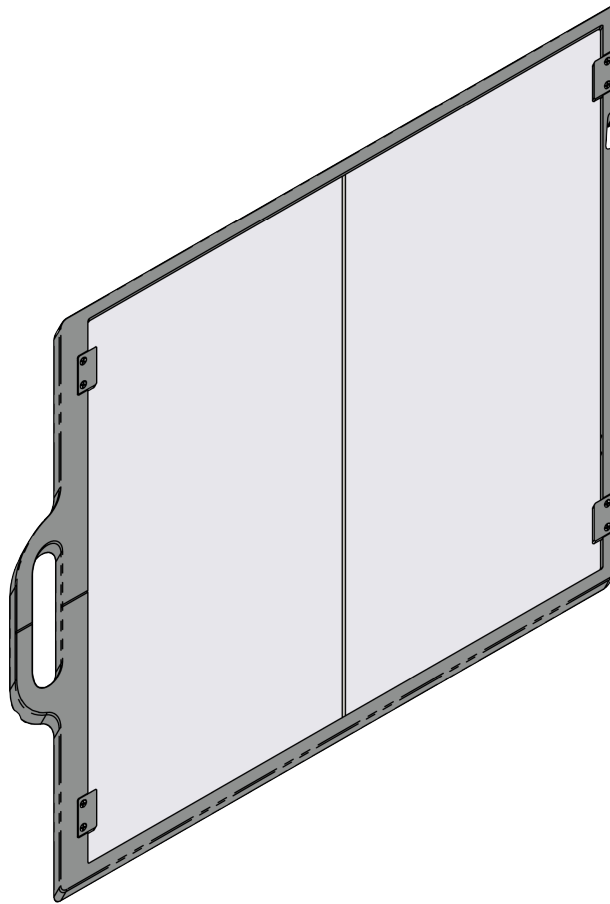


Figure 2-30: Full Assembled Grid Tray

2.15 Grid Cabinet with Flex Detector Tray (110-5185)



Note: This is also referred to as the "**Rotate Tray Cabinet with Optional Charging**", please be aware as these might be used interchangeably throughout this documentation

The Grid Cabinet with Flex Detector Tray is intended to support 17in x 17in detectors, portrait to landscape when paired with a 14in x 17in detector, in-tray charging (for select models), a removable grid frame (Grid Tray), and AEC.

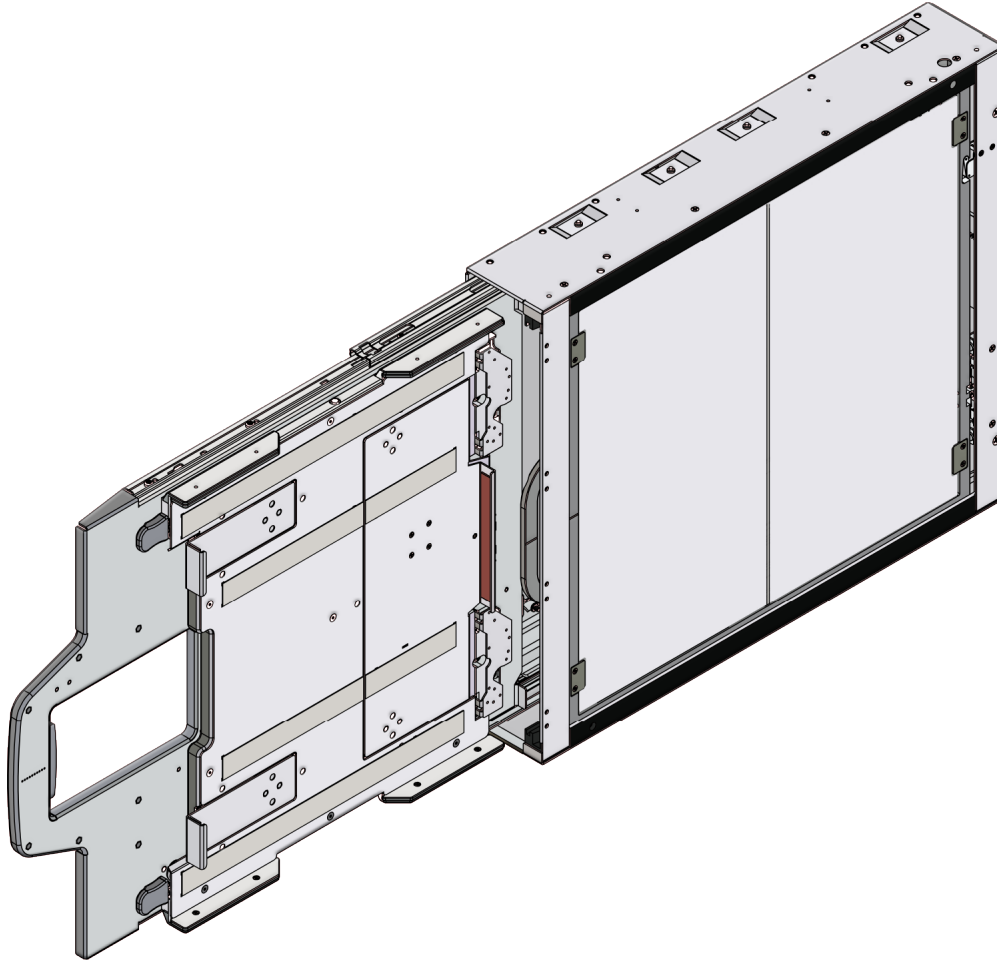


Figure 2-31: FLEXTRAY-5185 110-5185



Note: Each configuration of the tray is only compatible with their designated detector panel, using an incompatible detector panel may not work and can cause potential damage to the detector panel or cabinet and it's components.

2.16 Grid Cabinet with Flex Detector Tray Standard Features

- Compatible External Detector Panel Dimensions: 18 in x 18 in, 15 in x 18 in
- Compatible with Detector sizes that have Full, Landscape and Portrait sensing:
 - Full Sensing Area: 17 in x 17 in
 - Landscape Sensing Area: 17 in x 14 in
 - Portrait Sensing Area: 14 in x 17 in
- Tray equipped with charging feature for select models, see *"1.1 Specifications" on page 1*.
- Configurable to either direction when rotating
- Removable Grid Tray/ Detector
- ION Chamber (ICX-2122/3122)

2.17 How to Operate Cabinet

Follow the steps below to operate the Grid Cabinet with Flex Detector Tray.

Step by Step:

1. When inserting the detector, lay flat towards the cabinet bumpers (these bumpers will vary depending on the cabinet purchased).
2. Ensure it is secured with the cabinet clamps shown in *"Figure 2-32: Grid Cabinet with Flex Detector Tray Clamps and Bumpers" on the next page*.
3. Once secured, you can rotate the tray by swinging the tray either clockwise or counterclockwise by pressing onto the gray tabs shown in *"Figure 2-33: Grid Cabinet with Flex Detector Tray Tabs" on page 127*.



Note: For **Fuji** detectors, a lockout must occur meaning the detector should be pushed into the cabinet bumpers to lock out the tray and allow the detector to lay flat.

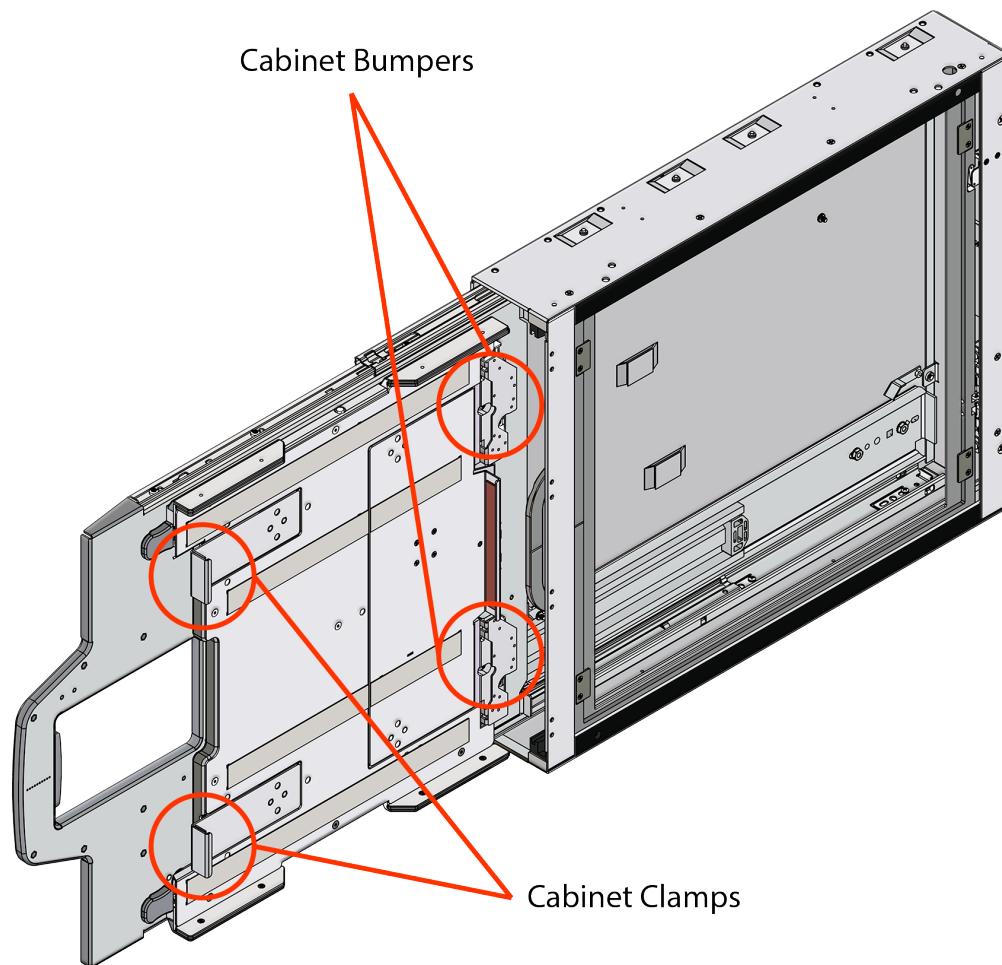


Figure 2-32: Grid Cabinet with Flex Detector Tray Clamps and Bumpers

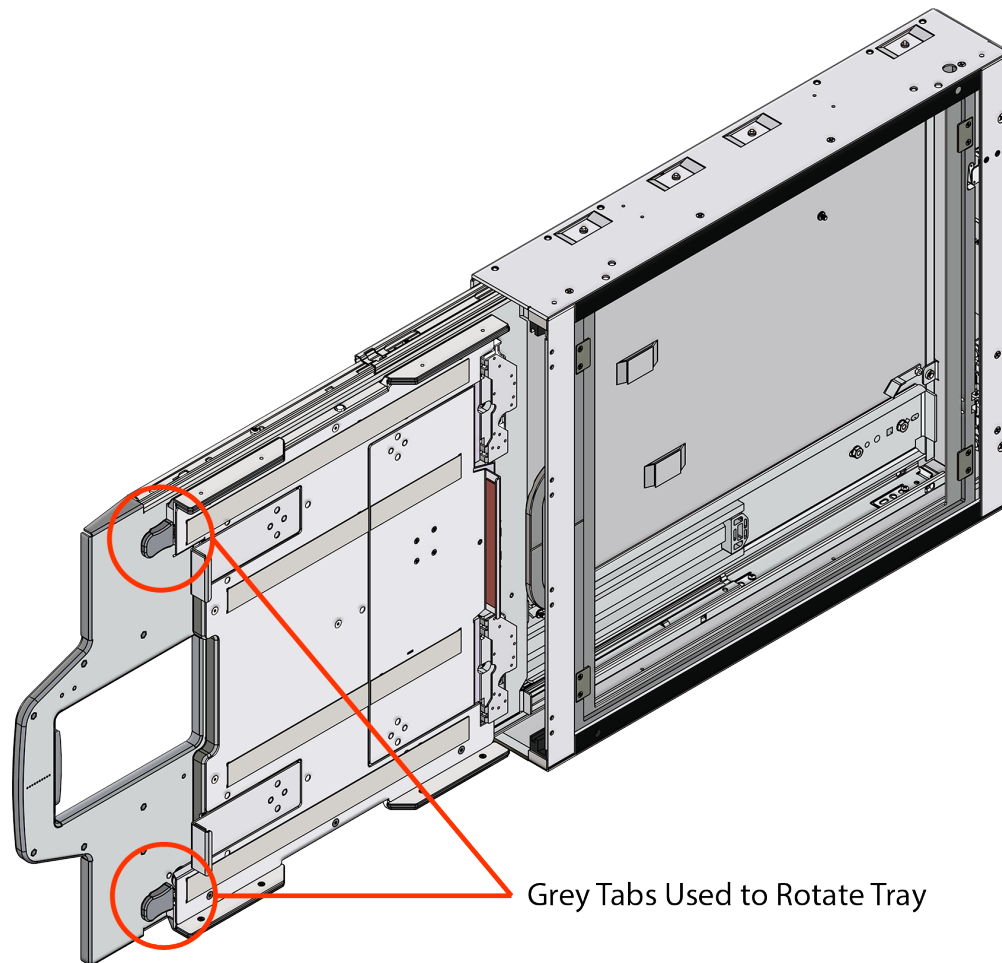


Figure 2-33: Grid Cabinet with Flex Detector Tray Tabs

2.18 Charging Detector Compatibility

This cabinet is designed to accommodate a range of X-ray detectors. Compatibility may vary depending on the specific detector model, and some adjustments may be required during installation. See the Grid Cabinet with Flex Detector Tray Operation Manual.

If you have questions about detector compatibility, consult your service provider or contact Del Medical Inc. at support@delmedical.com.



Note: Equipment damage may occur if an incorrect detector is placed into the FLEXTRAY-5185, please consult any accompanying cabinet manual for information or contact support.

2.19 Changing the Grid

The Grid within the Grid Cabinet with Flex Detector Tray is removable and can be replaced. Follow the instructions below on how to install it.

Tools Required:

- Phillips Screwdriver (PH1)

Instructions:

1. The Grid Cabinet with Flex Detector Tray is equipped with a removable Grid Tray.

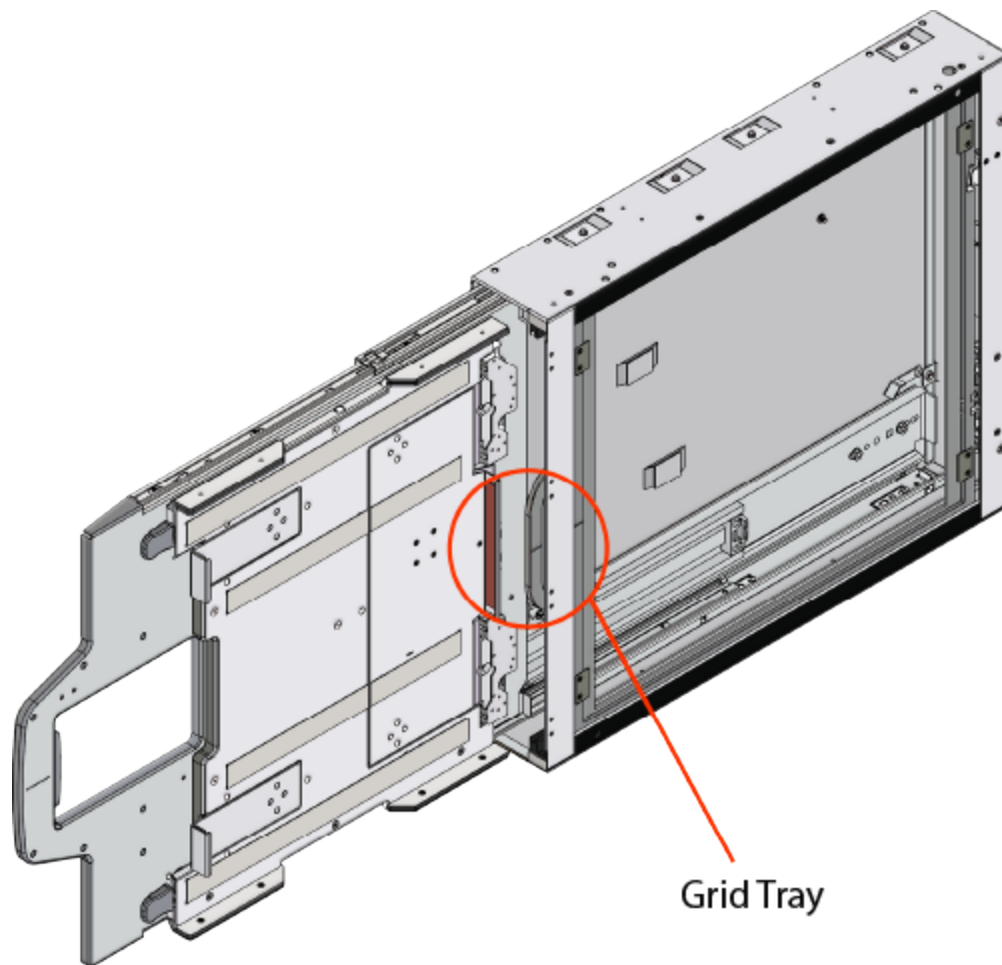


Figure 2-34: Grid Tray Location

2. To remove the Grid Tray, push the handle of the Grid Tray into the FLEXTRAY-5185 for the spring loaded latch to disengage, as shown below. An audible click can be heard. Once the click occurs, go ahead and pull out the Grid Tray.

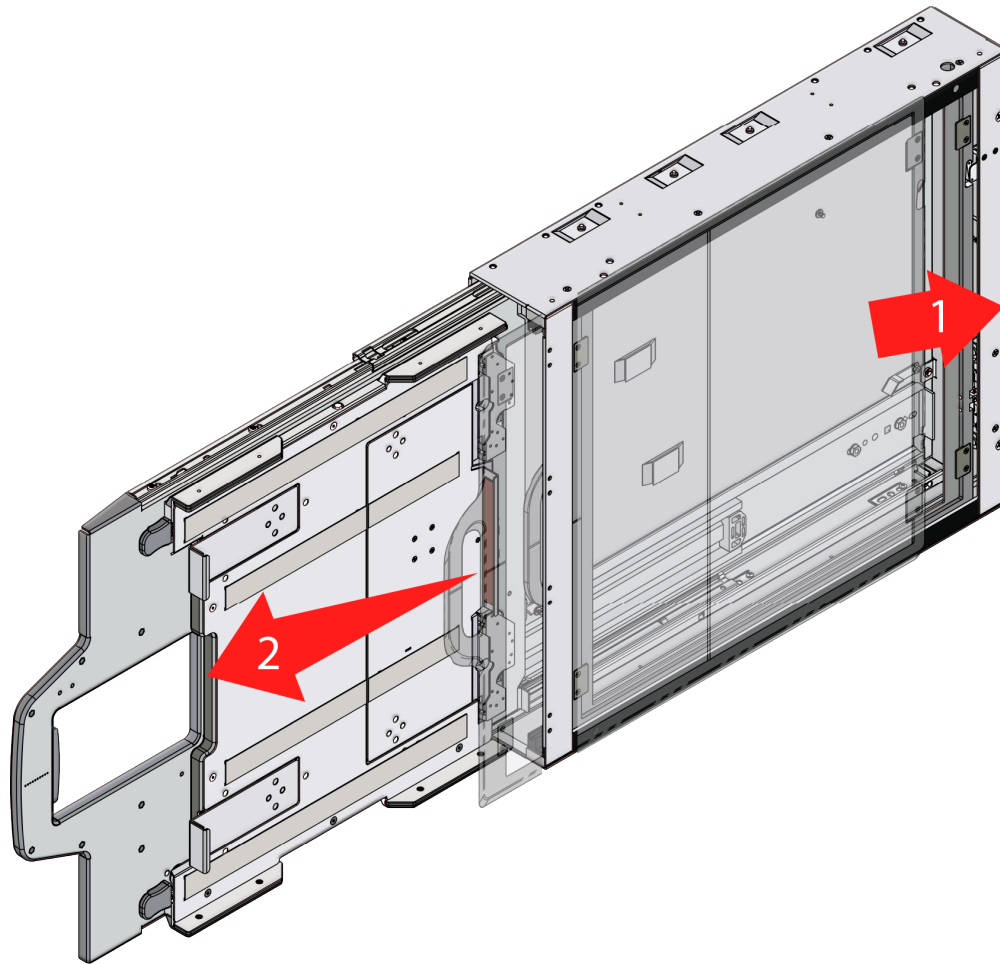
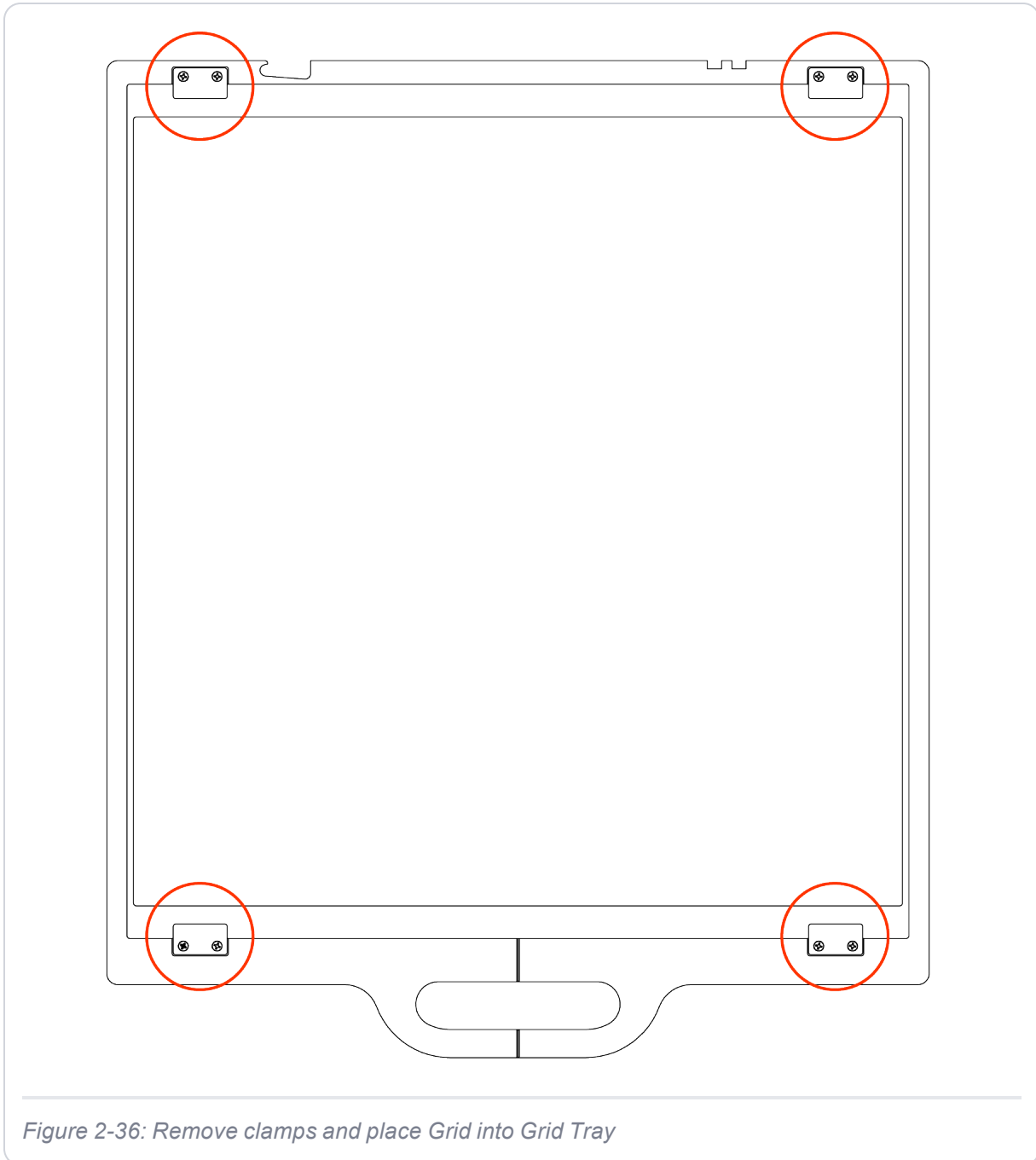


Figure 2-35: Push forward and then pull out Grid Tray

1. Once pulled out, locate the 4 tabs marked in red and place your Grid Tray after removing the clamps. There are two screws per clamp, use a Phillips Screwdriver (PH1) to remove the screws.



2. Once placed, screw back on the clamps. You can now reinsert the Grid Tray into the Rotate Tray Cabinet, ensure an audible click occurs after completing the installation.

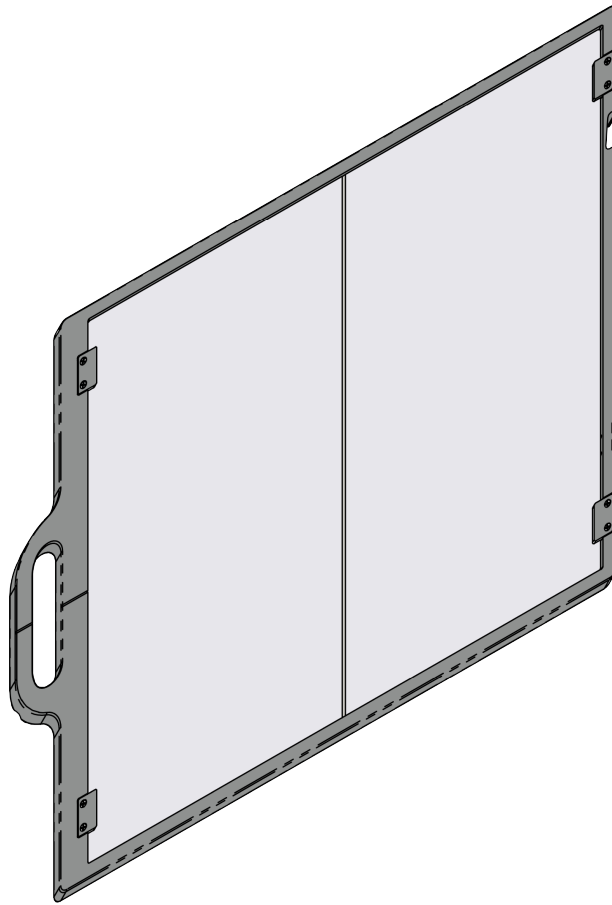


Figure 2-37: Full Assembled Grid Tray

Instructions:

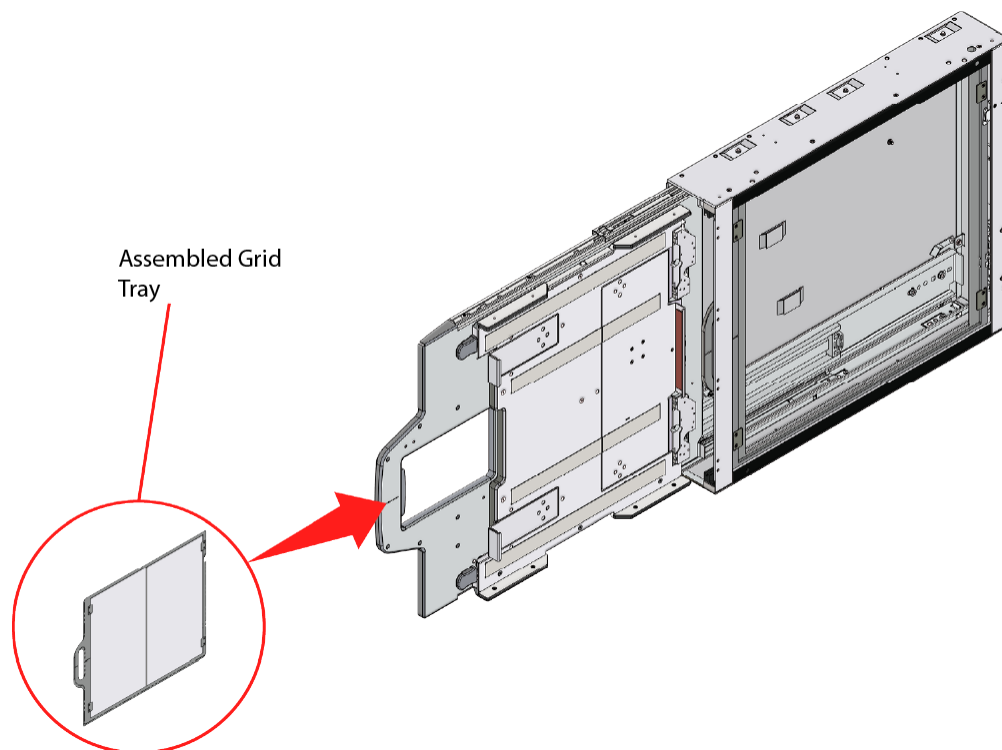


Figure 2-38: Place the assembled Grid Tray back into FLEXPTRAY-5185, an audible click should occur

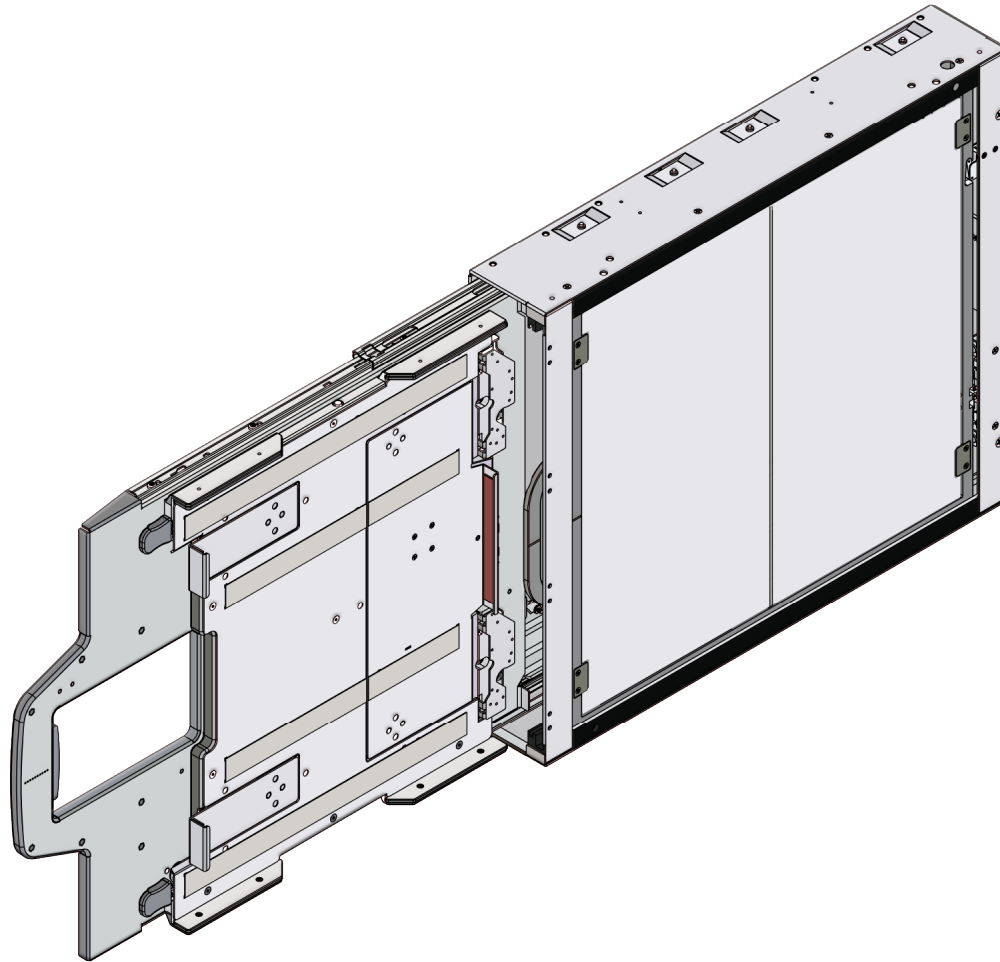


Figure 2-39: Installation Complete

3 Acquisition Software

3.1 Introduction

An Image Acquisition System refers to the system of components responsible for detecting the radiation which has passed through a patient and producing a diagnostic quality image. The DM-OTC18T Overhead Tube Crane System is compatible with various different brands of DR (direct radiography) image acquisition systems. Compatibility means the DR system has been tested and the system is configured to function with a specific DR. DR compatibility refers to automatic collimator control and / or generator communication and control. Not all DR systems that are compatible with generator communication are compatible with automatic collimator control.

3.1.1 DR Compatibility with the System Generator

Compatibility with the generator refers to communication with the generator to control technique parameters and exposure release. The DR does NOT directly control the prep or exposure commands of the generator. The exposure release restricts the exposure from taking place until DR system determines the panel is ready to receive radiation.

3.1.2 DR Compatibility with the System Collimator



Note: This section is only applicable when the AL02 automatic collimator is installed on the system. If the ML03 manual collimator is installed, this section does not apply.

Compatibility with the collimator refers to communication between the DR system and collimator so the collimator can size to preset collimator sizes set in the view.

3.1.3 DR Compatibility Matrix

The table below describes the compatibility between various DR systems which have been tested, the system offered generators, and automatic collimation.

Generator	DR Brand					
	DelWorks	Canon	Samsung	Fuji	Konica	Viewworks
CM Series	O	O	O	O	X	X*
RF Series	O	-	-	O	-	-
O = Compatible with generator and automatic collimator X = Compatible with generator only - = Not compatible * = One way communication						
<i>Table 3-1: DR Compatibility</i>						



Note: This table describes compatibility in regards to DR systems which have been tested with the DM-OTC18T Overhead Tube Crane System by Del Medical Inc.. Other DR systems may be installed with the system but Del Medical Inc. cannot guarantee functionality.

3.2 DR Operation and Workflow

The operation and work-flow of the DR system installed with the DM-OTC18T Overhead Tube Crane System is described in the documentation for the DR system. It is important that the documentation is read and understood prior to using the system.

3.3 Non-DR Applications

If the DM-OTC18T Overhead Tube Crane System is installed with a non-DR acquisition system, such as CR (computer radiography) or film, compatibility does not apply. These image acquisition systems do not integrate into the DM-OTC18T Overhead Tube Crane System.

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4 Image Quality

4.1 Overview

This chapter describes the functions of the DM-OTC18T Overhead Tube Crane System to achieve acceptable image quality. Selecting proper generator techniques to achieve a proper Deviation Index (DI) on the acquisition software is the goal to producing diagnostic quality images.

All information within this chapter assumes that the periodic maintenance schedule is followed. Calibration of the tube, AEC, and detector are part of normal periodic maintenance and need to be performed by qualified service personnel and at the intervals recommended in *"5 Periodic Maintenance" on page 145*.



Caution: Failure to follow these guidelines and failure to keep the equipment properly maintained could cause poor image quality and unnecessary retakes / over-exposure.

4.2 Radiographic Performance

4.2.1 CM Series Generator Radiographic Performance

"Table 4-1: CM Series Generator Radiographic Performance" on the next page shows the radiographic performance of the CM Series generator.

Description	Values
kVp range	40 kV to 150 kV
kVp steps	Variable in 1 kV steps
kVp accuracy	$\pm (5\% + 1 \text{ kV})$ measured 5 ms after the beginning of the exposure; $\pm 2\%$ between 70-80 kVp measured between the 75% points of the kV waveform.

Description	Values
Rise time (10%-90%)	<1.5 ms (typically <1.0ms) with 100 ft (30 m) Locaflex L3 or equivalent HV cables
Time range	1.0 to 6300 milliseconds
Exposure time steps	Variable in 1 ms steps via protocol
Exposure time accuracy	$\pm(2\% + 0.5 \text{ ms})$ from 5 ms to 6300 ms and $> 0.5 \text{ mAs}$ $\pm(10\% + 1 \text{ ms})$ for $>0.1 \text{ mAs}$ and for $< 5 \text{ ms}$ or $\leq 0.5 \text{ mAs}$ for 100 ft (30 m) HV cables Measured between the 75% points of the kV waveform
mAs range	0.1 to 630 mAs (50 kW) 0.1 to 800 mAs (65 kW) 0.1 to 1000 mAs (80 kW) Note for Minimum mAs: mAs Mode: 0.3 mAs ($> 60 \text{ kV}$, 28 mA, 11 ms) mA, ms Mode: 0.3 mAs ($> 60 \text{ kV}$, 10 mA, 30 ms) mAs or mA, ms Mode: 0.1 mAs (40 kV - 60 kV, 10 mA, 10 ms)
mAs accuracy	$\pm(10\% + 0.2 \text{ mAs})$ for $> 0.5 \text{ mAs}$ $\pm(10\% + 0.05 \text{ mAs})$: 0.1 mAs – 0.5 mAs (preliminarily specified for the range beyond IEC standard)
mA range	10 to 630 mA (50 kW) 10 to 800 mA (65 kW) 10 to 1000mA (80 kW)
mA steps	Variable in 0.1 mA steps via protocol

Table 4-1: CM Series Generator Radiographic Performance

4.2.2 RF Series Generator Radiographic Performance

"Table 4-2: RF Series Generator Radiographic Performance" on the next page shows the radiographic performance of the RF Series generator.

Description	Values
kVp range	40 kV to 150 kV
kVp accuracy	± 5%
Maximum tube current (RF)	@60 kV 500mA (30kW) 640 mA (55kW) @80 kV 375 mA (30kW) 550 mA (55kW) @100 kV 300 mA (30kW) 550mA (55kW) @125 kV 241 mA (30kW) 443 mA (55kW) @150 kV 200 mA (30kW) 366 mA (55kW)
Maximum tube current (RF ESU)	@60 kV 500mA (30kW) 640 mA (55kW) @80 kV 375 mA (30kW) 550 mA (55kW) @100 kV 300 mA (30kW) 550mA (55kW) @125 kV 240 mA (30kW) 324 mA (55kW) @150 kV 173 mA (30kW) 173 mA (55kW)
Exposure time	1-point technique: 1 ms to 5 s with mAs post indication 2-point technique: 1 ms to 5 s depending on mAs and kV 3-point technique: 20 ms to 5 s depending on mAs and kV
Maximum mAs (RF)	@60 kV 800mAs @80 kV 700mAs @100 kV 560 mAs @125 kV 444 mAs @150 kV 373 mAs
Maximum mAs (RF ESU)	@60 kV 500mAs @80 kV 375mAs @100 kV 300 mAs @125 kV 250 mAs @150 kV 200 mAs
mAs accuracy	± (10% + 0.2 mAs)

Description	Values
Automatic control	0 point with DR mode 1 point with continuously falling load 2 point with constant load 3 point with constant load either mAs or mA (not with Button console)

Table 4-2: RF Series Generator Radiographic Performance

4.3 Deviation Index (DI)

The Deviation Index (displayed as “DI”) is a value displayed within the acquisition software. The DI value refers to the difference between the required radiation dose needed by the detector for image quality and the actual dose received by the detector. The optimal DI value is 0.0, this determines the detector received the correct amount of dose to produce a high quality image. Each panel has different dose requirements and the DI value displayed will represent the value for the detector used with the system.

If the DI value is negative, this means the dose received was too low. If the DI value is positive, the dose was too high.

If the DI value is too low or too high, the technique will need to be adjusted accordingly. See the documentation for the image acquisition system for information on deviation index and other information on image quality.

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5 Periodic Maintenance

5.1 Introduction

This chapter describes Periodic Maintenance information for the DM-OTC18T Overhead Tube Crane System that are necessary to ensure the correct and reliable function of the device after installation.

5.2 Periodic Maintenance Schedule

To assure continued safe performance of this X-ray equipment, the owner must establish a preventive maintenance program. Maintenance procedures for the DM-OTC18T Overhead Tube Crane System are required within 30 days after installation and every 6 months thereafter (unless otherwise noted in *"5.2 Periodic Maintenance Schedule" above*). Safe equipment performance also requires that these procedures be performed by service personnel specifically trained and experienced with medical X-ray apparatus.

Refer to the schedule below for information on when to perform periodic maintenance on the DM-OTC18T Overhead Tube Crane System.

**Note:** Due to varying operating conditions, these procedures may need to be performed at longer or shorter intervals, depending on the equipment’s performance.

**Warning:** Failure to adhere to the manufacturer recommended periodic maintenance schedule may result in equipment malfunction. The impact of equipment malfunctions can range from undesired operation to severe personal injury.

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5.2.2 Inspecting Equipment Movement	147
5.2.3 General Periodic Maintenance	147
5.2.4 Inspect Other Components	147

5.2.1 Cleaning External Surfaces

Schedule: Daily

Supplies Required:

- Cleaning Wipes
- Non-abrasive, hospital-grade cleaner

Use cleaning wipes and non-abrasive, hospital-grade cleaner to clean external surfaces to reduce risk of infection



Warning: Before servicing this equipment, turn off electrical power to this component and all its peripheral equipment (generator, tubestand, etc.) at the power sources. Also, ensure that the power sources are locked out and tagged “Equipment Being Serviced”. Some components in the system/component have other power sources, which is why all peripheral equipment must be turned off to avoid injury.

Observe the required radiation protective measures for tasks that must be performed with radiation switched on. When needed, wear protective clothing.



Warning: This equipment is NOT classified as anesthetic-proof and may ignite inflammable anesthetics. Flammable agents used for skin cleaning or disinfecting may also produce an explosion hazard.

- Ensure the power has been disconnected and that the emergency switches have been activated before starting any cleaning operation.
- Ensure no liquid enters the unit.
- Do not immerse the equipment, including any components or accessories in liquid.
- Do not autoclave the equipment, including any component or accessories.
- Do not use water. Water can short-circuit the electrical installation and cause corrosion to mechanical parts.
- Do not use acid or abrasive products.
- Use only a dry cloth to clean chrome-plated parts.

- Only the surface areas of unit parts, including accessories and connection cables, should be disinfected using a gaseous disinfectant. For safety reasons, do not spray disinfectants. Pour onto a cloth to moisten.
- Clean painted parts with a cloth and use products appropriate for cleaning plastic materials. After cleaning, wipe the surfaces with a clean, dry cloth.
- Do not spray cleaning or disinfection solution directly on the equipment. To disinfect, moisten a cloth with a 70% Isopropyl alcohol solution or equivalent and wipe the surface of the equipment.
- When disinfecting the examination room, ensure the unit is covered with plastic sheets.

5.2.2 Inspecting Equipment Movement

Schedule: Daily

Tools Required:

- None

Instructions:

1. Inspect all moving parts for unobstructed movement.
2. Test all aspects of tube crane movement and braking.

5.2.3 General Periodic Maintenance

Periodic maintenance should be performed by qualified service personnel only. Periodic maintenance should be scheduled to be performed at intervals of Every 6 Months.



Warning: Do not attempt to disassemble or service the DM-OTC18T Overhead Tube Crane System. This should be done by qualified service personnel only. Any attempt to disassemble or service the equipment by an untrained individual could result in severe injury.

5.2.4 Inspect Other Components

Please refer to any corresponding documentation for the Tube, Generator, Collimator, Cabinet, etc. for Periodic maintenance Information.

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